Quantitative Finance Problems & Solutions

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Quantitative Interview Problems

English Edition

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1

Probability

1.1 Basic and Conditional Probability

Key Ideas and Methods. Conditional and unconditional independence must be distinguished. A latent coin type, a common failure, or selection can alter the joint observation law. Write likelihoods for the information actually recorded, then normalize using total probability and Bayes' formula. When an action is required, compare conditional expected payoffs with the stated losses included.

Symmetry can simplify probabilities and motivate dependent samples that reduce estimation variance, but any atom at zero must be retained. Under stopped sampling, use the stopping-count distribution to check unbiasedness. For jointly Gaussian variables, an orthogonal change of coordinates can isolate the direction affected by selection; correlations after selection must then be computed from the conditional moments.

Example P02. Antithetic Estimation under Symmetry

Suppose S and $-S$ have the same distribution, and write $\tau = \mathbb{P}(S = 0)$. To estimate $p = \mathbb{P}(S > 0)$, compare

$$A = \frac{\mathbf{1}\{S_1 > 0\} + \mathbf{1}\{S_2 > 0\}}{2}, \qquad B = \frac{\mathbf{1}\{S > 0\} + \mathbf{1}\{-S > 0\}}{2},$$

where S_1, S_2 are independent copies of S. Prove that both estimators are unbiased, find their variances, and determine when the antithetic estimator has no random error.

Solution. Symmetry gives $p = \mathbb{P}(S < 0) = (1 - \tau)/2$, so $\mathbb{E}A = \mathbb{E}B = p$. Independence of the two indicators in A yields

$$\operatorname{Var} A = \frac{p(1-p)}{2} = \frac{1 - \tau^2}{8}.$$

The variables S and $-S$ cannot both be positive. Unless $S = 0$, exactly one is positive. Consequently,

$$B = \frac{1 - \mathbf{1}\{S = 0\}}{2}, \qquad \operatorname{Var} B = \frac{\tau(1-\tau)}{4},$$

and hence

$$\boxed{\operatorname{Var} A - \operatorname{Var} B = \frac{(1-\tau)^2}{8} \geq 0.}$$

For $0 \le \tau < 1$, the variance ratio is $2\tau/(1+\tau)$. In particular, $\tau = 0$ implies $B \equiv 1/2$: pairing removes all random error. If $\tau = 1$, both estimators are identically zero, so their variance ratio is undefined.

Method. The improvement comes from negative dependence. Two evaluations with the same marginal law need not be independent; pairing opposite signs leaves only the uncertainty of the atom at zero.

Example P03. Estimating a Probability after Stopping

Independent Bernoulli trials have success probability $p \in (0,1)$. Sampling stops at the rth success, where $r \ge 2$, and T denotes the number of trials. Prove that $(r-1)/(T-1)$ is unbiased for p, and determine whether the observed success proportion r/T is unbiased.

Solution. For $T = t$, the first $t-1$ trials must contain exactly $r-1$ successes and the last trial must succeed. Thus

$$\mathbb{P}(T = t) = \binom{t-1}{r-1} p^r (1-p)^{t-r}, \quad t \ge r.$$

The identity

$$\frac{r-1}{t-1}\binom{t-1}{r-1} = \binom{t-2}{r-2}$$

and the negative binomial series give, with $j = t - r$,

$$\begin{aligned}\mathbb{E}\frac{r-1}{T-1} &= p^r \sum_{j=0}^{\infty} \binom{j+r-2}{r-2}(1-p)^j \\ &= p^r p^{-(r-1)} = p.\end{aligned}$$

Moreover, for every possible t,

$$\frac{r}{t} - \frac{r-1}{t-1} = \frac{t-r}{t(t-1)} \ge 0.$$

Since $\mathbb{P}(T > r) > 0$, the inequality is strict after taking expectations. Hence $\boxed{\mathbb{E}(r/T) > p}$: the natural proportion is upward biased. At the boundary $p = 1$, both estimates equal 1 because $T = r$.

Method. The stopping rule forces the last observation to be a success. Removing that forced success and using the preceding $T-1$ trials corrects the bias; fixed-sample intuition alone does not justify a stopped-sample estimator.

Example P30. A Fixed Source or a Fresh Source

Two coin types have head probabilities $1/4$ and $3/4$. Protocol F selects a type with equal probabilities once and keeps it. Protocol R independently selects a type with equal probabilities before every toss. Tosses are independent conditional on their selected types. The two protocols have equal prior probabilities. Find the posterior probability of F after the ordered outcomes HH and HT. Also find the

covariance of the first two head indicators under each protocol and explain why one observation cannot distinguish them.

Solution. Both protocols have marginal head probability $1/2$ on every toss. A single outcome therefore has the same likelihood under either protocol and leaves the prior unchanged. For two outcomes,

	HH	HT
F	$\frac{1}{2}[(1/4)^2 + (3/4)^2] = 5/16$	$3/16$
R	$1/4$	$1/4$

With equal priors, Bayes' formula gives

$$\boxed{\mathbb{P}(F \mid HH) = \frac{5}{9}, \qquad \mathbb{P}(F \mid HT) = \frac{3}{7}.}$$

Let I_1, I_2 be the head indicators. Under F, the latent head probability P is equally likely to be $1/4$ or $3/4$. Conditional independence implies

$$\mathrm{Cov}(I_1, I_2) = \mathbb{E}(P^2) - (\mathbb{E}P)^2 = \mathrm{Var}(P) = \frac{1}{16}.$$

Under R, independent selection and tossing make I_1, I_2 independent, so their covariance is zero.

Method. Identical one-observation distributions can conceal different dependence structures. Repeated outcomes favor a fixed source, whereas alternating outcomes relatively favor a fresh source.

Example P31. Competing Signals with Simultaneous Arrivals

In each round, signals A and B occur independently with probabilities α and β, where $0 < \alpha, \beta < 1$. Different rounds are independent. Stop at the first round containing at least one signal. Let T be that round and let J record whether only A, only B, or both occurred. Find the joint law of (T, J), determine whether they are independent, and compute the mean and variance of T.

Solution. Put $q = (1-\alpha)(1-\beta)$ and $s = 1 - q$. The three stopping causes have one-round probabilities

$$w_A = \alpha(1-\beta), \quad w_B = (1-\alpha)\beta, \quad w_{AB} = \alpha\beta,$$

whose sum is $s = \alpha + \beta - \alpha\beta$. Stopping in round n requires $n-1$ rounds with neither signal, so

$$\mathbb{P}(T = n, J = j) = q^{n-1} w_j = (s q^{n-1}) \frac{w_j}{s}, \quad n \geq 1.$$

The two factors are precisely the marginal probabilities. Thus T and J are independent. Even conditioning on simultaneous signals at termination leaves the waiting-time law unchanged. Differentiating the geometric series gives

$$\boxed{\mathbb{E}T = \frac{1}{s}, \qquad \mathrm{Var}\, T = \frac{q}{s^2}, \qquad \mathbb{P}(J = j) = \frac{w_j}{s}.}$$

Caution. The total stopping probability is not $\alpha + \beta$, because both signals occur with probability $\alpha\beta$. Independence of time and cause must be established by factorization, not by their different interpretations.

Example P33. Majority Voting with a Common Failure

A binary truth T has prior $\mathbb{P}(T = 1) = \pi \in (0, 1)$. With probability $c \in [0, 1]$, a common failure makes all three sensors report the opposite truth. Otherwise, the sensors independently flip the truth with probability $\varepsilon \in (0, 1/2)$. The failure mechanism is independent of T. Find the majority error rate, the posterior after a majority report of 1, and the improvement over one sensor. Also compute the covariance of two sensors' error indicators.

Solution. Without a common failure, the majority is wrong when exactly two or three sensors err. This has probability

$$3\varepsilon^2(1 - \varepsilon) + \varepsilon^3 = 3\varepsilon^2 - 2\varepsilon^3.$$

Thus the majority and single-sensor error rates are

$$e = c + (1 - c)(3\varepsilon^2 - 2\varepsilon^3), \quad e_1 = c + (1 - c)\varepsilon.$$

Their difference is

$$e_1 - e = (1 - c)\varepsilon(1 - \varepsilon)(1 - 2\varepsilon) > 0$$

when $c < 1$. The common-failure component cannot be removed by voting. If M is the majority report, symmetry of the error mechanism gives

$$\boxed{\mathbb{P}(T = 1 \mid M = 1) = \frac{\pi(1 - e)}{\pi(1 - e) + (1 - \pi)e}.}$$

For error indicators E_i, E_j, the probability of two errors is $c + (1 - c)\varepsilon^2$. Therefore

$$\boxed{\operatorname{Cov}(E_i, E_j) = c + (1 - c)\varepsilon^2 - e_1^2 = c(1 - c)(1 - \varepsilon)^2.}$$

Method. Condition first on the common failure and apply a binomial calculation only within the no-failure case. Treating the sensors as independent misses a source of error that averaging cannot remove.

Example P12. Correlation Induced by Selection

Let X, Z be independent standard normal variables. Retain only observations with $X + Z > c$, where $c \geq 0$. Find the retained population's means, variances, covariance, and correlation. Prove that the correlation is negative and evaluate the case $c = 0$. Write ϕ, Φ for the standard normal density and distribution function.

Solution. Rotate the coordinates:

$$S = \frac{X + Z}{\sqrt{2}}, \quad D = \frac{X - Z}{\sqrt{2}}, \quad a = \frac{c}{\sqrt{2}}.$$

Then S, D are independent standard normals. Selection restricts only $S > a$, leaving D standard normal and independent of the truncated S. Integration by parts using $\phi'(s) = -s\phi(s)$ yields

$$\lambda := \mathbb{E}(S \mid S > a) = \frac{\phi(a)}{1 - \Phi(a)}, \quad \mathbb{E}(S^2 \mid S > a) = 1 + a\lambda.$$

Put $v = 1 + a\lambda - \lambda^2$. Since $X = (S + D)/\sqrt{2}$ and $Z = (S - D)/\sqrt{2}$, all moments below are conditional on selection:

$$\boxed{\begin{aligned} &\mathbb{E}X = \mathbb{E}Z = \lambda/\sqrt{2}, \qquad \operatorname{Var} X = \operatorname{Var} Z = (v+1)/2, \\ &\operatorname{Cov}(X, Z) = (v-1)/2, \qquad \operatorname{Corr}(X, Z) = \frac{v-1}{v+1}. \end{aligned}}$$

Under the conditional law $S > a$, we have $\lambda > a$. As $\lambda > 0$, it follows that $v = 1 - \lambda(\lambda - a) < 1$, proving negative correlation. Nondegeneracy also gives $v > 0$.

For $c = 0$, $\lambda = \sqrt{2/\pi}$ and $v = 1 - 2/\pi$. Each mean is $1/\sqrt{\pi}$, each variance is $1 - 1/\pi$, the covariance is $-1/\pi$, and the correlation is $-1/(\pi - 1)$.

Method. Selection based on a sum destroys the original independence. Coordinates along the sum and difference reduce the calculation to one truncated normal variable.

Example P13. Bayesian Prediction with Abstention

A coin's head probability p has the uniform prior on $(0, 1)$. Conditional on p, tosses are independent. After observing h heads in $k \geq 0$ tosses, one may predict the next outcome or abstain. A correct prediction earns 1, an incorrect prediction loses $\lambda > 1$, and abstention earns zero. Find the optimal rule and its expected payoff before observing the data. Evaluate $k = 4, \lambda = 2$.

Solution. The posterior density is proportional to $p^h(1 - p)^{k-h}$. Hence the predictive head probability is

$$r = \frac{\int_0^1 p^{h+1}(1-p)^{k-h}\,\mathrm{d}p}{\int_0^1 p^h(1-p)^{k-h}\,\mathrm{d}p} = \frac{h+1}{k+2}.$$

The conditional payoffs from predicting heads, predicting tails, and abstaining are respectively

$$(\lambda + 1)r - \lambda, \qquad 1 - (\lambda + 1)r, \qquad 0.$$

Predict heads if $r > \lambda/(\lambda + 1)$ and tails if $r < 1/(\lambda + 1)$; otherwise abstain. At either boundary, the corresponding prediction and abstention tie.

The prior predictive count is uniform:

$$\mathbb{P}(H = h) = \binom{k}{h} \int_0^1 p^h(1-p)^{k-h}\,\mathrm{d}p = \frac{1}{k+1}, \quad 0 \leq h \leq k.$$

Using the symmetry between h and $k-h$, and writing $[z]_+ = \max(z, 0)$, gives

$$\boxed{V_k = \frac{2}{(k+1)(k+2)} \sum_{h=0}^{k} [(\lambda+1)(h+1) - \lambda(k+2)]_+.}$$

For $k = 4, \lambda = 2$, only $h = 0, 4$ make prediction strictly profitable; each yields conditional payoff $1/2$. Abstention is optimal otherwise, with zero-payoff ties at $h = 1, 3$. Thus $V_4 = 1/5$.

Method. First obtain a predictive probability, then compare action payoffs. Asymmetric losses create an abstention region; choosing the more probable outcome alone is not the optimal rule.

Section Exercises

OP02. Comparing Two Head Counts

Player A tosses a fair coin n times and player B tosses a fair coin $2n+1$ times, where $n \geq 1$. All tosses are independent. If their respective head counts are X and Y, find $\mathbb{P}(Y > 2X)$.

OP30. A Coin Type after Two Heads

One coin is fair, with one head and one tail; another has heads on both sides. Choose either coin with equal probability and toss that same coin twice, independently conditional on its type. Given two heads, find the probability that the selected coin is the double-headed coin.

1.2 Counting and Expectation

Key Ideas and Methods. For a full distribution, count equally likely permutations, subsets, or gap configurations. For a mean, a sum of indicators often avoids the full law and needs no independence. Variances require joint probabilities, which can be classified by overlap. In circular counting, specify the marking rule and how often each object is counted.

Exchangeability is weaker than independence. Independence created by a Poisson sample size must be proved by factoring the joint law. If the number of terms depends on observations, retain that dependence in a first-step calculation.

A high-dimensional volume can be expressed as a probability for independent uniform samples. The law of large numbers locates a threshold; when the mean cancels and variance is finite and nonzero, the central limit theorem describes fluctuations near it.

Example P05. The First Repeated Type

Draw independently and uniformly from $m \geq 1$ types. Let T be the first draw whose type has appeared before, and let L be that repeated type. Find the distribution and mean of T, determine whether L and T are independent, and find the number of distinct types observed at stopping.

Solution. Write $(m)_k = m(m-1)\cdots(m-k+1)$, with $(m)_0 = 1$. The event $T > k$ means that the first k types are distinct, so

$$\mathbb{P}(T > k) = \frac{(m)_k}{m^k}, \quad 0 \le k \le m,$$

and the tail is zero for $k \ge m+1$. There must be $t-1$ distinct draws followed by a repeat of one of them. Therefore

$$\boxed{\mathbb{P}(T = t) = \frac{(m)_{t-1}}{m^{t-1}} \frac{t-1}{m}, \quad 2 \le t \le m+1.}$$

The tail-sum identity gives

$$\boxed{\mathbb{E}T = \sum_{k=0}^{m} \frac{(m)_k}{m^k}.}$$

For fixed t, relabeling types preserves every sequence probability. Thus $\mathbb{P}(T = t, L = \ell)$ is the same for every ℓ, and these m probabilities sum to $\mathbb{P}(T = t)$. Consequently,

$$\mathbb{P}(T = t, L = \ell) = \mathbb{P}(T = t)/m.$$

This proves that L is uniform and independent of T. Exactly $T-1$ distinct types have appeared at stopping, with mean $\mathbb{E}T - 1$. The formulas also cover $m = 1$, when $T = 2$.

Method. Count collision-free prefixes first. This avoids tracking both the repeated type and all possible locations of its earlier occurrence.

Example P17. Fixed Points in a Random Assignment

Assign $n \ge 2$ tasks uniformly at random to n people, one task per person. Task i originally belongs to person i. Let F count people receiving their own task. Find the distribution, mean, and variance of F, and explain why $F = n-1$ is impossible.

Solution. An assignment is a uniform permutation. To have exactly k fixed points, choose their positions and derange the remaining $n-k$ positions. Inclusion–exclusion gives the number of derangements of r positions:

$$D_r = \sum_{j=0}^{r} (-1)^j \binom{r}{j} (r-j)! = r! \sum_{j=0}^{r} \frac{(-1)^j}{j!}, \quad D_0 = 1.$$

Hence

$$\boxed{\mathbb{P}(F = k) = \frac{\binom{n}{k} D_{n-k}}{n!} = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}, \quad 0 \le k \le n.}$$

Let I_i indicate that person i receives their own task. Fixing one or two prescribed positions leaves $(n-1)!$ or $(n-2)!$ permutations, respectively. Thus

$$\mathbb{E}I_i = \frac{1}{n}, \qquad \mathbb{E}(I_i I_j) = \frac{1}{n(n-1)} \quad (i \ne j).$$

It follows that $\mathbb{E}F = 1$ and

$$\mathbb{E}F^2 = \sum_i \mathbb{E}I_i + 2\sum_{i<j} \mathbb{E}(I_i I_j) = 2, \qquad \boxed{\operatorname{Var} F = 1.}$$

If $n-1$ people receive their own tasks, the last person must do so as well. Equivalently, $D_1 = 0$ in the probability formula.

Method. The full distribution calls for inclusion–exclusion. The first two moments need only one- and two-position probabilities, avoiding a sum over that distribution.

Example P18. Short Cycles in a Random Permutation

In a uniform permutation of n elements, let C_k be the number of cycles of length k, where $1 \le k \le n$. Find $\mathbb{E}C_k$, $\operatorname{Var} C_k$, and $\operatorname{Cov}(C_j, C_k)$ for $j \ne k$. Explain the effect of the finite supply of elements.

Solution. For each k-element subset A, let I_A indicate that its elements form one cycle. There are $(k-1)!$ internal cycles and $(n-k)!$ permutations of the other elements. Hence

$$\mathbb{P}(I_A = 1) = \frac{(k-1)!(n-k)!}{n!}, \qquad \boxed{\mathbb{E}C_k = \frac{1}{k}.}$$

The product $C_k(C_k - 1)$ counts ordered pairs of distinct k-cycles. Their supports must be disjoint. If $2k \le n$, its expectation is

$$\binom{n}{k}\binom{n-k}{k}\frac{[(k-1)!]^2(n-2k)!}{n!} = \frac{1}{k^2}.$$

If $2k > n$, two such cycles cannot coexist, so this factorial moment is zero. Therefore

$$\boxed{\operatorname{Var} C_k = \frac{1}{k} - \frac{\mathbf{1}\{2k > n\}}{k^2}.}$$

Similarly, for $j \ne k$ and $j + k \le n$, choosing disjoint supports and cyclic orders gives $\mathbb{E}(C_j C_k) = 1/(jk)$. When $j + k > n$, the product is identically zero. Thus

$$\boxed{\operatorname{Cov}(C_j, C_k) = -\frac{\mathbf{1}\{j + k > n\}}{jk}, \quad j \ne k.}$$

Caution. Zero covariance does not imply independence: all cycle counts still satisfy $\sum_k kC_k = n$. Factorial moments are useful because they count distinct cycles without counting a cycle twice.

Example P19. Selecting the Best after a Sample

There are $n \ge 2$ candidates of distinct quality in uniformly random order. Only relative ranks among those seen are observable, and rejection is irreversible. Reject the first r candidates, then select the first candidate better than everyone previously seen; if none appears, the rule fails. Here $0 \le r \le n-1$, with $r = 0$

meaning selection of the first candidate. Find the success probability, describe how to choose the optimal r, and obtain its large-sample limit.

Solution. For $r = 0$, success has probability $1/n$. Suppose $r \geq 1$ and the best candidate arrives at position $k > r$. The rule reaches that candidate exactly when the best of the first $k-1$ candidates lies among the rejected first r. That position is uniform, giving conditional probability $r/(k-1)$. Thus

$$\boxed{P_r = \frac{r}{n} \sum_{j=r}^{n-1} \frac{1}{j}, \quad 1 \leq r \leq n-1.}$$

Adjacent probabilities differ by

$$P_{r+1} - P_r = \frac{1}{n} \left(\sum_{j=r+1}^{n-1} \frac{1}{j} - 1 \right), \quad 1 \leq r \leq n-2.$$

The expression in parentheses decreases strictly with r. Increase r until the tail harmonic sum is at most 1; equality produces two adjacent maximizers. Also compare $r = 0$: since $P_1 = H_{n-1}/n \geq 1/n$, it cannot strictly outperform $r = 1$.

If $r/n \to x \in (0,1)$, the harmonic sum tends to $-\log x$, so $P_r \to -x \log x$. This function has its unique maximum $1/e$ at $x = 1/e$. Together with the discrete difference criterion, this shows that the optimal rejected fraction and the optimal success probability both converge to $1/e$.

Method. Condition on the arrival position of the best candidate. The entire preceding history then reduces to the position of its maximum, and optimization reduces to a monotone adjacent difference.

Example P25. Regions from Diameters and an Offset Chord

Draw $n \geq 1$ diameters of the unit disk with independent directions uniform on $[0, \pi)$. Add a fixed chord whose supporting line is at distance $d \in (0,1)$ from the center. Find the distribution, mean, and variance of the final number R of regions.

Solution. Almost surely the diameter directions are distinct. They meet at the center and create $2n$ regions. Rotate coordinates so that the chord lies on $x = d$. A diameter at angle θ intersects the chord inside the disk exactly when

$$\frac{d}{|\cos \theta|} < 1.$$

The probability is therefore $p = 2 \arccos(d)/\pi$. Independent directions imply that the number K of crossings is $\mathrm{Bin}(n, p)$. Since $d > 0$, the chord misses the common center, so distinct diameters give distinct crossings. Endpoint coincidences have probability zero.

The K crossings split the new chord into $K+1$ pieces, each splitting one existing region. Hence

$$\boxed{R = 2n + K + 1.}$$

For $k = 0, \dots, n$, this gives

$$\mathbb{P}(R = 2n + k + 1) = \binom{n}{k} p^k (1-p)^{n-k},$$

and

$$\boxed{\mathbb{E}R = 2n + 1 + np, \qquad \operatorname{Var} R = np(1-p).}$$

Caution. The diameters are concurrent, so a general-position formula that counts every pairwise intersection separately is inapplicable. The restriction $d > 0$ also matters: a chord through the center would merge all its crossings.

Example P26. The Span of Target Cards

Uniformly permute N distinct cards, of which K are targets, with $2 \le K \le N$. Let A, B be the first and last target positions, and put $R = B - A$. Find the distribution and mean of R.

Solution. The target positions form a uniform K-element subset of $\{1, \dots, N\}$. For span r, the left endpoint has $N - r$ choices. Both endpoints are targets, and the remaining $K - 2$ targets occupy $K - 2$ of the $r - 1$ internal positions. Thus

$$\boxed{\mathbb{P}(R = r) = \frac{(N-r)\binom{r-1}{K-2}}{\binom{N}{K}}, \quad K - 1 \le r \le N - 1.}$$

For the mean, let $G_0, \dots, G_K$ count ordinary cards in the $K + 1$ gaps before, between, and after targets. Every nonnegative integer vector with sum $N - K$ corresponds to one target-position set. The gaps are therefore exchangeable, and

$$\mathbb{E}G_i = \frac{N-K}{K+1}.$$

Since $A = G_0 + 1$ and $B = N - G_K$,

$$\boxed{\mathbb{E}R = N - 1 - 2\frac{N-K}{K+1} = \frac{(K-1)(N+1)}{K+1}.}$$

For $K = N$, the span is deterministically $N - 1$, consistently with both formulas.

Method. Count extreme positions for the distribution, but use exchangeable boundary gaps for the mean. The simplest method may differ with the quantity being requested.

Example P27. The Number of Isolated Red Cards

Uniformly arrange $m \ge 2$ red cards and $n \ge 2$ black cards. Let J count red runs of length exactly one, including runs at either end. Find its mean and variance.

Solution. Put $N = m + n$. The black cards define $n + 1$ gaps, including both end gaps. Let G_i be the red count in gap i. Each nonnegative integer vector with sum m corresponds to one color sequence, so the $\binom{N}{n}$ such vectors are equally likely. A singleton red run is precisely a gap with $G_i = 1$.

Fixing one gap at one leaves $m-1$ cards in n gaps; fixing two leaves $m-2$ cards in $n-1$ gaps. Hence

$$p := \mathbb{P}(G_i = 1) = \frac{\binom{N-2}{n-1}}{\binom{N}{n}} = \frac{mn}{N(N-1)},$$

$$p_2 := \mathbb{P}(G_i = G_j = 1) = \frac{\binom{N-4}{n-2}}{\binom{N}{n}}$$

$$= \frac{m(m-1)n(n-1)}{N(N-1)(N-2)(N-3)}, \quad i \neq j.$$

Expanding $J = \sum_{i=0}^{n} \mathbf{1}\{G_i = 1\}$ and its second moment yields

$$\boxed{\begin{aligned} \mathbb{E}J &= (n+1)p, \\ \operatorname{Var} J &= (n+1)p(1-p) + n(n+1)(p_2 - p^2). \end{aligned}}$$

Caution. Uniform compositions do not arise by placing each red card independently into a gap. Gap sizes are dependent, and the two end gaps must be treated on the same footing as the internal gaps.

Example P28. Color Changes around a Circle

Place m red and n black cards uniformly in $N = m+n$ labeled circular positions, where $m, n \geq 1$. Let C count adjacent pairs of different colors, including positions N and 1. Find its full distribution and mean. For $N = 2$, count the two circular adjacencies separately.

Solution. Every red run is followed by a black run. Thus $C = 2k$ means that there are k runs of each color, where $1 \leq k \leq \min(m, n)$.

Mark the start of one red run. There are N possible marked positions. The successive red run lengths are a positive composition of m into k parts, and the black lengths are a positive composition of n into k parts. The number of marked arrangements is

$$N\binom{m-1}{k-1}\binom{n-1}{k-1}.$$

Each unmarked arrangement is counted exactly k times. Since all $\binom{N}{m}$ color arrangements are equally likely,

$$\boxed{\mathbb{P}(C = 2k) = \frac{N\binom{m-1}{k-1}\binom{n-1}{k-1}}{k\binom{N}{m}}.}$$

For the mean, any specified adjacency has differently colored endpoints with probability $2mn/[N(N-1)]$. Summing over all N adjacencies gives

$$\boxed{\mathbb{E}C = \frac{2mn}{N-1}.}$$

Method. Marking one run converts cyclic structure into linear compositions. Divide by the actual number of possible marks on each object to undo overcounting.

Example P29. Coverage after Poisson Sampling

First draw $N \sim \mathrm{Pois}(\lambda)$, where $\lambda > 0$, then make N independent category draws with probabilities $p_i \geq 0$, $i = 1, \ldots, d$, and $\sum_i p_i = 1$. The total N is independent of the sampling sequence. Let X_i count category i, and let D count categories observed at least once. Prove independence of the X_i and find $\mathbb{E}D$, $\mathrm{Var}\, D$, and $\mathrm{Cov}(D, N)$.

Solution. For nonnegative integers $x_1, \ldots, x_d$, put $s = \sum_i x_i$. Conditioning on the total and using the multinomial law gives

$$\begin{aligned}\mathbb{P}(X_i = x_i,\ 1 \leq i \leq d) &= \frac{e^{-\lambda}\lambda^s}{s!}\,\frac{s!}{\prod_i x_i!}\prod_i p_i^{x_i} \\ &= \prod_{i=1}^{d} \frac{e^{-\lambda p_i}(\lambda p_i)^{x_i}}{x_i!}.\end{aligned}$$

Thus the counts are independent Poisson variables of means λp_i; if $p_i = 0$, the corresponding count is identically zero. Set $I_i = \mathbf{1}\{X_i > 0\}$ and $q_i = 1 - e^{-\lambda p_i}$. Since $D = \sum_i I_i$ is a sum of independent indicators,

$$\boxed{\mathbb{E}D = \sum_i q_i, \qquad \mathrm{Var}\, D = \sum_i q_i(1 - q_i).}$$

Also $N = \sum_i X_i$, and cross-category covariances vanish. Using $I_i X_i = X_i$,

$$\mathrm{Cov}(I_i, X_i) = \lambda p_i - q_i \lambda p_i = \lambda p_i e^{-\lambda p_i}.$$

Consequently,

$$\boxed{\mathrm{Cov}(D, N) = \sum_i \lambda p_i e^{-\lambda p_i}.}$$

Method. A fixed total constrains the category counts. A Poisson total makes their joint law factor, simplifying both coverage fluctuations and dependence on sample size.

Example P32. A Sum Stopped by Its Observations

Let X_i be independent $\mathrm{Unif}(0, 1)$ variables, and fix $0 \leq c < 1$. Stop at the first $X_i > c$, writing N for the stopping index and $S = \sum_{i=1}^{N} X_i$. Find $\mathbb{E}S$, $\mathrm{Var}\, S$, and $\mathrm{Cov}(S, N)$. Explain why the random-sum variance formula for an independent number of terms is inapplicable.

Solution. Put $q = 1 - c$. Then N is geometric with mean $1/q$. Since $0 \leq S \leq N$, all required second moments are finite. With (S', N') an independent copy of the continuation process,

$$S = X + \mathbf{1}\{X \leq c\}S', \qquad N = 1 + \mathbf{1}\{X \leq c\}N'.$$

Let $\mu = \mathbb{E}S$ and $m_2 = \mathbb{E}S^2$. Using $\mathbb{E}X = 1/2$, $\mathbb{E}X^2 = 1/3$, and $\mathbb{E}[X\mathbf{1}\{X \le c\}] = c^2/2$, we obtain

$$\mu = \frac{1}{2} + c\mu, \qquad m_2 = \frac{1}{3} + c^2\mu + cm_2.$$

Solving these equations gives

$$\boxed{\mathbb{E}S = \frac{1}{2q}, \qquad \operatorname{Var} S = \frac{6c^2 - 4c + 1}{12q^2}.}$$

For $h = \mathbb{E}(SN)$, multiply the two first-step identities and take expectations:

$$h = \frac{1}{2} + \frac{c^2}{2q} + c\mu + ch.$$

Thus $h = (1 + c^2)/(2q^2)$ and

$$\boxed{\operatorname{Cov}(S, N) = h - \mu/q = \frac{c^2}{2q^2}.}$$

Caution. Before stopping, observations are at most c; the terminal observation exceeds c. Conditioning on N therefore changes their laws. The independent-count random-sum formula loses this dependence, whereas first-step decomposition retains it.

Example P14. A Threshold for High-Dimensional Volume

Define the volume

$$V_n(\alpha) = \int_{[0,1]^n} \mathbf{1}\left\{\sum_{i=1}^n x_i^2 \le \alpha \sum_{i=1}^n x_i\right\} \mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

Find its limit as $n \to \infty$ for fixed real α. Then find the limit when $\alpha_n = 2/3 + c/\sqrt{n}$ for fixed real c.

Solution. Let U_i be independent uniforms on $(0, 1)$. The volume is the probability of the indicated event, and

$$\mathbb{E}(U_i^2 - \alpha U_i) = \frac{1}{3} - \frac{\alpha}{2}.$$

The law of large numbers makes the sample average converge to this constant. A positive constant gives limiting probability zero, and a negative constant gives one. The threshold is therefore $\alpha = 2/3$.

At the threshold, put $Y_i = U_i^2 - 2U_i/3$. Then $\mathbb{E}Y_i = 0$ and

$$\sigma^2 = \mathbb{E}Y_i^2 = \frac{1}{5} - \frac{1}{3} + \frac{4}{27} = \frac{2}{135} > 0.$$

The central limit theorem gives $\sum_i Y_i/\sqrt{n} \Rightarrow N(0, \sigma^2)$. For the moving threshold, the event becomes

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i \le c\left(\frac{1}{n} \sum_{i=1}^n U_i\right).$$

The right side converges in probability to $c/2$. Continuity of the limiting normal law therefore gives

$$\boxed{\lim_{n\to\infty} V_n(\alpha_n) = \Phi\left(c\sqrt{\frac{135}{8}}\right).}$$

Taking $c = 0$ gives $1/2$ at the fixed threshold. For fixed α, the limiting volumes are consequently 0, $1/2$, and 1 according as $\alpha < 2/3$, $\alpha = 2/3$, or $\alpha > 2/3$.

Method. The law of large numbers locates the threshold. When the mean cancels, fluctuations on the $n^{-1/2}$ scale determine the probability; a zero limiting mean alone does not answer the question.

Section Exercises

OP18. Randomly Joining Noodles

There are $n \geq 1$ noodles, each with two ends. Repeatedly choose an unordered pair uniformly from the remaining free ends and join them, until all $2n$ ends have been paired. Find the expected number of closed loops formed.

OP25. Intersections of Random Chords

Choose $2n$ points independently and uniformly on a circle and pair them in a fixed, predetermined way to obtain $n \geq 1$ chords. Find the expected number of distinct intersection points strictly inside the circle.

OP26. Drawing All Four Aces

Shuffle a standard 52-card deck uniformly and draw cards without replacement until all four aces have appeared. Let T be the number of cards drawn, including the final ace. Find $\mathbb{E}T$.

1.3 Waiting Times

Key Ideas and Methods. Specify whether stopping includes the triggering observation and whether inspection occurs after each step or each round. For a nonnegative integer-valued stopping count T, tails give

$$\mathbb{E}T = \sum_{n\geq 0} \mathbb{P}(T > n),$$

allowing infinite values. Adjacent tail differences give the probability mass function.

Negative logarithms of independent uniform factors are exponential increments, turning multiplicative shrinkage into an arrival-time problem. For comparisons between successive observations, equally likely ranks of independent identically distributed continuous samples may be simpler than densities. Positive-increment crossing problems can be analyzed by summing the densities of successive partial sums.

Before manipulating first- or second-moment recursions, establish finiteness of the relevant moments. For limits of infinite random sums or products, justify exchanging limits and expectations.

Example P06. Stopping at an Unobserved Threshold

Let $A, U_1, U_2, \ldots$ be independent uniforms on $(0, 1)$, with the threshold A unobserved. Stop when the sum of increments first exceeds A, and let N be the number of increments. Find the distribution, mean, and variance of N. If only $N = n$ is observed, find the posterior density and mean of A.

Solution. Conditional on $A = a \le 1$, the event that the first k increments sum to at most a is a simplex of volume $a^k/k!$. Hence

$$\mathbb{P}(N > k) = \int_0^1 \frac{a^k}{k!}\,\mathrm{d}a = \frac{1}{(k+1)!}, \quad k \ge 0.$$

Subtracting successive tails gives

$$\boxed{\mathbb{P}(N = n) = \frac{n}{(n+1)!}, \quad n \ge 1.}$$

The integer tail identities yield

$$\mathbb{E}N = \sum_{k\ge 0} \frac{1}{(k+1)!} = e - 1, \quad \mathbb{E}N^2 = \sum_{k\ge 0} \frac{2k+1}{(k+1)!} = e + 1.$$

Thus $\boxed{\operatorname{Var} N = 3e - e^2}$.

The likelihood of observing $N = n$ at threshold a is

$$\mathbb{P}(N = n \mid A = a) = \frac{a^{n-1}}{(n-1)!} - \frac{a^n}{n!}.$$

Multiplying by the uniform prior density and normalizing gives

$$\boxed{f_{A|N=n}(a) = (n+1)a^{n-1} - \frac{n+1}{n}a^n, \quad 0 < a < 1.}$$

Integrating a against this density gives

$$\boxed{\mathbb{E}(A \mid N = n) = 1 - \frac{n+1}{n(n+2)}.}$$

For example, stopping after one increment gives posterior mean $1/3$. A longer observed waiting time provides evidence for a larger threshold.

Example P20. Crossing after Two Shrinks per Round

Each round multiplies a length by two fresh independent $\mathrm{Unif}(0, 1)$ variables; all variables are independent. Starting from $R_0 = 1$, let $R_n = \prod_{i=1}^n U_i V_i$ and stop at the first $R_n < x$, where $0 < x < 1$. Find the law and mean of the stopping round N, and the density of $Y = R_N/x$.

Solution. Put $t = -\log x$. Each negative logarithm is an independent rate-one exponential increment. Unpack the pairs into individual increments, and let K count arrivals completed by time t. For $k \geq 1$, integrating over the kth arrival time gives

$$\mathbb{P}(K = k) = \int_0^t \frac{s^{k-1}e^{-s}}{(k-1)!} e^{-(t-s)}\, \mathrm{d}s = e^{-t}\frac{t^k}{k!}.$$

The probability for $k = 0$ is e^{-t}, so $K \sim \mathrm{Pois}(t)$.

Lengths are checked only after pairs of increments. Thus $N = \lfloor K/2 \rfloor + 1$, giving

$$\boxed{\mathbb{P}(N = n) = e^{-t}\left[\frac{t^{2n-2}}{(2n-2)!} + \frac{t^{2n-1}}{(2n-1)!}\right], \quad n \geq 1.}$$

The Poisson generating function gives $\mathbb{P}(K \text{ odd}) = (1 - e^{-2t})/2$. Therefore

$$\boxed{\mathbb{E}N = 1 + \frac{\mathbb{E}K - \mathbb{P}(K \text{ odd})}{2} = \frac{3 + 2t + e^{-2t}}{4}.}$$

By exponential memorylessness, the distance $O = -\log Y$ from t to the next inspection is one exponential waiting time if K is odd, and the sum of two if K is even. With $w_o = (1 - e^{-2t})/2$ and $w_e = (1 + e^{-2t})/2$,

$$f_O(o) = e^{-o}(w_o + w_e o), \quad o > 0.$$

Transforming by $y = e^{-o}$ yields

$$\boxed{f_Y(y) = w_o - w_e \log y, \quad 0 < y < 1.}$$

Method. Inspecting in batches changes the residual waiting time. Depending on parity, one or two exponential increments remain, so the terminal ratio is no longer uniform.

Example P21. A Renewal Process with Two Step Sizes

Independent increments X_i equal 1 or 2 with probabilities $1 - q$ and q, where $0 \leq q \leq 1$. For integer $m \geq 1$, let $T_m = \min\{n : \sum_{i=1}^n X_i \geq m\}$ and $O_m = \sum_{i=1}^{T_m} X_i - m$. Find $\mathbb{E}T_m$ and the law of O_m.

Solution. Write $e_m = \mathbb{E}T_m$, with $e_0 = e_{-1} = 0$. Conditioning on the first step gives

$$e_m = 1 + (1 - q)e_{m-1} + qe_{m-2}, \quad m \geq 1.$$

For $d_m = e_m - e_{m-1}$, we have $d_1 = 1$ and $d_m = 1 - qd_{m-1}$. Summing the resulting geometric progression yields

$$d_m = \frac{1 - (-q)^m}{1 + q}, \qquad \boxed{e_m = \frac{m}{1+q} + \frac{q[1 - (-q)^m]}{(1+q)^2}.}$$

Each step is at least one, so $T_m \le m$. The event $\{T_m \ge i\}$ depends only on the first $i-1$ increments. Consequently,

$$\mathbb{E}\sum_{i=1}^{T_m} X_i = \sum_{i=1}^{m} \mathbb{E}[X_i \mathbf{1}\{T_m \ge i\}] = (1+q)e_m.$$

Since the overshoot is either zero or one,

$$\boxed{\mathbb{P}(O_m = 1) = \mathbb{E}O_m = \frac{q[1-(-q)^m]}{1+q}, \quad \mathbb{P}(O_m = 0) = 1 - \mathbb{P}(O_m = 1).}$$

At $q = 0$, exactly m steps are needed. At $q = 1$, the count is $\lceil m/2 \rceil$ and the overshoot depends on the parity of m, consistently with the formulas.

Method. The alternating term $(-q)^m$ records the lattice structure. An approximation based only on the mean step size would miss it.

Example P22. The First Local Peak

Let $X_1, X_2, \ldots$ be independent observations from any continuous distribution. Define

$$N = \min\{n \ge 3 : X_{n-2} < X_{n-1} > X_n\}.$$

Thus stopping includes the descending observation that confirms the first local peak. Find the distribution, mean, and variance of N.

Solution. A sequence of n distinct observations has no local peak exactly when its adjacent comparison signs never change from an increase to a decrease. It must therefore first decrease and then increase, allowing either part to be empty.

Place the smallest rank at the valley. Each of the other $n-1$ ranks can be assigned to either side. The left side must be decreasing and the right side increasing, so the order is then unique. There are 2^{n-1} peak-free permutations. All $n!$ ranks are equally likely, giving

$$\boxed{\mathbb{P}(N > n) = \frac{2^{n-1}}{n!}, \quad n \ge 1,}$$

with $\mathbb{P}(N > 0) = 1$. Hence, for $n \ge 3$,

$$\mathbb{P}(N = n) = \frac{2^{n-2}}{(n-1)!} - \frac{2^{n-1}}{n!}.$$

The tail tends to zero, so stopping occurs almost surely. Tail sums and the exponential series give

$$\mathbb{E}N = 1 + \frac{1}{2}\sum_{n\ge 1} \frac{2^n}{n!} = \frac{e^2+1}{2},$$

$$\mathbb{E}N^2 = 1 + \frac{1}{2}\sum_{n\ge 1} (2n+1)\frac{2^n}{n!} = \frac{5e^2+1}{2}.$$

Therefore

$$\boxed{\operatorname{Var} N = \frac{5e^2+1}{2} - \left(\frac{e^2+1}{2}\right)^2.}$$

Method. The actual observation density is irrelevant; only rank patterns matter. The stopping convention must explicitly include or exclude the final observation that confirms a peak.

Example P23. The Gaps before and after Crossing

Let U_i be independent uniforms on $(0,1)$, $S_k = \sum_{i=1}^{k} U_i$, $S_0 = 0$, and $N = \min\{k : S_k > 1\}$. Define the pre-crossing gap $A = 1 - S_{N-1}$ and overshoot $O = S_N - 1$. Find their joint density, the density of the final increment $J = U_N$, and $\mathbb{P}(O > A)$.

Solution. For $0 < s < 1$, the density of the kth positive partial sum is $s^{k-1}/(k-1)!$. Summing over all positive partial sums gives the renewal density

$$u(s) = \sum_{k\geq 1} \frac{s^{k-1}}{(k-1)!} = e^s.$$

The first increment cannot exceed 1, so the last pre-crossing sum is positive; there is no contribution from an atom at $S_0 = 0$. A pre-crossing sum s followed by increment v contributes density $e^s\,\mathrm{d}s\,\mathrm{d}v$ on $s + v > 1$. Positive increments ensure that this crossing step is unique.

The change of variables $a = 1 - s$, $o = s + v - 1$ has absolute Jacobian one. Therefore

$$\boxed{f_{A,O}(a,o) = e^{1-a}, \quad 0 < a < 1, \quad 0 < o < 1 - a.}$$

Its integral is $\int_0^1 e^{1-a}(1-a)\,\mathrm{d}a = 1$. Since $J = A + O$,

$$\boxed{f_J(j) = \int_0^j e^{1-a}\,\mathrm{d}a = e - e^{1-j}, \quad 0 < j < 1.}$$

The inequality $O > A$ requires $a < 1/2$, giving

$$\boxed{\mathbb{P}(O > A) = \int_0^{1/2} e^{1-a}(1-2a)\,\mathrm{d}a = 2\sqrt{e} - e.}$$

Method. Sum over possible pre-crossing locations and append the final increment. This produces the joint law on both sides of the boundary without enumerating stopping times individually.

Example P24. Total Remaining Distance

Starting at $R_0 = x \in [0, 1)$, choose R_j uniformly from $(R_{j-1}, 1)$ using fresh independent randomness. Define

$$Q = \sum_{j \geq 1} (1 - R_j).$$

Prove that Q is almost surely finite, find its mean and variance, and obtain the Laplace transform $L(t) = \mathbb{E}e^{-tD}$ of $D = Q/(1 - x)$ for $t \geq 0$.

Solution. For independent uniform variables U_j, the remaining distances have the representation

$$1 - R_j = (1 - x) \prod_{i=1}^{j} U_i.$$

Nonnegativity permits summation under expectation:

$$\mathbb{E}Q = (1 - x) \sum_{j \geq 1} 2^{-j} = 1 - x.$$

Finite expectation of a nonnegative variable implies almost sure finiteness.

The normalized sum satisfies $D = U(1 + D')$, where D' has the law of D and is independent of U. Its second moment is finite: the L^2 norm of every partial sum is bounded by $\sum_{j \geq 1} 3^{-j/2} < \infty$, and monotone convergence then applies. Thus

$$\mathbb{E}D = 1, \qquad \mathbb{E}D^2 = \frac{1}{3}(1 + 2\mathbb{E}D + \mathbb{E}D^2) = \frac{3}{2}.$$

Consequently,

$$\boxed{\mathbb{E}Q = 1 - x, \qquad \operatorname{Var} Q = \frac{(1 - x)^2}{2}.}$$

For $t > 0$, the distributional recursion yields

$$L(t) = \int_0^1 e^{-tu} L(tu)\, \mathrm{d}u, \qquad tL(t) = \int_0^t e^{-s} L(s)\, \mathrm{d}s.$$

The right side is differentiable and $L(t) > 0$, so

$$\frac{L'(t)}{L(t)} = \frac{e^{-t} - 1}{t}.$$

Integration with $L(0) = 1$ gives

$$\boxed{L(t) = \exp\left\{ \int_0^t \frac{e^{-s} - 1}{s}\, \mathrm{d}s \right\}.}$$

The integrand extends continuously to -1 at zero, so the expression also covers $t = 0$.

Method. Although individual gaps shrink multiplicatively, their total satisfies a simple recursive sum. Establish finiteness of moments before manipulating the moment equations.

Section Exercises

OP22. Stopping an Alternating Sequence

Generate independent $U_1, U_2, \ldots \sim \mathrm{Unif}(0,1)$ and require

$$U_1 < U_2 > U_3 < U_4 > U_5 < \cdots .$$

Stop as soon as a newly generated value violates the required alternating inequality. Let T count all generated values, including the final violating one. Find $\mathbb{E}T$.

OP24. A Product of Increasing Uniform Draws

First draw $R_1 \sim \mathrm{Unif}(0,1)$. Given the past, draw R_{k+1} uniformly on $(R_k, 1)$ using fresh independent randomness at each step. Evaluate

$$\lim_{n\to\infty} \mathbb{E}\left[\prod_{k=1}^{n} R_k\right].$$

1.4 Distributions and Geometry

Key Ideas and Methods. Ratios, extremes, and relative positions often call for a change of variables in a joint density. State the support and Jacobian explicitly. Conditional rescaling may remove the conditioning location and prove independence. Uniform cut points produce a constant density on the spacing simplex, but the spacings are dependent.

Geometric probability requires specified position and direction laws. A distance threshold can become a binomial neighborhood count, while intersection and pairing counts admit indicator decompositions. Concurrence and boundary cases need separate attention. Symmetry alone does not ensure that a mean exists.

For random colorings, an expected nonnegative integer count of bad events below one proves existence. For secrecy, compare ciphertext distributions conditional on messages; uniform ciphertext marginals alone do not establish joint secrecy.

Example P04. Conditioning on a Product

Let U, V be independent uniforms on $(0,1)$, and define $P = UV$ and $Q = \log U / \log P$. Prove that P, Q are independent. Find the conditional density and mean of U given $P = p \in (0,1)$, and compare that mean with $\sqrt{p}$.

Solution. Put $X = -\log U$ and $Y = -\log V$. These are independent rate-one exponentials. With $S = X+Y$ and $Q = X/S$, the inverse transformation is $X = QS$, $Y = (1-Q)S$, with Jacobian s. Hence

$$f_{S,Q}(s,q) = se^{-s}, \quad s > 0, \quad 0 < q < 1.$$

This factors into the gamma density of S and the uniform density of Q. Thus Q is independent of S, and therefore of $P = e^{-S}$. The transformation also gives $f_P(p) = -\log p$.

Given $P = p$, we have $U = p^Q$ with Q still uniform. Since p^q decreases with q, for $p < u < 1$,

$$\mathbb{P}(U \le u \mid P = p) = 1 - \frac{\log u}{\log p} = \frac{\log(u/p)}{\log(1/p)}.$$

Consequently,

$$\boxed{f_{U|P=p}(u) = \frac{1}{u \log(1/p)}, \quad p < u < 1,}$$

and

$$\boxed{\mathbb{E}(U \mid P = p) = \int_0^1 p^q \, \mathrm{d}q = \frac{1-p}{\log(1/p)} > \sqrt{p}.}$$

The strict inequality follows from Jensen's inequality for the strictly convex function $q \mapsto p^q$, since $\mathbb{E}Q = 1/2$ and Q is nondegenerate.

Caution. Conditioning on an exact continuous value requires a conditional density, rather than division by a zero event probability. Fixing a product does not force either factor to equal its geometric mean.

Example P08. The Relative Difference of Two Gaussians

Let X, Y be independent standard normals and define

$$R = \frac{X - Y}{X + Y}, \qquad H = X^2 + Y^2.$$

The denominator is nonzero almost surely. Find the density of R, prove that R, H are independent, and evaluate $\mathbb{P}(|R| \le c)$ for $c \ge 0$. Does R have mean zero?

Solution. Set $D = (X - Y)/\sqrt{2}$ and $S = (X + Y)/\sqrt{2}$. Orthogonal transformation preserves the joint standard normal density, so D, S remain independent standard normals, with $H = D^2 + S^2$. Write

$$S = \rho \cos\theta, \qquad D = \rho \sin\theta.$$

The polar density is $(2\pi)^{-1} \rho e^{-\rho^2/2}$, which factors into a radial density and a uniform angular density. Angle and radius are independent, so $R = \tan\theta$ is independent of $H = \rho^2$.

Reducing the angle modulo π gives density $1/\pi$ on $(-\pi/2, \pi/2)$. The transformation $\theta = \arctan r$ yields

$$\boxed{f_R(r) = \frac{1}{\pi(1 + r^2)}, \quad r \in \mathbb{R},} \qquad \boxed{\mathbb{P}(|R| \le c) = \frac{2}{\pi} \arctan c.}$$

Although the density is symmetric, $\int_0^\infty r/[\pi(1 + r^2)] \, \mathrm{d}r = \infty$. Both positive and negative parts have infinite expectation, so $\mathbb{E}R$ does not exist; it is not zero.

Method. Rotation turns the relative difference into a ratio of independent Gaussians, and polar coordinates separate direction from magnitude. Symmetry and existence of a mean are distinct requirements.

Example P10. Dependent Binary Choices

Starting from the unit interval, choose its left or right half at step j according as $B_j = 0$ or 1. The first bit is fair. Thereafter the bit stays unchanged with probability $p \in [0,1]$ and flips with probability $1-p$, according to a Markov transition. Let X be the limiting position. Find its mean and variance, describe its law for $p = 0, 1/2, 1$, and bound the error from using the first n bits.

Solution. The nested interval lengths tend to zero, and their unique limiting point is

$$X = \sum_{j \geq 1} 2^{-j} B_j.$$

The chain is stationary with $\mathbb{E}B_j = 1/2$. Put $Z_j = B_j - 1/2$ and $\lambda = 2p - 1$. Since $\mathbb{E}(Z_{j+1} \mid Z_j) = \lambda Z_j$, iteration gives

$$\operatorname{Cov}(B_i, B_j) = \frac{1}{4}\lambda^{|i-j|}.$$

Absolute convergence permits rearrangement. Using $\sum_i 4^{-i} = 1/3$,

$$\begin{aligned} \operatorname{Var} X &= \frac{1}{4}\sum_i 4^{-i} + \frac{1}{2}\sum_{k \geq 1} \lambda^k \sum_{i \geq 1} 2^{-i} 2^{-(i+k)} \\ &= \frac{1}{12} + \frac{1}{6}\sum_{k \geq 1} (\lambda/2)^k = \boxed{\frac{2+\lambda}{12(2-\lambda)}}. \end{aligned}$$

Also $\boxed{\mathbb{E}X = 1/2}$.

For $p = 0$, the bits alternate, giving $X = 1/3, 2/3$ with equal probabilities. For $p = 1$, all bits agree, giving $X = 0, 1$ equally. For $p = 1/2$, the bits are independent; every binary interval of length 2^{-n} has probability 2^{-n}, proving that X is uniform. For $p = 0, 1/2, 1$, the variances are respectively $1/36, 1/12, 1/4$.

For $X_n = \sum_{j=1}^{n} 2^{-j} B_j$, we always have $0 \leq X - X_n \leq 2^{-n}$. Using the midpoint of the selected interval reduces the absolute error bound to 2^{-n-1}.

Method. Fair marginal bits do not ensure a uniform limiting point. Dependence between digits accumulates through the geometric weights and changes the variance.

Example P11. Median Error for a Laplace Sample

Let X_1, X_2, X_3 be independent with Laplace density $e^{-|x-c|/b}/(2b)$, where $b > 0$. Compare the sample median M and sample mean $\overline{X}$ as estimators of c. Prove unbiasedness and compute their mean squared errors.

Solution. Reflection about c preserves the joint law, so $M - c$ has a symmetric distribution. Also $|M - c| \leq \sum_i |X_i - c|$ is integrable, proving $\mathbb{E}M = c$. Each observation has mean c by symmetry and integrability, hence $\mathbb{E}\overline{X} = c$.

For $t \geq 0$, set $q(t) = \mathbb{P}(X_i > c + t) = e^{-t/b}/2$. The median exceeds $c + t$ precisely when at least two observations do, so

$$\mathbb{P}(M > c + t) = 3q(t)^2 - 2q(t)^3.$$

By symmetry,

$$\mathbb{P}(|M - c| > t) = \frac{3}{2}e^{-2t/b} - \frac{1}{2}e^{-3t/b}.$$

The tail integral for a square gives

$$\begin{aligned}\mathbb{E}(M - c)^2 &= 2\int_0^\infty t\left(\frac{3}{2}e^{-2t/b} - \frac{1}{2}e^{-3t/b}\right)\mathrm{d}t \\ &= \frac{3b^2}{4} - \frac{b^2}{9} = \boxed{\frac{23b^2}{36}}.\end{aligned}$$

Meanwhile, $\operatorname{Var} X_i = b^{-1}\int_0^\infty t^2 e^{-t/b}\,\mathrm{d}t = 2b^2$. Independence therefore gives

$$\boxed{\mathbb{E}(\overline{X} - c)^2 = \frac{2b^2}{3} = \frac{24b^2}{36}.}$$

For this distribution and sample size, the median improves mean squared error by $b^2/36$.

Method. Comparing estimators requires a specified sampling law and loss. A binomial tail for the median, followed by a tail integral, avoids a piecewise density calculation.

Example P35. Relative Position in the Remaining Interval

Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of n independent uniforms on $(0, 1)$. For $1 \leq i < j \leq n$, define

$$W = \frac{U_{(j)} - U_{(i)}}{1 - U_{(i)}}.$$

Prove that W and $U_{(i)}$ are independent. Find the distribution, mean, and variance of W, and evaluate $\mathbb{P}(U_{(j)} > (1 + U_{(i)})/2)$.

Solution. Conditional on $U_{(i)} = u$, the unordered observations above u have the law of $n-i$ independent uniforms on $(u, 1)$. This follows by factoring the constant joint order-statistic density into the ordered regions below and above u. Rescaling by $(v - u)/(1 - u)$ gives $m = n - i$ standard uniforms.

The variable W is their rth order statistic, where $r = j - i$. Placing $r - 1$ observations below w, one near w, and the rest above gives

$$f_W(w) = \frac{m!}{(r-1)!(m-r)!}w^{r-1}(1-w)^{m-r}, \quad 0 < w < 1.$$

This conditional density does not depend on u, proving independence. It is the $\mathrm{Beta}(r, m-r+1)$ density. Beta integrals give its first two moments and hence

$$\boxed{\mathbb{E}W = \frac{r}{m+1}, \qquad \operatorname{Var} W = \frac{r(m-r+1)}{(m+1)^2(m+2)}.}$$

The event $W > 1/2$ means that at most $r-1$ of the m scaled observations lie in the left half. Therefore

$$\boxed{\mathbb{P}\left(U_{(j)} > \frac{1+U_{(i)}}{2}\right) = 2^{-m}\sum_{k=0}^{r-1}\binom{m}{k}.}$$

Method. Absolute positions are dependent, while a conditionally scaled relative position can be independent. Removing the conditional scale reveals more than a covariance calculation alone.

Example P36. The Shortest Spacing and Exceedance Counts

Cut the unit interval at $n-1$ independent uniform points, producing lengths $D_1, \ldots, D_n$, where $n \geq 2$. Let $Z = \min_i D_i$ and, for $t \geq 0$, let $K_t = \sum_i \mathbf{1}\{D_i > t\}$. Find the distribution, mean, and variance of Z, and the probability generating function and mean of K_t.

Solution. The ordered cut points have constant joint density $(n-1)!$. The linear change to spacings shows that the spacing vector is uniform on the simplex $D_i \geq 0$, $\sum_i D_i = 1$.

If r specified spacings exceed t, subtract t from those coordinates. The remaining simplex has total length $1-rt$ and dimension $n-1$, so the event has probability $(1-rt)_+^{n-1}$. Taking $r = n$ gives

$$\boxed{\mathbb{P}(Z > z) = (1-nz)^{n-1}, \quad 0 \leq z \leq 1/n.}$$

Outside this interval the tail follows from $Z \in [0, 1/n]$. Tail integration with $u = nz$ yields

$$\mathbb{E}Z = \frac{1}{n^2}, \qquad \mathbb{E}Z^2 = \frac{2}{n^3(n+1)}, \qquad \boxed{\operatorname{Var} Z = \frac{n-1}{n^4(n+1)}.}$$

For the count, expand

$$z^{K_t} = \prod_{i=1}^{n}[1 + (z-1)\mathbf{1}\{D_i > t\}].$$

Every subset of size r has the same joint exceedance probability. Hence

$$\boxed{\mathbb{E}z^{K_t} = \sum_{r=0}^{n}\binom{n}{r}(z-1)^r(1-rt)_+^{n-1}.}$$

The term $r = 0$ is interpreted as one. Differentiating at $z = 1$ gives

$$\boxed{\mathbb{E}K_t = n(1-t)_+^{n-1}.}$$

Method. Simultaneous lower bounds on spacings translate and shrink one simplex. A single joint-probability calculation handles both the minimum spacing and the count above a threshold.

Example P37. The kth Nearest Distance to a Fixed Point

Draw $n \geq 1$ independent uniform points on the unit interval. Fix a location $a \in [0, 1]$ that is not one of the random observations. Order the sample distances from a and let $D_{(k)}$ be the kth smallest, where $1 \leq k \leq n$. Find its distribution for general a, and its density, mean, and variance when $a = 0$ and $a = 1/2$.

Solution. For $r \geq 0$, the neighborhood of radius r has length

$$\ell_a(r) = \min(1, a + r) - \max(0, a - r).$$

This equals one once r exceeds all possible distances. Each sample independently falls inside with probability $\ell_a(r)$, so the neighborhood count is binomial. The kth distance exceeds r exactly when at most $k - 1$ samples lie inside. Thus

$$\boxed{\mathbb{P}(D_{(k)} > r) = \sum_{j=0}^{k-1} \binom{n}{j} \ell_a(r)^j [1 - \ell_a(r)]^{n-j}, \quad r \geq 0.}$$

At $a = 0$, each individual distance is uniform on $(0, 1)$. Consequently,

$$f_{D_{(k)}}(r) = \frac{n!}{(k-1)!(n-k)!} r^{k-1}(1-r)^{n-k}, \quad 0 < r < 1.$$

This is the Beta$(k, n - k + 1)$ density, whose beta integrals yield

$$\boxed{\mathbb{E}D_{(k)} = \frac{k}{n+1}, \qquad \operatorname{Var} D_{(k)} = \frac{k(n-k+1)}{(n+1)^2(n+2)}.}$$

At $a = 1/2$, each distance has distribution function $2r$ on $[0, 1/2]$, so $2D_{(k)}$ has the same beta law. Its density is

$$f_{D_{(k)}}(r) = \frac{2n!}{(k-1)!(n-k)!} (2r)^{k-1}(1-2r)^{n-k}, \quad 0 < r < 1/2.$$

The mean is half the endpoint value and the variance one quarter of it.

Method. Replace an ordered-distance event by a count in a fixed neighborhood. This handles arbitrary locations and ranks; boundary effects arise from neighborhood growth, without pairing sample points with one another.

Example P38. A Zero-Sum Bonus for the Middle Rank

Three participants draw distinct numbers uniformly without replacement from $1, \ldots, M$, where $M \geq 3$. Write the sorted numbers as $A < B < C$. The middle participant receives $C - A$, and each endpoint participant pays $(C - A)/2$. Given that your number is x, find your conditional expected net payoff $g(x)$ and determine its maximizers. Evaluate the optimal number and payoff when $M = 9$.

Solution. The other two numbers form a uniform unordered pair among $D = \binom{M-1}{2}$ possibilities. You are in the middle for $(x-1)(M-x)$ pairs. Conditional on this event, the low and high numbers are uniform on their respective sides, so their mean range is

$$\frac{x+1+M}{2} - \frac{x}{2} = \frac{M+1}{2}.$$

If you are smallest, both other numbers exceed x, giving $\binom{M-x}{2}$ pairs. Put $L = M - x$. The maximum of two numbers sampled from $1, \ldots, L$ has mean

$$\sum_{b=2}^{L} b(b-1) \Big/ \binom{L}{2} = \frac{2(L+1)}{3}.$$

Your conditional payoff is therefore $-(M - x + 1)/3$. By reflection, if you are largest the conditional payoff is $-x/3$, with $\binom{x-1}{2}$ pairs. Impossible rank cases contribute zero pairs and require no conditional mean.

Combining the three cases gives

$$g(x) = \frac{1}{D}\left[\frac{M+1}{2}(x-1)(M-x) - \frac{M-x+1}{3}\binom{M-x}{2} - \frac{x}{3}\binom{x-1}{2}\right].$$

Simplification yields

$$\boxed{g(x) = \frac{2M(x-1)(M-x)}{(M-1)(M-2)} - \frac{M}{3}.}$$

The product $(x-1)(M-x)$ is largest at integers nearest $(M+1)/2$. There is one maximizer for odd M and two adjacent maximizers for even M. For $M = 9$, the optimal number is $x = 5$ and $g(5) = 15/7$.

Method. Both the probability of each rank and its conditional mean range matter. Although the three payoffs sum to zero in every outcome, observing a central number creates a positive conditional expectation.

Example CP01. A Random Needle across Parallel Lines

Parallel lines in the plane are spaced $h > 0$ apart. A needle has length $l > 0$. Its center's distance to the nearest line is uniform on $[0, h/2]$, and its acute angle to the lines is uniform on $[0, \pi/2]$, independently. Find the probability of hitting at least one line and the expected number of intersections. The needle may be longer than the spacing.

Solution. Write Θ for the angle, D for the center distance, and $r = l/h$. The perpendicular projection has length $l \sin\Theta$, so a hit occurs exactly when $D \le l \sin\Theta/2$. Conditioning on the angle gives

$$p(r) = \frac{2}{\pi} \int_0^{\pi/2} \min(1, r \sin\theta)\, \mathrm{d}\theta.$$

For $r \le 1$, this is $2r/\pi$. For $r > 1$, split the integral at $\theta_0 = \arcsin(1/r)$ to obtain

$$\boxed{p(r) = \begin{cases} 2r/\pi, & 0 < r \le 1, \\ \dfrac{2}{\pi}\left(r - \sqrt{r^2 - 1} + \arccos(1/r)\right), & r > 1. \end{cases}}$$

For the intersection count J, fix the angle. The needle projects to an interval of length $b = l \sin\theta$, while the lines form a lattice of spacing h. Reflection of the center across a line leaves the count unchanged, so the uniform-distance model gives the same count law as a uniform shift over one period. Under that shift, the expected number of lattice points in any interval of length b is b/h: integrating and summing the coverage indicators gives a total shift length of b. Therefore

$$\boxed{\mathbb{E}J = \mathbb{E}\left(\frac{l \sin\Theta}{h}\right) = \frac{2l}{\pi h}.}$$

When $l \le h$, at most one intersection occurs almost surely, so the probability and expectation coincide. A long needle can cross several lines, and the two quantities differ.

Method. A random-needle model must specify both position and direction laws. Conditioning on the angle reduces the problem to length ratios and handles short and long needles alike.

Example CP02. A Coloring with No Monochromatic Set

A finite vertex set V has m specified subsets, each containing at least $k \ge 2$ vertices. If $m < 2^{k-1}$, prove that the vertices can be colored red and blue so that every specified subset contains both colors. Explain how to choose colors one vertex at a time without enumerating all colorings.

Solution. First color vertices independently with equal probabilities. A subset of size s is monochromatic with probability $2^{1-s} \le 2^{1-k}$. If Z counts monochromatic specified subsets, then

$$\mathbb{E}Z \le m 2^{1-k} < 1.$$

If every coloring had a monochromatic subset, we would have $Z \ge 1$ always, contradicting this expectation. Thus some coloring has $Z = 0$. The union bound also gives success probability at least $1 - m2^{1-k} > 0$.

To make deterministic choices, maintain the conditional expectation after fixing some colors. The average of the two conditional expectations obtained by coloring the next vertex red or blue equals the current value. At least one choice does not increase it; select that color.

These conditional expectations are explicit. A subset already containing both colors has monochromatic probability zero. If its fixed vertices all have one color and u vertices remain uncolored, the probability is 2^{-u}. If no vertex has been fixed, it is 2^{1-s}. Sum these probabilities over subsets. After every vertex is fixed, the conditional expectation is the integer Z, still less than one, and must therefore be zero.

Method. Randomness can prove the existence of a deterministic object. Choosing the smaller conditional expectation at each step converts that proof into a construction. The numerical condition is sufficient, not necessarily necessary.

Example CP03. A One-Time Pad and Key Reuse

Messages and keys lie in $\mathbb{Z}_q = \{0, \dots, q-1\}$, where $q \geq 2$, with encryption $C = M + K \pmod{q}$. The key K is independent of the message M. Prove that a uniform key makes C independent of M. Conversely, if every message value has positive probability, prove that this independence requires a uniform key. Then encrypt two independent uniform messages using the same independent uniform key. Find their conditional joint distribution given both ciphertexts and identify the information that is revealed exactly.

Solution. For uniform K, any m with $\mathbb{P}(M = m) > 0$ and any c satisfy

$$\mathbb{P}(C = c \mid M = m) = \mathbb{P}(K = c - m) = \frac{1}{q}.$$

The conditional distribution does not depend on m, so the ciphertext is uniform and independent of the message, even if the message prior is not uniform.

Conversely, independence and full message support imply, for fixed c, that $\mathbb{P}(K = c - m) = \mathbb{P}(C = c)$ for every m. As m runs through $\mathbb{Z}_q$, so does $c - m$. All key probabilities are equal and must be $1/q$. Full support matters: a constant message is independent of any ciphertext.

With reuse, each key k and fixed ciphertext pair (c_1, c_2) determine the unique message pair $(c_1 - k, c_2 - k)$. Every original triple (M_1, M_2, K) has probability q^{-3}, so the ciphertext pair has probability $q \cdot q^{-3} = q^{-2}$. Consequently,

$$\boxed{\begin{aligned}&\mathbb{P}(M_1 = m_1, M_2 = m_2 \mid C_1 = c_1, C_2 = c_2)\\ &\qquad = \begin{cases} 1/q, & m_2 - m_1 = c_2 - c_1 \pmod{q},\\ 0, & \text{otherwise.}\end{cases}\end{aligned}}$$

Each individual message remains uniform conditionally, but their difference is revealed exactly by $C_2 - C_1$. The message pair is reduced from q^2 possibilities to q, so joint perfect secrecy fails.

Method. Secrecy of individual ciphertexts does not guarantee joint secrecy. Reusing a key creates dependence, and subtraction removes the key while exposing a relation between messages.

Section Exercises

OP35. Correlation of the Sample Extremes

Let $X_1, \ldots, X_n$ be independent $\mathrm{Unif}(0,1)$ variables, where $n \geq 2$. Put $L = \min_i X_i$ and $H = \max_i X_i$. Find $\mathrm{Corr}(L, H)$.

OP36. A Randomly Broken Stick as a Polygon

Choose $n-1$ independent uniform cut points on a unit segment, where $n \geq 3$, producing n pieces. Find the probability that these pieces can be the sides of a nondegenerate n-gon. All cut points are sampled on the original unit segment; pieces are not selected and cut recursively.

OP37. Points in Mutual Nearest-Neighbor Pairs

Sample $N \geq 3$ points independently and uniformly from $(0,1)$. The neighbor of a point is its nearest other sampled point; ties, which have probability zero, may be resolved arbitrarily. A point is called good if its neighbor also chooses it as its neighbor. Find the expected number of good points.

1.5 Exercises

Key Ideas and Methods. These exercises vary the observation scheme or sampling rule. For conditional probabilities, write the likelihood of the complete record and check whether missingness or selection changes the law. Conditional expectation is the optimal squared-loss predictor when the target has a finite second moment.

Classify pairs of counted objects by their overlap when computing variances. For stopped costs, ask whether the event that a draw occurs depends only on earlier observations, then use independence of the fresh increment from the past. In distribution calculations, distinguish uniform selection of a spacing from selection proportional to its length, and check endpoints, normalization, and scaling. Marginal sign probabilities alone cannot determine independence of joint sign events.

PG1-1. Observing Only Whether a Head Appeared

Choose a coin with head probability p_1 with probability π, otherwise one with probability p_2, where $0 < \pi, p_1, p_2 < 1$. Toss the chosen coin independently $n \geq 1$ times. The only recorded information is that at least one head occurred; neither the count nor the order is available. Find the posterior type probability and the predictive probability of a head next. Explain why the record is not generally equivalent to observing one head.

PG1-2. Competing Clocks before a Deadline

Independent clocks A, B ring after exponential waiting times with rates $\alpha, \beta > 0$. Let T be the first ringing time. Given that A rings first and $T < t$, for fixed $t > 0$, find the conditional density and mean of T, including the limit as $t \to \infty$.

PG1-3. Posterior Odds with Missing Reports

A binary truth has prior $\mathbb{P}(T = 1) = \pi \in (0, 1)$. Conditional on the truth, reports are independent: each is correct with probability a, flipped with probability b, and missing with probability $1-a-b$, where $a, b > 0$ and $a+b < 1$. Observe h ones, z zeros, and e missing reports. Find the posterior odds and explain whether missingness is informative. Does that conclusion persist if the missing probability depends on the truth?

PG1-4. Selecting a Sum near Zero

Let X, Z be independent standard normals. Retain only observations with $|X+Z| < a$, where $a > 0$. Find the retained population correlation and its limits as $a \downarrow 0$ and $a \to \infty$.

PG2-1. The First Repeat with Unequal Probabilities

Independent draws have three possible types, with probabilities $1/2, 1/4, 1/4$. Let T be the first draw whose type has already appeared. Find $\mathbb{E}T$ and $\mathbb{P}(T = 4)$.

PG2-2. Red Runs on a Circle

Place $m \geq 1$ red and $n \geq 1$ black cards uniformly in $N = m+n$ labeled circular positions. Find the expected number of red runs and the probability that all red cards form one circular run.

PG2-3. Variance of Triple Collisions

Independently place $m \geq 3$ objects into d buckets with probabilities $p_i \geq 0$, $\sum_i p_i = 1$. Let C count unordered triples whose three objects all enter one bucket. With $s_r = \sum_i p_i^r$, find the mean and variance of C.

PG2-4. Weighted Difference across Targets

Uniformly permute $K \geq 1$ target cards and $m \geq 0$ ordinary cards. Ordinary card i has fixed real weight w_i. Let D be the total ordinary-card weight before the first target minus the total weight after the last target. Find $\mathbb{E}D$ and $\operatorname{Var} D$, allowing negative weights.

PG3-1. Sampling Costs Tied to Increments

Let U_i be independent uniforms on $(0, 1)$ and stop when their cumulative sum first exceeds one, at time N. The cost of draw j is its observed squared increment U_j^2. Find $\mathbb{E}C$ for $C = \sum_{j=1}^{N} U_j^2$, explaining why the required independence does not mean that N is independent of the increments.

PG3-2. Multiplicative Stopping with Fixed Shrinkage

Let U_i be independent uniforms on $(0, 1)$, $R_n = 2^{-n} \prod_{i=1}^{n} U_i$, and $R_0 = 1$. Stop at the first $R_n < x$, where $0 < x < 1$, and call this time N. Give a deterministic upper bound, the tail probabilities, and a finite-sum expression for its mean. Find the full distribution for $x = 1/4$.

PG3-3. Renewal Counts in an Interval

Let U_i be independent uniforms on $(0, 1)$ and $S_k = \sum_{i=1}^{k} U_i$. For $0 < a < b \leq 1$, let K count positive partial sums in $(a, b]$. Find its full distribution and mean.

PG3-4. Optimal Prediction of the Remaining Total

In Problem P24, observe $R_1, \ldots, R_n$, where $n \geq 0$, and put $Q_n = \sum_{j=1}^{n}(1 - R_j)$. Predict the full total Q under squared loss. Find the optimal predictor, its conditional error variance, and its unconditional mean squared error. Explain why the partial sum alone is not optimal.

PG4-1. The Relative Share of Two Uniforms

Let U, V be independent uniforms on $(0, 1)$, and define $W = U/(U+V)$. Find its density, mean, and variance, and determine whether it is uniformly distributed.

PG4-2. The Midpoint of the Sample Extremes

Take $n \geq 2$ independent uniforms on $(0, 1)$, with minimum L and maximum H. Find the density, mean, and variance of their midpoint $C = (L + H)/2$.

PG4-3. The Spacing Containing a Random Point

Cut the unit interval at $n - 1$ independent uniform points, producing $n \geq 2$ spacings. Independently choose a uniform point on the interval, and let L be the length of the spacing containing it. Find the density of L, $\mathbb{E}L$, and $\mathbb{E}(1/L)$, and compare with choosing one of the n spacings with equal probabilities.

PG4-4. Sign Changes in Three Directions

Let X, Y be independent standard normals and $0 < \theta < \pi/2$. Let $A, B, C \in \{-1, 1\}$ be the signs of the projections of (X, Y) onto directions $0, \theta, 2\theta$. Define $D_1 = \mathbf{1}\{A \neq B\}$ and $D_2 = \mathbf{1}\{B \neq C\}$. Find their joint distribution, covariance, and the probability that all three signs agree.

2

Stochastic Processes

2.1 Markov Chains

Key Ideas and Methods. A Markov state must retain the history that affects the next transition: useful suffixes for pattern searches, task types for recursive work, and machine status for repairs. In continuous time, include the holding time as well as the next destination. Write first-step equations with their absorbing boundaries. A uniform positive chance of completion within a fixed number of steps gives a geometric tail and justifies moment calculations.

A stationary law must satisfy balance and normalization. For a finite irreducible chain it determines long-run occupation fractions; convergence of the distribution requires additional conditions, such as aperiodicity. Clearing or overwriting mechanisms can instead give direct couplings. For counts over a fixed window, retain covariances; for coverage, consider inclusion–exclusion and tail sums.

Example R01. An Alarm in a Moving Window

Independent trials succeed with probability $p \in (0, 1)$ and fail with probability $q = 1 - p$. Starting with no observations, an alarm sounds as soon as the last at most three trials contain two successes. Thus either HH or HTH triggers it. Find the mean alarm time and the probability that the final trigger is HH.

Solution. Before the alarm, retain three states: 0 means no H in the last two trials, A means the last trial is H, and B means the suffix is HT. The empty record also has state 0. Write the remaining means as e_0, e_A, e_B. First-step analysis gives

$$e_0 = 1 + qe_0 + pe_A, \quad e_A = 1 + qe_B, \quad e_B = 1 + qe_0.$$

Two consecutive H's always finish the task, so the waiting time has a geometric tail and these means are finite. Substitution yields

$$(1 - q^2)e_0 = \frac{1}{p} + 1 + q, \qquad \boxed{e_0 = \frac{1/p + 2 - p}{p(2 - p)}}.$$

Let u_0, u_A, u_B denote the probabilities of an HH trigger. Then $u_0 = u_A$, $u_A = p + qu_B$, and $u_B = qu_0$, giving

$$\boxed{u_0 = \frac{p}{1 - q^2} = \frac{1}{2 - p}}.$$

For fair trials the mean is $14/3$ and the HH probability is $2/3$.

Method. One failure does not erase all useful information. Recording only the current success-run length would confuse HT with an empty useful prefix.

Example P16. Collecting Both Repeated Patterns

Toss a fair coin independently until the record has contained both HH and TT. Patterns may overlap, and the record is never cleared. Find the mean and variance of the total number T of tosses.

Solution. First wait for the first equal adjacent pair. After the initial toss, indicators of matching the preceding toss are independent fair Bernoulli variables: the first toss and the switch indicators uniquely determine the entire coin string, and every string is equally likely. Thus this stage has length $D = 1 + G$, with G geometric of parameter $1/2$ on the positive integers. Therefore $\mathbb{E}D = 3$ and $\operatorname{Var} D = 2$.

After HH first appears, wait for TT; after TT first appears, the symmetric task remains. Let U be this second-stage length. States 0 and 1 mean respectively no trailing target symbol and one trailing target symbol. Their means obey

$$a_0 = 1 + \tfrac{1}{2}a_0 + \tfrac{1}{2}a_1, \qquad a_1 = 1 + \tfrac{1}{2}a_0,$$

so $a_0 = 6, a_1 = 4$. Two consecutive target symbols give a geometric tail, ensuring all moments are finite. Second moments satisfy

$$b_0 = 1 + a_0 + a_1 + \tfrac{1}{2}b_0 + \tfrac{1}{2}b_1, \quad b_1 = 1 + a_0 + \tfrac{1}{2}b_0.$$

Substitution gives $b_0 = 58$, hence $\operatorname{Var} U = 58 - 36 = 22$. Future tosses are independent of the first-stage history. By H–T symmetry, the conditional second-stage law depends neither on the first pattern nor on D, proving independence of D, U. Consequently,

$$\boxed{\mathbb{E}T = 9}, \qquad \boxed{\operatorname{Var} T = 24}.$$

Method. Collecting both patterns is a coverage condition. Their two waiting times from an empty record cannot simply be added; the first completion determines the suffix at the start of the second stage.

Example R02. A Queue with Random Clearing

A queue length $X_n \in \{0, 1, \ldots\}$ independently receives one of three operations at each step: increase by 1 with probability a, decrease by 1 with probability b (remain at zero if empty), or clear the queue with probability c. Assume $a, b, c > 0$ and $a+b+c = 1$. Find the stationary law and mean length, and bound convergence to stationarity in total variation from any initial state.

Solution. At $k \geq 1$, stationarity requires $\pi_k = a\pi_{k-1} + b\pi_{k+1}$. Put

$$r = \frac{1 - \sqrt{1 - 4ab}}{2b} \in (0, 1), \qquad \boxed{\pi_k = (1 - r)r^k}.$$

Indeed, $br^2 - r + a = 0$. This polynomial is positive at 0 and equals $-c$ at 1, so the chosen root lies in $(0, 1)$. Substitution verifies the positive-state equations.

The zero-state equation is $\pi_0 = b(\pi_0 + \pi_1) + c$, which follows from the same root identity. The geometric series normalizes the probabilities. Consequently,

$$\boxed{\mathbb{E}_\pi X = \frac{r}{1-r}}.$$

Couple the chain from an arbitrary state with a stationary chain by using the same operations. After the first clearing, they coincide forever. The probability of no clearing in the first n steps is $(1-c)^n$, so the coupling inequality gives

$$\boxed{\|\mathcal{L}(X_n) - \pi\|_{\mathrm{TV}} \le (1-c)^n}.$$

This also proves uniqueness. The chain is irreducible and positive recurrent, so the mean positive-time return to 0 is $1/\pi_0 = 1/(1-r)$; clearing or attempting service at an empty queue counts as a return.

Method. Clearing erases the initial condition regardless of queue size. The stationary tail parameter and the convergence bound describe different properties.

Example R03. Work Generated by Two Task Types

After processing an A task, with equal probabilities it produces either no new tasks or one A and one B. A B task produces no new tasks with probability $2/3$ and two A tasks with probability $1/3$. All choices are independent. Processing one task counts as one operation. Starting from one A, prove that all tasks are eventually processed almost surely, and find the expected total number of each type. If their unit costs are $c_A, c_B \ge 0$, find the expected total cost.

Solution. Assign weights 9 and 7 to pending A and B tasks. Processing A produces expected child weight $(9+7)/2 = 8$, while processing B produces $2\cdot 9/3 = 6$. Both reduce expected pending weight by 1. Let T be the completion time. Nonnegativity of the weight after $n \wedge T$ operations and successive conditional expectations give

$$\mathbb{E}(n \wedge T) \le 9.$$

Monotone convergence yields $\mathbb{E}T \le 9$, hence $T < \infty$ almost surely. The total counts are therefore integrable. Denote their means by A, B. Counting all children of each type gives

$$A = 1 + \tfrac{1}{2}A + \tfrac{2}{3}B, \qquad B = \tfrac{1}{2}A.$$

Thus $A = 1 + 5A/6$, and

$$\boxed{A = 6, \quad B = 3, \quad \mathbb{E}T = 9}, \qquad \boxed{\mathbb{E}C = 6c_A + 3c_B}.$$

Method. Offspring type affects future work. A single unweighted offspring average loses this information; positive weights establish integrability before the count equations are solved.

Example R04. Competing Consensus States

There are $m \geq 2$ binary bits. Each step chooses one uniformly and flips it. Initially k bits are 1, where $1 \leq k < m$. Stop when all bits are 0 or all are 1. Find the probability h_k of reaching all 1's first. Also find the mean stopping time for $m = 4, k = 2$.

Solution. The number of 1's is sufficient. From any nonabsorbing state, a prescribed sequence that flips each existing 1 to 0 reaches all 0's within m steps with probability at least m^{-m}. Hence the stopping time has a geometric tail. For $1 \leq k < m$,

$$h_k = \frac{k}{m} h_{k-1} + \frac{m-k}{m} h_{k+1}, \qquad h_0 = 0, \quad h_m = 1.$$

Let $d_k = h_k - h_{k-1}$. Then $(m-k)d_{k+1} = k d_k$, so

$$d_k = \frac{d_1}{\binom{m-1}{k-1}}, \qquad \boxed{h_k = \frac{\displaystyle\sum_{j=0}^{k-1} \binom{m-1}{j}^{-1}}{\displaystyle\sum_{j=0}^{m-1} \binom{m-1}{j}^{-1}}}.$$

For $m = 4$, symmetry gives remaining means $e_1 = e_3$, and

$$e_1 = 1 + \tfrac{3}{4} e_2, \qquad e_2 = 1 + e_1.$$

Thus $e_1 = 7$ and $\boxed{e_2 = 8}$. The absorption probabilities are $h_1 = 3/8, h_2 = 1/2, h_3 = 5/8$.

Method. The interior drift points toward the middle, so absorption probabilities are generally not k/m. The harmonic function must reflect the actual transitions.

Example R05. A Race Between Overlapping Patterns

Independent signals are H with probability $p \in (0, 1)$ and T with probability $q = 1 - p$. Starting with no observations, stop on the first occurrence of substring HHT or HTH. Find the probability that HHT appears first and the mean stopping time for fair signals.

Solution. Keep the longest suffix that is a prefix of either target. The nonterminal states are $0, H, HH, HT$. In particular, H from HH leaves state HH, while T from HT returns to 0. Let u_s be the probability that HHT wins. Then

$$u_0 = u_H, \quad u_{HH} = 1, \quad u_{HT} = q u_0, \quad u_H = p + q u_{HT}.$$

From HH a T eventually occurs because $q > 0$, explaining $u_{HH} = 1$. Solving gives

$$\boxed{u_0 = \frac{1}{2 - p}}.$$

For fair signals, write e_s for the remaining means. Independent blocks of length 3 equal HHT with probability $1/8$, ensuring finiteness. First-step equations are

$$e_0 = 2 + e_H, \quad e_{HH} = 2, \quad e_{HT} = 1 + \tfrac{1}{2}e_0, \quad e_H = 1 + \tfrac{1}{2}e_{HH} + \tfrac{1}{2}e_{HT}.$$

Hence $e_0 = 9/2 + e_0/4$, giving $\boxed{e_0 = 6}$.

Method. Prefix overlaps determine what information survives a failed match. Equal pattern lengths do not imply equal winning probabilities.

Example R06. Counting Runs in Markov Signals

A stationary two-state chain takes values H and T, with $P(H,T) = a$ and $P(T,H) = b$, where $0 < a, b < 1$. Observe $n \geq 2$ consecutive signals. A run is a maximal consecutive string of equal signals. Find the mean and variance of the number R_n of runs, and simplify when $a = b$.

Solution. Let $I_j = \mathbf{1}_{\{X_{j+1} \neq X_j\}}$ and $m = n - 1$. Then $R_n = 1 + \sum_{j=1}^{m} I_j$. The stationary law is $(b/(a+b), a/(a+b))$. Put

$$r = \frac{2ab}{a+b}, \quad \lambda = 1 - a - b, \quad C = \frac{ab(a-b)^2}{(a+b)^2}.$$

Thus $\mathbb{E}I_j = r$. Conditional on a switch, the new state is equally likely to be H or T, since both stationary directional flows equal $ab/(a+b)$. After $k-1$ further steps, the deviation in the next-switch probability is multiplied by λ^{k-1}. Therefore

$$\mathbb{P}(I_{j+k} = 1 \mid I_j = 1) = r + \lambda^{k-1}\left(\frac{a+b}{2} - r\right),$$

so $\operatorname{Cov}(I_j, I_{j+k}) = C\lambda^{k-1}$. Summing gives

$$\boxed{\mathbb{E}R_n = 1 + mr},$$

$$\boxed{\operatorname{Var} R_n = mr(1-r) + 2C\frac{m - 1 - m\lambda + \lambda^m}{(1-\lambda)^2}}.$$

The fraction is the sum $\sum_{k=1}^{m-1}(m-k)\lambda^{k-1}$ and equals zero when $m = 1$. If $a = b$, independent Bernoulli variables with parameter a can determine the switches, so $R_n - 1 \sim \operatorname{Bin}(n-1, a)$. This agrees with $C = 0$.

Method. Counts over a fixed window require covariances. Knowing the one-step switch rate alone does not establish a binomial law.

Example P15. Exact Mixing in a Shift Register

A register retains $r \geq 1$ binary bits, including leading zeroes. Each step discards the leftmost bit, shifts the rest left, and appends an independent bit that equals 1 with probability $p \in (0,1)$. As an integer, the state obeys $S_{n+1} = (2S_n + \xi_{n+1}) \bmod 2^r$. Find its stationary law. Starting from all zeroes, find the total variation distance to stationarity after n steps.

Solution. After r updates, all old bits have been overwritten, so the state consists of the last r independent bits. If $w(s)$ counts the 1's in state s, then

$$\boxed{\pi(s) = p^{w(s)}(1-p)^{r-w(s)}}.$$

Every initial state has exactly this law after r steps, proving stationarity and uniqueness. For $n < r$, the all-zero start leaves the first $r - n$ bits fixed at zero, while the last n bits are independent fresh bits. The law is therefore π conditioned on $A_n =$ "the first $r - n$ bits are zero." For an event A with $\pi(A) > 0$, the total variation distance between π and its conditional law on A is $1 - \pi(A)$: sum the absolute probability differences separately on A and its complement. Hence

$$\boxed{d_{\mathrm{TV}}(\mathcal{L}(S_n), \pi) = \begin{cases} 1 - (1-p)^{r-n}, & 0 \le n < r, \\ 0, & n \ge r. \end{cases}}$$

Method. The initial information is completely overwritten, yielding exact mixing in finitely many steps. The stationary law is uniform only when $p = 1/2$.

Example R07. Two Machines and One Repairer

Each of two working machines fails independently at rate $\lambda > 0$. A single repairer works whenever any machine is broken. Each repair takes an independent exponential time of rate $\mu > 0$; the repaired machine immediately returns to service. All failure and repair times are independent. The system is available if at least one machine works. Find its long-run availability and the mean time from both working to the first time both are broken.

Solution. Let $K_t = 0, 1, 2$ count broken machines. The upward transition rate is $(2 - k)\lambda$, and the downward rate is μ. Adjacent balance gives

$$2\lambda\pi_0 = \mu\pi_1, \qquad \lambda\pi_1 = \mu\pi_2.$$

With $D = \mu^2 + 2\lambda\mu + 2\lambda^2$, normalization yields

$$(\pi_0, \pi_1, \pi_2) = \frac{1}{D}(\mu^2, 2\lambda\mu, 2\lambda^2), \qquad \boxed{\text{availability} = 1 - \frac{2\lambda^2}{D}}.$$

The finite irreducible continuous-time chain has these long-run proportions. Let e_0, e_1 be the remaining means before state 2. Exponential holding times and the first jump give

$$e_0 = \frac{1}{2\lambda} + e_1, \qquad e_1 = \frac{1}{\lambda + \mu} + \frac{\mu}{\lambda + \mu} e_0.$$

Rearranging gives

$$\boxed{e_0 = \frac{3\lambda + \mu}{2\lambda^2}}.$$

Finiteness follows from a uniform positive probability, over a fixed time block, of enough failures to reach state 2 without a completed repair.

Method. With one repairer, the total repair rate remains μ even when both machines are broken. The machines cannot be treated as independent two-state systems.

Example R08. Hitting Times on a Weighted Triangle

A triangle with vertices A, B, C has positive edge weights $w_{AB} = a, w_{BC} = b, w_{CA} = c$. At each step, choose a neighbor in proportion to the incident edge weights. Find the mean hitting times from A to B and from B to A, and their mean round-trip duration. With $a = b = 1$, does increasing c speed up the trip from A to B?

Solution. Let u_A, u_C be mean times to B. First-step analysis gives

$$u_A = 1 + \frac{c}{a+c}u_C, \qquad u_C = 1 + \frac{c}{b+c}u_A.$$

The finite irreducible chain has finite hitting means. Put $D = ab + ac + bc$. Eliminating u_C yields

$$\boxed{\mathbb{E}_A T_B = \frac{(a+2c)(b+c)}{D}}.$$

Interchanging A and B interchanges b, c. If C denotes the time to reach B and subsequently return to A, the strong Markov property gives

$$\boxed{\mathbb{E}_B T_A = \frac{(a+2b)(b+c)}{D}}, \qquad \boxed{\mathbb{E}_A C = \frac{2(a+b+c)(b+c)}{D}}.$$

When $a = b = 1$, the first mean simplifies to $\boxed{1+c}$. Increasing the A–C weight makes reaching B slower, because the walk spends more steps shuttling between A and C.

Method. Increasing one edge weight also decreases the probabilities of other exits. Graph distance alone does not determine the duration of a weighted random walk.

Example R09. Covering a Reflecting Segment

A walk lives on $\{0, 1, \ldots, N\}$, where $N \geq 2$. Interior steps move left or right with equal probability. From 0 the next step must be to 1, and from N it must be to $N-1$. Starting from $1 \leq i < N$, stop when both endpoints have been visited; call the time C. Find the probability that N is visited last and find $\mathbb{E}_i C$.

Solution. Let σ be the first visit to either endpoint. Before this time the walk is the usual absorbing interval walk. Its harmonic and mean-time equations give

$$\mathbb{P}_i(X_\sigma = 0) = \frac{N-i}{N}, \qquad \mathbb{E}_i \sigma = i(N-i).$$

For example, $v_i = i(N-i)$ satisfies $v_i = 1 + (v_{i-1} + v_{i+1})/2$ and zero endpoint values. Visiting N last is exactly the event of visiting 0 first.

Let u_j be the mean time from j to N, retaining reflection at 0. Then $u_N = 0$, $u_0 = 1 + u_1$, and the interior mean-time equation is unchanged. The quadratic

$u_j = N^2 - j^2$ satisfies all these conditions. Hence a full crossing from 0 to N has mean N^2, as does the reverse crossing. The strong Markov property now gives

$$\boxed{\mathbb{P}_i(\text{last endpoint } N) = \frac{N-i}{N}}, \qquad \boxed{\mathbb{E}_i C = i(N-i) + N^2}.$$

A uniform positive probability of crossing within a fixed number of steps ensures integrability of these times. Visiting both endpoints necessarily visits every intermediate vertex.

Example R10. Coverage by Pair Sampling

There are $n \geq 3$ labels. Each round independently chooses a uniformly random unordered pair of distinct labels. Stop when all labels have appeared, and let T be the number of rounds. Give finite sums for its tail probability, mean and second moment, then compute the mean and variance for $n = 4$.

Solution. For a specified set of k labels, the probability that a round avoids it is

$$q_k = \frac{\binom{n-k}{2}}{\binom{n}{2}},$$

where the numerator is zero if its upper argument is less than 2. Failure to cover after t rounds means at least one label is absent. Inclusion–exclusion gives

$$\mathbb{P}(T > t) = \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} q_k^t, \qquad t = 0, 1, \ldots,$$

with $0^0 = 1$ at $t = 0$. Every $q_k < 1$, so each geometric series in this finite sum converges. Integer tail-sum identities yield

$$\boxed{\mathbb{E}T = \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} \frac{1}{1-q_k}},$$

$$\boxed{\mathbb{E}T^2 = \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} \frac{1+q_k}{(1-q_k)^2}}.$$

For $n = 4$, $(q_1, q_2, q_3, q_4) = (1/2, 1/6, 0, 0)$. Substitution gives

$$\boxed{\mathbb{E}T = \frac{19}{5}}, \quad \mathbb{E}T^2 = \frac{423}{25}, \quad \boxed{\operatorname{Var} T = \frac{62}{25}}.$$

Method. One round may discover two labels, so successive discovery stages are not separate geometric waits. Events specifying missing labels are better suited to inclusion–exclusion.

Section Exercises

OR01. Waiting for Consecutive Heads

A fair coin is tossed independently until n consecutive heads have appeared, where $n \geq 1$ is an integer. Find the expected total number T_n of tosses.

OR07. The Long-Run Frequency of Getting Wet

A commuter owns $n \geq 1$ umbrellas. Initially, k are at home and $n - k$ are at the office, where $0 \leq k \leq n$. Each morning the commuter travels from home to the office and returns in the evening. Independently on each trip, it rains with probability $p \in [0, 1]$. An umbrella is carried only when it rains and one is available at the departure location. The commuter gets wet if it rains and no umbrella is available. Find the long-run fraction of trips on which the commuter gets wet.

2.2 Random Walks

Key Ideas and Methods. Specify absorption, reflection and restarting before writing a walk's equations. If direction has memory or the boundary moves with past records, include direction or drawdown in the state. Hitting probabilities satisfy harmonic equations, whereas mean durations satisfy first-step equations with an added time cost. Conditioning on success reweights transitions by subsequent success probabilities; merely deleting the unsuccessful endpoint does not define the conditioned walk.

Martingales offer another route to boundary calculations, but optional stopping needs hypotheses. First stop at $\tau \wedge n$, then justify passage to the limit through uniform integrability or an integrable dominating variable. Almost-sure termination alone is insufficient. Jumps may also overshoot a target, so use the actual terminal value rather than automatically substituting the boundary level.

Example R11. Emptying with Random Restarts

An integer process starts at 1 and stops at 0. At each nonterminal step it resets to 1 with probability $\rho \in (0, 1)$; otherwise it moves left or right with equal probability. A reset counts as a step, and all choices are independent. Find the mean stopping time and the choice of ρ minimizing it.

Solution. From any positive state, two successive operations "reset, then move left" finish with probability $\rho(1-\rho)/2$. Thus the remaining means have a uniform finite bound. Writing e_i for the mean from i gives

$$e_i = 1 + \rho e_1 + \frac{1-\rho}{2}(e_{i-1} + e_{i+1}), \qquad e_0 = 0.$$

The characteristic roots are reciprocal. The smaller one is

$$r = \frac{1 - \sqrt{1 - (1-\rho)^2}}{1-\rho}.$$

A constant particular solution is $(1 + \rho e_1)/\rho$. Boundedness discards the larger root, and the zero boundary gives

$$e_i = \frac{1 + \rho e_1}{\rho}(1 - r^i), \qquad \boxed{e_1 = \frac{1 - r}{\rho r}}.$$

The root equation gives $1 - \rho = 2r/(1 + r^2)$. Reparameterizing by $r \in (0, 1)$,

$$\rho = \frac{(1 - r)^2}{1 + r^2}, \qquad e_1 = \frac{1 + r^2}{r(1 - r)}.$$

The derivative has the sign of $r^2 + 2r - 1$, and the mean diverges at both endpoints. The unique minimizer is therefore $r = \sqrt{2} - 1$, giving

$$\boxed{\rho_* = 1 - \frac{1}{\sqrt{2}}}, \qquad \boxed{\min \mathbb{E}T = 2 + 2\sqrt{2}}.$$

Method. Restarting retrieves paths that have wandered far away but also wastes progress near the target. The optimal rate balances these two effects.

Example R12. Occupation Counts on Successful Paths

A fair integer walk starts at $i \in \{1, \dots, N - 1\}$ and stops at 0 or N, at time τ. Conditional on absorption at N, find the expected number of visits to each interior state j before stopping, counting the initial time, and hence find the conditional mean duration.

Solution. First find the unconditional occupation function

$$G(i, j) = \mathbb{E}_i \sum_{t=0}^{\tau-1} \mathbf{1}_{\{X_t = j\}}.$$

For fixed j, it is harmonic and linear away from $i = j$. At j it obeys $G(j, j) = 1 + [G(j - 1, j) + G(j + 1, j)]/2$, and it vanishes at both endpoints. Matching the two linear pieces and their slope jump at j gives

$$G(i, j) = \frac{2 \min(i, j)[N - \max(i, j)]}{N}.$$

The success probability is $h_i = i/N$. Following a visit to j, the future success probability is h_j. Conditional expectation term by term in the nonnegative visit count gives success-weighted mean $h_j G(i, j)$. Dividing by h_i yields

$$\boxed{G^*(i, j) = \frac{2j \min(i, j)[N - \max(i, j)]}{Ni}}.$$

Split the sum into $j \le i$ and $j > i$, and use the sum of squares to obtain

$$\boxed{\mathbb{E}_i(\tau \mid X_\tau = N) = \sum_{j=1}^{N-1} G^*(i, j) = \frac{N^2 - i^2}{3}}.$$

Alternatively, the success-weighted duration $v_i = i(N^2 - i^2)/(3N)$ satisfies $v_i = h_i + (v_{i-1} + v_{i+1})/2$ and zero boundaries. The geometric exit tail on a finite interval ensures all these expectations are finite.

Method. Conditioning reweights visits to j by h_j/h_i. Removing the unsuccessful endpoint without changing the transition probabilities would not produce the conditioned law.

Example R13. A Walk with Directional Memory

A walk on $\{0, \dots, N\}$ stops at an endpoint, where $N \geq 2$. At an interior state, it keeps its preceding direction with probability $\alpha \in (0, 1)$ or reverses it with probability $1 - \alpha$, then moves one unit in the chosen direction. The initial position is $1 \leq i < N$, and the preceding direction is known. Find the probabilities h_i^+, h_i^- of reaching N first when that direction is right or left.

Solution. Position alone is not a Markov state; direction must be included. The interior equations are

$$h_i^+ = \alpha h_{i+1}^+ + (1 - \alpha) h_{i-1}^-, \qquad h_i^- = (1 - \alpha) h_{i+1}^+ + \alpha h_{i-1}^-.$$

The only reachable absorbing states are $h_N^+ = 1, h_0^- = 0$. Try $h_i^- = Ai$ and $h_i^+ = Ai + B$. Substitution gives $(1 - \alpha)B = (2\alpha - 1)A$, while the upper boundary gives $AN + B = 1$. With

$$D = N(1 - \alpha) + 2\alpha - 1 = (N - 2)(1 - \alpha) + 1 > 0,$$

we obtain

$$\boxed{h_i^- = \frac{i(1 - \alpha)}{D}, \qquad h_i^+ = \frac{i(1 - \alpha) + 2\alpha - 1}{D}}.$$

These candidates satisfy every equation. Consecutive rightward steps have a uniform positive probability of absorption within N steps, so the finite-state boundary problem has a unique probability solution. When $\alpha = 1/2$, both expressions reduce to i/N. When $\alpha > 1/2$, a rightward preceding direction increases the success probability.

Method. Averaging left and right steps discards directional persistence. The initial direction is essential information.

Example R14. A Triangle Walk with a Fixed Endpoint

A walk on triangle A, B, C chooses either other vertex with equal probability and starts at A. Conditional on being at A after $n \geq 2$ steps, find the probability of being at B at an integer time $0 \leq k \leq n$, and the expected total number of visits to A from time 0 through n. Evaluate these for $n = 4, k = 2$.

Solution. Put $r = -1/2$. The recurrence $a_{t+1} = (1 - a_t)/2, a_0 = 1$ gives the return probability and the probability of any specified different vertex:

$$a_t = \frac{1 + 2r^t}{3}, \qquad b_t = \frac{1 - r^t}{3}.$$

The Markov property at time k and transition symmetry give

$$\boxed{\mathbb{P}(X_k = B \mid X_n = A) = \frac{b_k b_{n-k}}{a_n}}.$$

For $L = \sum_{k=0}^{n} \mathbf{1}_{\{X_k = A\}}$,

$$\mathbb{E}(L \mid X_n = A) = \frac{1}{a_n} \sum_{k=0}^{n} a_k a_{n-k}.$$

Expanding the product as $[1 + 2r^k + 2r^{n-k} + 4r^n]/9$ and summing the geometric series yields

$$\boxed{\mathbb{E}(L \mid X_n = A) = \frac{(n+1)(1+4r^n) + 4(1-r^{n+1})/(1-r)}{3(1+2r^n)}}.$$

For $n = 4, k = 2$, $a_4 = 3/8, b_2 = 1/4$, so the probability is $\boxed{1/6}$ and the visit mean is $\boxed{8/3}$. The conditioned chain is generally time-inhomogeneous: at time k, its transition from u to v is multiplied by $P^{n-k-1}(v, A)/P^{n-k}(u, A)$, used only on conditionally reachable states with positive denominator.

Example R15. Alternating Additions to a Threshold

Two players take turns generating independent $U[0, 1]$ increments. All increments are added to one common total, initially zero. The player whose increment first makes the total exceed $t \in (0, 1]$ wins. Find the first player's winning probability, the fair threshold, and the mean and variance of the total number N of turns.

Solution. For $n \geq 1$, no crossing means that the first n increments sum to at most t. Since $t \leq 1$, this is a simplex on which the individual upper bounds do not bind. Its volume is

$$\mathbb{P}(N > n) = \frac{t^n}{n!}; \qquad \mathbb{P}(N > 0) = 1.$$

This tail ensures all moments are finite. The probability generating function follows from the tail-sum identity:

$$G(z) = \mathbb{E}z^N = 1 + (z-1) \sum_{n=0}^{\infty} z^n \mathbb{P}(N > n) = 1 + (z-1)e^{zt}.$$

The first player wins exactly when N is odd. Therefore

$$\boxed{\mathbb{P}(\text{first player wins}) = \frac{1 - G(-1)}{2} = e^{-t}}, \qquad \boxed{t_{\text{fair}} = \log 2}.$$

Differentiation gives $G'(1) = e^t, G''(1) = 2te^t$, hence

$$\boxed{\mathbb{E}N = e^t}, \qquad \boxed{\operatorname{Var} N = (2t+1)e^t - e^{2t}}.$$

Method. Monotone accumulation turns the winner into a question about the parity of a stopping count. The mean number of turns alone does not determine first-player advantage.

Example R16. Drawdown from the Running Maximum

A fair integer walk S_n starts at 0, and $M_n = \max_{0\le j\le n} S_j$. For an integer $d \ge 1$, stop at the first time τ when $M_n - S_n = d$. Find $\mathbb{E}\tau$, $\operatorname{Var}\tau$ and the distribution of the terminal running maximum M_τ.

Solution. The gap $Y_n = M_n - S_n$ is a finite-state chain. At 0 it stays or moves to 1 with equal probability; at $1, \dots, d-1$ it moves one unit in either direction; d is absorbing. A block of d downward walk steps ends the process, so all stopping moments are finite. With $m_i = \mathbb{E}_i\tau$, we have $m_0 = 1 + (m_0 + m_1)/2$, the interior equation $m_i = 1 + (m_{i-1} + m_{i+1})/2$, and $m_d = 0$. The quadratic solution is

$$m_i = D - i(i+1), \qquad D = d(d+1), \qquad \boxed{\mathbb{E}\tau = D}.$$

Let v_i be the second moment. Expanding $(1+\tau')^2$ gives

$$v_0 - v_1 = 4D - 2, \qquad v_{i-1} - 2v_i + v_{i+1} = 2 - 4m_i, \qquad v_d = 0.$$

Successive summation of first differences yields

$$v_0 = \sum_{j=0}^{d-1} (d-j)(4m_j - 2) = \frac{5D^2 - D}{3}, \qquad \boxed{\operatorname{Var}\tau = \frac{D(2D-1)}{3}}.$$

For the maximum, restart at each new high h. The probability of reaching $h+1$ before $h-d$ is $d/(d+1)$. Repeating this competition by the strong Markov property gives

$$\boxed{\mathbb{P}(M_\tau \ge k) = \left(\frac{d}{d+1}\right)^k, \quad k = 0, 1, \dots}.$$

Method. Absolute position and the running maximum form a two-dimensional state. Their difference turns the moving boundary into reflection with a self-loop.

Example R17. The Limit of Fractional Betting Wealth

Starting with wealth $x > 0$, bet a fixed fraction $f \in (0,1)$ on independent fair rounds. Wealth is multiplied by $1+f$ after a win and by $1-f$ after a loss. Write it as Z_n. Show that $\mathbb{E}Z_n = x$ for every n, yet $Z_n \to 0$ almost surely. For a target $A > x$, bound the probability of ever reaching or exceeding A, and explain why terminal wealth cannot simply be set equal to A.

Solution. The conditional mean multiplier is 1, so Z_n is a nonnegative martingale and $\mathbb{E}Z_n = x$ at every finite time. However, the strong law gives

$$\frac{1}{n}\log\frac{Z_n}{x} \longrightarrow \tfrac{1}{2}\log(1+f) + \tfrac{1}{2}\log(1-f) = \tfrac{1}{2}\log(1-f^2) < 0.$$

Thus wealth tends to zero almost surely. Its constant mean shows that the martingale family is not uniformly integrable.

Let $\tau = \inf\{n : Z_n \geq A\}$. Bounded optional stopping gives $\mathbb{E}Z_{n\wedge\tau} = x$. The stopped family is bounded by $(1+f)A$: before hitting it is below A, and a crossing starts below A. Its limit is Z_τ on hitting and zero on nonhitting. Dominated convergence yields

$$x = \mathbb{E}[Z_\tau \mathbf{1}_{\{\tau<\infty\}}].$$

Using $A \leq Z_\tau < (1+f)A$ gives

$$\boxed{\frac{x}{(1+f)A} \leq \mathbb{P}(\tau < \infty) \leq \frac{x}{A}}.$$

Method. Fair arithmetic returns do not ensure growth on typical paths. Multiplicative jumps overshoot the target, so the terminal value is not a fixed boundary value.

Example R18. Costs Until the First Collision

Each independent draw has type $j \in \{1, \ldots, m\}$ with probability $p_j > 0$, where the probabilities sum to 1, and cost $c_j \geq 0$. Stop at the first repeated type. Let N include that final draw. Find the expected total cost and the distribution of the final type. Evaluate the case of probabilities $2/3, 1/3$ and costs 1 and 4.

Solution. Let $e_k(p)$ be the elementary symmetric polynomial of degree k in $p_1, \ldots, p_m$, with $e_0 = 1$. The first k draws are distinct with probability $k!e_k(p)$, so

$$\boxed{\mathbb{E}N = \sum_{k=0}^{m} k!e_k(p)}.$$

Here $2 \leq N \leq m+1$. If draw t costs C_t and $\bar{c} = \sum_j p_j c_j$, the event $N \geq t$ depends only on earlier draws and is independent of C_t. A finite sum gives

$$\boxed{\mathbb{E}\sum_{t=1}^{N} C_t = \bar{c}\,\mathbb{E}N}.$$

For the final type to be j at time $k+1$, the first k distinct types must include j. Ordering these types and then drawing j once more has probability $k!p_j^2 e_{k-1}(p_{-j})$, where p_{-j} omits entry j. Thus

$$\boxed{\mathbb{P}(J_{\text{last}} = j) = p_j^2 \sum_{k=1}^{m} k!e_{k-1}(p_{-j})}.$$

For two types with $p = 2/3, q = 1/3$, $\mathbb{E}N = 2 + 2pq = 22/9$ and $\bar{c} = 2$, so the mean total cost is $\boxed{44/9}$. The final type is the first one with probability $p^2(1+2q) = 20/27$. The final draw's mean cost is $\boxed{16/9}$, not its unconditional single-draw mean 2.

Method. Cumulative cost uses a continuation event measurable from the past. The terminal type is selected by the stopping rule and does not satisfy the same independence argument.

Section Exercises

OR11. Hitting Zero with Asymmetric Jumps

A random walk starts at a positive integer n. Its independent increments equal -1 with probability $p \in (0,1)$ and $+2$ with probability $1-p$. Find the probability u_n that the walk ever reaches 0.

OR16. The Second Moment of an Exit Time

Let $S_0 = 0$ and let S_n be a random walk with independent increments, each equal to $+1$ or -1 with probability $1/2$. For positive integers a, b, define

$$\tau = \inf\{n \geq 0 : S_n = -a \text{ or } S_n = b\}.$$

Find $\mathbb{P}(S_\tau = b)$, $\mathbb{E}\tau$, and $\mathbb{E}\tau^2$.

2.3 Optimal Stopping

Key Ideas and Methods. Optimal stopping compares the payoff available now with the value of continuing and then acting optimally. The state must include the remaining opportunities, any offer that can be recalled, and the posterior for an unknown parameter. Do not charge costs already paid a second time. Compare feasible actions at each observation before averaging over observations; equality may allow several optimal decisions.

Backward induction verifies finite-horizon policies. With unlimited opportunities, establish a finite value, justify the limit of the recursion, and check that the proposed policy attains it. Discounting can provide a uniform bound on truncation error. Replication and superhedging also require nodewise checks of self-financing, borrowing and short-selling restrictions, not just comparison of terminal payoffs.

Example R19. A Digital Payoff under Funding Constraints

A stock starts at 100. In each of two periods its price is multiplied by 2 or $1/2$, and every path has positive probability. Cash earns zero interest; arbitrary fractional stock positions can be traded without friction. Trading must be self-financing: no funds are added or withdrawn after the initial investment. The terminal payoff is $\mathbf{1}_{\{S_2 \geq 100\}}$. First allow borrowing and short selling, and find the exact replication cost and holdings. Then require both stock holdings and cash to be nonnegative after each rebalance. Find the smallest initial wealth covering the payoff on every path.

Solution. At the up node with stock price 200, both terminal payoffs are 1, replicated by cash 1. At the down node with price 50, successor prices are 100 and 25. Holdings (Δ, B) solve

$$100\Delta + B = 1, \qquad 25\Delta + B = 0,$$

so $\Delta = 1/75$, $B = -1/3$, with node value $1/3$. Initial holdings solve $200\Delta_0 + B_0 = 1$ and $50\Delta_0 + B_0 = 1/3$, giving

$$\boxed{\Delta_0 = \frac{1}{225}, \quad B_0 = \frac{1}{9}, \quad V_0 = \frac{5}{9}}.$$

At either node the old portfolio value equals the new portfolio cost, so the strategy is self-financing.

Now impose nonnegative holdings. At the low node, covering the payoff at price 100 requires $100\Delta + B \geq 1$, hence the cost $50\Delta + B \geq 1/2$. Holdings $\Delta = 1/100$, $B = 0$ attain this bound. At the high node, $100\Delta + B \geq 1$ implies cost $200\Delta + B \geq 1$, attained by cash 1. Initial wealth must therefore cover node values 1 and $1/2$. Every feasible initial portfolio satisfies

$$100\Delta_0 + B_0 = \tfrac{1}{3}(200\Delta_0 + B_0) + \tfrac{2}{3}(50\Delta_0 + B_0) \geq \frac{2}{3}.$$

Holdings $\Delta_0 = 1/300$, $B_0 = 1/3$ reach exactly those node values and permit the stated continuations. Thus

$$\boxed{V_0^{\text{super}} = \frac{2}{3}}.$$

Method. Exact replication borrows at the low node. Once borrowing is prohibited, the cost must be recomputed by nodewise superhedging rather than taken from the unconstrained price.

Example R20. Stopping with Recall of an Offer

Observe $X \sim U[0,1]$ for free. Either accept it, or pay $c > 0$ to observe an independent $Y \sim U[0,1]$, after which you receive $\max(X, Y)$. Only one further observation is allowed. Find the optimal rule and initial net value. Evaluate $c = 1/8$, and compare with the rule requiring acceptance of Y after payment, with no recall of X.

Solution. Given $X = x$, the expected gain from the additional observation rather than stopping is

$$\mathbb{E}(Y - x)^+ - c = \int_x^1 (y - x)\,\mathrm{d}y - c = \frac{(1-x)^2}{2} - c.$$

For $0 < c < 1/2$, continue exactly when $x < 1 - \sqrt{2c}$; either action is optimal at equality. For $c \geq 1/2$, always stop. The optimal value is

$$V = \frac{1}{2} + \int_0^{1-\sqrt{2c}} \left(\frac{(1-x)^2}{2} - c \right) \mathrm{d}x = \frac{2}{3} - c + \frac{(2c)^{3/2}}{3}$$

for $c < 1/2$, and $V = 1/2$ otherwise. With $c = 1/8$, the threshold is $1/2$ and $\boxed{V = 7/12}$.

Without recall, the continuation value is the constant $\mathbb{E}Y - c = 3/8$. Continue only for $x < 3/8$. The net value is

$$\boxed{\mathbb{E}\max(X, 3/8) = \frac{1}{2} + \frac{1}{2}\left(\frac{3}{8}\right)^2 = \frac{73}{128}}.$$

Recall gives the higher value because the old offer remains a floor on the eventual prize.

Example R21. Stopping while Learning a Win Rate

An unknown parameter P is drawn from $U[0, 1]$. Conditional on P, successive coin tosses are independent with H probability P. Each H pays $+1$ and each T pays -1. You may stop at any time, including before the first toss, and may toss at most three times. Past rewards are retained. Find the maximum expected total reward and a complete optimal policy. Compare with a coin known to be fair.

Solution. After h H's and t T's, the posterior density is proportional to $p^h(1-p)^t$, so the next H probability is $q_{h,t} = (h+1)/(h+t+2)$. Let $V_k(h, t)$ be the optimal additional reward with k tosses left. Then $V_0 = 0$ and

$$\begin{aligned} C_k(h,t) &= q_{h,t}[1 + V_{k-1}(h+1,t)] \\ &\quad + (1 - q_{h,t})[-1 + V_{k-1}(h,t+1)], \\ V_k(h,t) &= \max\{0, C_k(h,t)\}. \end{aligned}$$

Stopping gives additional reward zero. The continuation term includes the next outcome and subsequent optimal decisions, so backward induction proves the recursion. With one toss left,

$$V_1(2,0) = \tfrac{1}{2}, \qquad V_1(1,1) = V_1(0,2) = 0.$$

The preceding stages give

$$V_2(1,0) = \tfrac{2}{3}, \quad V_2(0,1) = 0, \qquad \boxed{V_3(0,0) = \tfrac{1}{3}}.$$

Toss initially. Stop after a first T and continue after a first H. After HH, continue once more. After HT, the final toss has conditional mean zero, so stopping and continuing tie. For a coin known to be fair, bounded optional stopping makes the expected reward zero under every rule using at most three tosses.

Method. Each toss has marginal H probability $1/2$, but observations change the predictive probability. A shared unknown parameter makes the outcomes dependent before conditioning on it.

Example R22. Discounted Offers over an Unlimited Horizon

Offers $X_1, X_2, \ldots$ are independent $U[0,1]$ variables. After each observation, accept it or discard it permanently. Acceptance at observation N pays present value $\delta^{N-1}X_N$, where $0 < \delta < 1$. Never accepting pays zero. Find the optimal rule, initial value and mean observation count under a strict threshold rule.

Solution. Let V_k be the value with at most k observations, and $V_0 = 0$. Backward induction gives

$$V_{k+1} = \mathbb{E}\max(X, \delta V_k).$$

The map has Lipschitz constant at most $\delta < 1$, so iteration converges to its unique fixed point V. Truncating any infinite-horizon rule at observation k changes its payoff by at most δ^{k-1} in absolute value. Thus the fixed point is the infinite-horizon optimum.

Put $t = \delta V \in (0,1)$. Integration gives $V = (1+t^2)/2$, or $\delta t^2 - 2t + \delta = 0$. Taking the root in $(0,1)$ yields

$$\boxed{t = \frac{1-\sqrt{1-\delta^2}}{\delta}}, \qquad \boxed{V = \frac{1}{1+\sqrt{1-\delta^2}}}.$$

Accept at the first $X_n > t$. The observation count is geometric of parameter $1-t$, hence $\boxed{\mathbb{E}N = 1/(1-t)}$. Its actual discounted payoff is

$$\sum_{n=1}^{\infty} (\delta t)^{n-1}\mathbb{E}[X\mathbf{1}_{\{X>t\}}] = \frac{1-t^2}{2(1-\delta t)} = V,$$

verifying attainment.

Method. Discounting bounds the contribution of the distant future by a deterministic geometric tail, justifying finite-horizon approximation of the unlimited problem.

Example R23. Choosing Which Die to Reroll

Roll two independent fair six-sided dice. After seeing both values, take their product, or pay $c \geq 0$ to reroll exactly one chosen die once. The new value must be used; the payoff is the final product minus the fee. Find the complete optimal rule and the maximum expected net payoff for $c = 1$.

Solution. Sort the observed values as $x \leq y$. Rerolling the smaller die has net expectation $(7/2)y - c$, whereas rerolling the larger gives $(7/2)x - c$. The former is at least as large. Thus

$$\boxed{\text{reroll the smaller die} \iff y(7/2 - x) > c},$$

with either action optimal at equality; otherwise stop. Without intervention the product mean is $(7/2)^2 = 49/4$, so

$$V(c) = \frac{49}{4} + \frac{1}{36}\sum_{i=1}^{6}\sum_{j=1}^{6}\left(\tfrac{7}{2}\max(i,j) - ij - c\right)^{+}.$$

For $c = 1$, continue exactly when the smaller value is at most 3. Grouping the 36 equally likely ordered outcomes by smaller value 1, 2 or 3 gives total gains $183/2, 48, 19/2$, respectively. Equal values count once and unequal pairs twice. Hence

$$\boxed{V(1) = \frac{49}{4} + \frac{149}{36} = \frac{295}{18}}.$$

Method. Compare the two controlled actions first, then compare the better one with stopping. A low observed value alone does not justify a reroll without accounting for its fee.

Example R24. Ordering Tests with Early Stopping

Independent tests $i = 1, \ldots, n$ pass with probabilities $p_i \in (0, 1)$ and incur costs $c_i > 0$. To determine whether all pass, choose their order, possibly adaptively, and stop at the first failure; if all pass, perform every test. Find an order minimizing expected cost. How does the rule change when the goal is to determine whether at least one passes? Evaluate $p = (9/10, 1/2, 1/10)$ and $c = (1, 2, 5)$.

Solution. For the all-pass decision, every nonterminal history consists entirely of passes. Thus each deterministic adaptive policy is equivalent to a fixed permutation; randomized policies are mixtures of permutations. The expected cost of permutation π is

$$\sum_{k=1}^{n} c_{\pi_k} \prod_{j<k} p_{\pi_j}.$$

Adjacent tests i, j contribute $c_i + p_i c_j$ or $c_j + p_j c_i$. The probability of continuing past both is $p_i p_j$ in either order, so later costs are unchanged. Putting the cheaper order first is equivalent to

$$c_i(1 - p_j) \le c_j(1 - p_i) \quad \Longleftrightarrow \quad \frac{c_i}{1 - p_i} \le \frac{c_j}{1 - p_j}.$$

Successively removing inversions proves that increasing $c_i/(1 - p_i)$ is globally optimal.

For the at-least-one-pass decision, stop at the first pass; every surviving history consists of failures. Replacing the continuation probabilities by $1 - p_i$ in the same interchange proof gives increasing c_i/p_i. For the stated data,

$$\boxed{\text{all pass: } (2, 3, 1), \quad C = \frac{91}{20}},$$

$$\boxed{\text{at least one: } (1, 2, 3), \quad C = \frac{29}{20}}.$$

Method. The ordering index depends on the logical stopping objective. The same tests can require different optimal orders for different decisions.

Section Exercises

OR20. Optimal Stopping with Red and Black Cards

A deck contains 26 red cards and 26 black cards and is uniformly shuffled. Cards are drawn one at a time without replacement. The score starts at 0, increases by 1 for a red card, and decreases by 1 for a black card. You may stop before drawing or after any draw and receive the current cumulative score. Find an optimal stopping rule and the maximum expected score.

OR22. Rerolling a Die at a Cost

A fair six-sided die has faces $1, 2, \ldots, 6$. The first roll is free. After observing a face, you may accept it and stop, or pay 1 to roll again; every subsequent reroll also costs 1. Rolls are independent, and previous faces cannot be recalled. Admissible strategies must stop after finitely many rolls almost surely. The final payoff is the last face minus all reroll fees. Find an optimal strategy and the maximum expected payoff.

2.4 Brownian Motion

Key Ideas and Methods. Independent increments and Gaussian structure reduce many Brownian calculations to covariance integrals. Deterministic square-integrable kernels give Gaussian Itô integrals; exponentiation or evaluation at an independent random time generally does not preserve normality. Use orthogonal projection for jointly Gaussian observations and reflection for path maxima. Conditioning on an endpoint, a probability-zero event, requires densities or a continuous version of the conditional law.

An independent exponential failure time becomes a discount factor; a generator boundary problem can then be verified with Itô's formula. Interchanging integrals, expectations or parameter derivatives requires nonnegativity, integrability or a suitable domination argument. Poisson splitting requires independent marking. In the renewal model, uniform observation over a long time interval asymptotically selects intervals in proportion to their lengths.

Example R25. Predicting an Exponential Brownian Integral

Let W be standard Brownian motion starting at 0, let $T > 0$, and define

$$I = \int_0^T e^{W_t - t/2}\,\mathrm{d}t.$$

Find its mean and variance, and the coefficients and error of the best predictor $\alpha + \beta W_T$ under squared loss. Then find the conditional mean given $W_T = b$ and determine whether it equals that affine predictor.

Solution. Write $M_t = e^{W_t - t/2}$. The normal moment generating function gives $\mathbb{E}M_t = 1$ and, for $s \le t$, $\mathbb{E}M_s M_t = e^s$. Tonelli's theorem for the nonnegative integrands yields

$$\mathbb{E}I = T, \qquad \mathbb{E}I^2 = 2\int_0^T \int_0^t e^s \, \mathrm{d}s \, \mathrm{d}t = 2(e^T - 1 - T).$$

Finiteness also supplies the square integrability needed for subsequent covariance interchanges. Hence

$$\boxed{\operatorname{Var} I = 2(e^T - 1 - T) - T^2}.$$

Independent increments and differentiation of the normal moment generating function give $\mathbb{E}(M_t W_T) = \mathbb{E}(M_t W_t) = t$, so $\operatorname{Cov}(I, W_T) = T^2/2$. The normal equations for squared loss give

$$\boxed{\alpha = T, \qquad \beta = T/2}.$$

$$\boxed{\mathrm{MSE}_{\mathrm{affine}} = 2(e^T - 1 - T) - T^2 - \frac{T^3}{4}}.$$

Given $W_T = b$, W_t is normal with mean tb/T and variance $t - t^2/T$. This follows from the Gaussian orthogonal decomposition of $W_t - (t/T)W_T$ and W_T. Using the normal exponential moment and conditional Tonelli again,

$$\boxed{\mathbb{E}(I \mid W_T = b) = \int_0^T \exp\left(\frac{tb}{T} - \frac{t^2}{2T}\right) \mathrm{d}t}.$$

This function has strictly positive second derivative in b, so it is not affine. Allowing nonlinear prediction strictly reduces mean squared error.

Method. Exponentiation destroys the joint Gaussian structure. Covariance still determines the best linear coefficient, but it does not force the conditional mean to be linear.

Example R26. The Maximum Given a Brownian Endpoint

Let W be standard Brownian motion and $M_T = \max_{0 \le t \le T} W_t$, where $T > 0$. Find the conditional law of M_T given $W_T = b$. In particular, find its mean when $b = 0$.

Solution. For $a > \max(0, b)$, reflect the path after its first visit to a about that level. Paths that reach a and finish near b correspond to paths ending near $2a - b$. Writing φ_T for the $N(0, T)$ density, the conditional tail is

$$\boxed{\mathbb{P}(M_T \ge a \mid W_T = b) = \frac{\varphi_T(2a - b)}{\varphi_T(b)} = \exp\left(-\frac{2a(a - b)}{T}\right)}.$$

This density ratio defines a continuous version of the regular conditional law; it is not a ratio of probabilities of zero-probability events. For $a \le \max(0, b)$, the tail probability is 1.

When $b = 0$, the maximum is nonnegative and its distribution function is $1 - e^{-2a^2/T}$ for $a \geq 0$. The tail integral gives

$$\boxed{\mathbb{E}(M_T \mid W_T = 0) = \int_0^\infty e^{-2a^2/T}\,\mathrm{d}a = \sqrt{\frac{\pi T}{8}}}.$$

Method. An endpoint constraint does not constrain the whole path. Even a path ending at zero may have crossed a positive barrier; reflection computes that crossing probability.

Example R27. Conditioning on Both Endpoint and Area

Let W be standard Brownian motion, $T > 0$, and $A = \int_0^T W_t\,\mathrm{d}t$. Conditional on $W_T = b$ and $A = a$, find the mean and covariance functions of the entire process. Explain why two path values can have negative conditional covariance.

Solution. First remove the endpoint direction by setting $B_t = W_t - (t/T)W_T$. Joint Gaussianity makes B independent of W_T, with covariance

$$K(s,t) = \mathrm{Cov}(B_s, B_t) = \min(s,t) - \frac{st}{T}.$$

Next put $J = \int_0^T B_t\,\mathrm{d}t = A - TW_T/2$. The bounded continuous covariance ensures square integrability of the integral. Integrating the covariance gives

$$\mathrm{Cov}(B_t, J) = \frac{t(T-t)}{2}, \qquad \mathrm{Var}\,J = \frac{T^3}{12}.$$

Project B_t onto J. The residual is independent of both J, W_T, jointly for every finite selection of times. Thus the conditional process remains Gaussian, with mean

$$\boxed{m(t) = \frac{tb}{T} + \frac{6t(T-t)}{T^3}\left(a - \frac{Tb}{2}\right)},$$

and covariance

$$\boxed{K_*(s,t) = \min(s,t) - \frac{st}{T} - \frac{3s(T-s)t(T-t)}{T^3}}.$$

The two observations have a nonsingular covariance matrix because $\mathrm{Var}\,J > 0$, so these formulas define a continuous conditional version for every real a, b. Taking $s = T/4, t = 3T/4$ gives $K_*(s,t) = -11T/256 < 0$. Fixing total area allows an unusually high value at one time to be compensated by a lower value elsewhere, unlike the original positive Brownian covariances.

Example R28. Hitting a Boundary before Independent Failure

Let W be standard Brownian motion and $X_t = x + \sigma W_t$, where $0 < x < L$ and $\sigma > 0$. Let τ be its first hit of 0 or L. An independent exponential failure clock ζ has rate $\lambda > 0$. Success means reaching L before both 0 and failure. Find the success probability and the mean duration conditional on success, including their limits as $\lambda \downarrow 0$.

Solution. Independence of the failure clock gives

$$u_\lambda(x) = \mathbb{E}_x[e^{-\lambda\tau}\mathbf{1}_{\{X_\tau=L\}}].$$

Consider the boundary problem

$$\frac{\sigma^2}{2}u'' - \lambda u = 0, \qquad u(0) = 0, \quad u(L) = 1.$$

Put $k = \sqrt{2\lambda}/\sigma$. The general solution is spanned by $\sinh(kx), \cosh(kx)$, and the endpoints determine

$$\boxed{u_\lambda(x) = \frac{\sinh(kx)}{\sinh(kL)}}.$$

To verify the interpretation, apply Itô's formula to $e^{-\lambda(t\wedge\tau)}u(X_{t\wedge\tau})$; its drift is zero. Boundedness of u, u' on the interval makes the stochastic integral mean zero at each finite time. A Brownian increment large enough to cross the interval has a uniform positive probability over a fixed time block, so $\tau < \infty$ almost surely. Dominated convergence gives the discounted terminal-payoff identity.

For $\lambda > 0$, $\tau e^{-\lambda\tau}$ is bounded over nonnegative times, uniformly for λ in a sufficiently small neighborhood. Differentiation under the expectation is therefore justified, giving

$$\mathbb{E}_x(\tau \mid \text{success}) = -\partial_\lambda \log u_\lambda(x) = \boxed{\frac{L\coth(kL) - x\coth(kx)}{\sigma^2 k}}.$$

Using $\sinh z \sim z$ and $\coth z = 1/z + z/3 + O(z^3)$,

$$\boxed{u_\lambda(x) \to \frac{x}{L}}, \qquad \boxed{\mathbb{E}_x(\tau \mid \text{success}) \to \frac{L^2 - x^2}{3\sigma^2}}.$$

Method. Independent failure weights a path of duration τ by $e^{-\lambda\tau}$, favoring faster successes. The negative logarithmic derivative of success probability yields the conditional duration.

Example CR01. Poisson Splitting and a First Arrival

Events arrive as a Poisson process of rate $\lambda > 0$. Independently label each arrival A with probability p and B with probability $q = 1 - p$, where $0 < p < 1$. Write $N_A(t), N_B(t)$ for the two counts. Prove that they are independent Poisson processes, and find the A count conditional on a total of n arrivals. Then let τ be the first A arrival and $K = N_B(\tau)$. Find the distribution of K and the conditional distribution of τ given $K = k$.

Solution. Conditioning first on the total count, the joint probability generating function at a fixed time is

$$\mathbb{E}[z^{N_A(t)}w^{N_B(t)}] = \exp\{\lambda t(pz + qw - 1)\} = e^{\lambda pt(z-1)}e^{\lambda qt(w-1)}.$$

It factors into generating functions for Poisson counts with rates λp and λq. Original increments over disjoint intervals are independent, as are all labels. The same

factorization therefore holds jointly over any finite collection of disjoint intervals, proving independence of the two processes. Given a total of n arrivals, their independent labels directly give

$$\boxed{N_A(t) \mid N_A(t) + N_B(t) = n \sim \mathrm{Bin}(n, p)}.$$

Put $\alpha = \lambda p, \beta = \lambda q$. The first A arrival has distribution $\mathrm{Exp}(\alpha)$ and is independent of the entire B process. Hence

$$\mathbb{P}(K = k, \tau \in \mathrm{d}t) = \alpha e^{-(\alpha+\beta)t} \frac{(\beta t)^k}{k!}\,\mathrm{d}t, \qquad k = 0, 1, \ldots.$$

Using $\int_0^\infty t^k e^{-\lambda t}\,\mathrm{d}t = k!/\lambda^{k+1}$ gives

$$\boxed{\mathbb{P}(K = k) = pq^k}.$$

Dividing the joint density by this probability yields

$$\boxed{f_{\tau|K=k}(t) = \frac{\lambda^{k+1}}{k!} t^k e^{-\lambda t}, \quad t > 0}, \qquad \boxed{\mathbb{E}(\tau \mid K = k) = \frac{k+1}{\lambda}}.$$

This is the sum of $k+1$ independent exponential interarrival times of rate λ: exactly k B arrivals precede the first A.

Method. Independence of counts at fixed times does not imply independence after stopping at the first arrival of one type. A joint density makes the resulting conditioning explicit.

Example CR02. Residual Life at a Random Time

A renewal process starts at time zero. Independent intervals have lengths 1 and 3, each with probability $1/2$. Independently choose U_t uniformly in $[0, t]$. Let L_t be the length of the complete renewal interval containing U_t, A_t the time since its left endpoint and R_t the time to its right endpoint. Find their joint limiting distribution as $t \to \infty$. Under that limit, compute $\mathbb{E}R$, $\mathbb{P}(R > 2)$ and $\mathrm{Cov}(A, R)$.

Solution. A uniformly chosen time selects intervals in proportion to their duration. More precisely, among the first n complete intervals, the fraction of total time spent in intervals of length ℓ converges by the strong law to

$$\frac{\ell \mathbb{P}(L = \ell)}{\mathbb{E}L}, \qquad \mathbb{E}L = 2.$$

The unfinished interval at the right endpoint has length at most 3, so its contribution to the time fraction vanishes. The limiting selected length L_* therefore satisfies

$$\boxed{\mathbb{P}(L_* = 1) = \frac{1}{4}, \qquad \mathbb{P}(L_* = 3) = \frac{3}{4}}.$$

Within an interval of length ℓ, a uniformly selected time has a uniform location. Conditional on $L_* = \ell$,

$$\boxed{A \sim U[0, \ell], \qquad R = \ell - A}.$$

This mixture specifies the joint distribution. All variables are bounded, so the distributional limit also gives the required moment limits. Since $\mathbb{E}L^2 = 5$ and $\mathbb{E}L^3 = 14$,

$$\boxed{\mathbb{E}R = \mathbb{E}A = \frac{\mathbb{E}L^2}{2\mathbb{E}L} = \frac{5}{4}}, \qquad \boxed{\mathbb{P}(R > 2) = \frac{\mathbb{E}(L-2)^+}{\mathbb{E}L} = \frac{1}{4}}.$$

The integral of the product of age and residual life over an interval of length ℓ is $\int_0^\ell u(\ell - u)\,\mathrm{d}u = \ell^3/6$. Consequently,

$$\mathbb{E}(AR) = \frac{\mathbb{E}L^3}{6\mathbb{E}L} = \frac{7}{6}, \qquad \boxed{\mathrm{Cov}(A,R) = \frac{7}{6} - \left(\frac{5}{4}\right)^2 = -\frac{19}{48}}.$$

Method. Sampling an interval uniformly and sampling the interval containing a uniform time are different experiments. The latter first weights by length, then selects a uniform location within that interval.

Example CR03. A Single-Server Busy Period

Customers arrive as a Poisson process of rate λ. Independent service times are exponential with rate μ, independently of arrivals, and $0 < \lambda < \mu$. One server works in first-come, first-served order. Starting with one customer, let B be the first time the system is empty. Find the Laplace transform, mean and variance of B, and explain what changes in the mean when $\lambda = \mu$.

Solution. Regard arrivals during a customer's service as that customer's offspring. Conditional on the service time S, the offspring count is $\mathrm{Pois}(\lambda S)$, with mean $\rho = \lambda/\mu < 1$. The expected size of generation j is ρ^j, so the expected total population is $1/(1-\rho)$. Its finiteness implies that the total population is finite almost surely. Existence of a customer does not depend on that customer's own future service time. Tonelli's theorem gives

$$\mathbb{E}B = \frac{1}{\mu}\sum_{j=0}^{\infty}\rho^j = \frac{1}{\mu - \lambda}.$$

Let $b(s) = \mathbb{E}e^{-sB}$ for $s \geq 0$. Arrivals during the first service generate independent copies of the descendant service total. The busy period is the sum of all service times in the tree, regardless of processing order. Conditioning on S therefore gives

$$b(s) = \mathbb{E}\left[e^{-sS}b(s)^{N(S)}\right] = \frac{\mu}{\mu + s + \lambda[1 - b(s)]}.$$

Thus $\lambda b^2 - (\lambda + \mu + s)b + \mu = 0$. Choose the root in $(0,1)$ for $s > 0$ and extend continuously to zero:

$$\boxed{b(s) = \frac{\lambda + \mu + s - \sqrt{(\lambda+\mu+s)^2 - 4\lambda\mu}}{2\lambda}}.$$

Differentiation for $s > 0$ is valid because $B^j e^{-sB}$ is bounded. Differentiating the quadratic equation twice and writing $D_s = 2\lambda b(s) - \lambda - \mu - s$ yields

$$D_s b'(s) = b(s), \qquad D_s b''(s) = 2b'(s) - 2\lambda[b'(s)]^2.$$

As $s \downarrow 0$, $b(s) \to 1$ and $D_s \to \lambda - \mu$. Monotone convergence for the nonnegative moment integrands identifies the finite derivative limits:

$$\boxed{\mathbb{E}B = \frac{1}{\mu - \lambda}}, \qquad \mathbb{E}B^2 = \frac{2\mu}{(\mu - \lambda)^3},$$

$$\boxed{\operatorname{Var} B = \frac{\lambda + \mu}{(\mu - \lambda)^3}}.$$

At criticality, the offspring generating function is $f(z) = \mu/[\mu + \lambda(1 - z)]$. Iteration of f from zero gives extinction probabilities within finitely many generations; the limit is the smallest fixed point in $[0, 1]$. When $\lambda = \mu$, that fixed point is 1. The busy period still ends almost surely, but every generation has expected size 1. The nonnegative series for expected total service therefore diverges, giving $\mathbb{E}B = \infty$.

Method. A busy period includes every generation of arrivals during service. Counting only the initial service and its immediate arrivals omits later generations.

Section Exercises

OR25. Correlation of Brownian Integrals

Let $(W_t)_{t \geq 0}$ be standard Brownian motion and let $T > 0$ be deterministic. Define

$$X = \int_0^T W_t \, dt, \qquad Y = \int_0^T t \, dW_t,$$

where the second integral is an Itô integral. Find the correlation of X and Y.

OR26. Where Planar Brownian Motion First Hits an Axis

Standard planar Brownian motion starts at $(1, 1)$, so its position is $(1 + B_t^{(1)}, 1 + B_t^{(2)})$, where the two coordinates are independent standard one-dimensional Brownian motions. Stop when the path first hits the y-axis. Find the probability that the stopping point lies on its positive half-axis.

2.5 Exercises

Key Ideas and Methods. Start by fixing the counting conventions: initial visits, the terminal step, charges and self-loops at an empty state. Then identify the needed prefix, direction, posterior parameter or conditional covariance, rather than selecting a formula from the problem's wording.

Check candidate values against boundaries, normalization and feasible actions. In sums stopped at a random time, determine which continuation events depend

only on past information. Conditional durations and distributions require the appropriate reweighting; unconditional formulas need not remain valid.

RS1. Waiting for a Self-Overlapping Word

Toss a fair coin independently until substring HTHT first appears. Starting with an empty record, use prefix states to find the mean number of tosses. Explain which state survives when H follows HTH.

RS2. A Clearing Queue's Tail

Each queue operation independently adds one item with probability $1/4$, serves one with probability $1/2$ (doing nothing if empty), or clears the queue with probability $1/4$. Find the stationary probability of length at least 2 and the mean positive-time return to an empty queue starting empty.

RS3. Subsampling and Identifiability

A symmetric two-state Markov chain switches with unknown one-step probability $\alpha \in (0, 1)$. Its two-step switch probability is known exactly to be $3/8$. Find all possible values of α. Can observations at even times alone distinguish them?

RS4. Variance of Recursive Work

An A task produces two B tasks with probability $1/2$ and none otherwise. A B task produces one A with probability $1/3$ and none otherwise. Choices are independent. Starting from one A, find the mean, second moment and variance of the total task count, and justify finiteness.

RH1. Restart Counts and Costs

An integer process starts at 1 and stops at 0. Each step resets to 1 with probability $1/2$; otherwise it moves one unit left or right with equal probability. A reset costs 3 and a move costs 1. Find the mean duration, reset count and total cost.

RH2. A Three-Unit Drawdown

A fair integer walk starts at 0 and stops when its drawdown from the running maximum first reaches 3. Find the mean and variance of the time, and the probability that its maximum before stopping is at least 4.

RH3. Information in the Initial Direction

A persistent walk on $\{0, \ldots, 6\}$ stops at either endpoint. Each step keeps the preceding direction with probability $3/4$ and reverses otherwise. Starting from 3, find the probability of reaching 6 first for a rightward and a leftward preceding direction. What if that unobserved initial direction is equally likely to be either?

RH4. Costs until a Repeat

Independent draws have three equally likely types with costs 1, 2 and 4. Stop at the first repeated type, charging for the final draw as well. Find the mean total cost and the mean number of distinct types seen by stopping.

RC1. A Put Payoff without Short Selling

A stock costs 1 and is worth 2 or $1/2$ one period later. Cash earns zero interest and fractional holdings are allowed. The payoff is $(1 - S_1)^+$. Find the replication price and holdings when borrowing and short selling are allowed. If stock holdings must be nonnegative, find the smallest initial wealth covering the payoff on every path.

RC2. Learning with Two Opportunities

A coin's unknown H probability has prior $U[0, 1]$; conditional on it, tosses are independent. H pays $+1$ and T pays -1. Past earnings are kept. You may toss at most twice and may stop at any time. Find the optimal rule and initial value.

RC3. Replacing a Correlated Offer

Receive $X \sim U[0, 1]$ for free. Accept it, or pay $1/8$ to replace it by $1 - X$. Only one replacement is allowed, with no recall. The relationship between the offers is known. Find the optimal rule and expected net payoff.

RC4. Two Testing Objectives

Three independent tests have pass probabilities $(1/2, 3/4, 1/4)$ and costs $(1, 2, 1)$. Find the optimal order and expected cost for deciding whether all pass, and for deciding whether at least one passes. Stop as soon as the required conclusion is known.

RG1. Quadratic Prediction of an Exponential Integral

For standard Brownian motion, let $I = \int_0^1 e^{W_t - t/2}\,\mathrm{d}t$. Find the predictor minimizing mean squared error among $\alpha + \beta W_1 + \gamma(W_1^2 - 1)$, and its error.

RG2. Negative Covariance under Two Constraints

For standard Brownian motion, put $A = \int_0^2 W_t\,\mathrm{d}t$. Condition on $W_2 = 0$ and $A = 1$. Find the conditional law of W_1 and the conditional covariance of $W_{1/2}, W_{3/2}$.

RG3. A Bridge's Maximum

Condition standard Brownian motion on $W_2 = 0$. Find the conditional probability that $\max_{0 \le t \le 2} W_t \ge 1$ and the conditional mean of this maximum.

RG4. A Successful Hit before Failure

Standard Brownian motion starts at 1 and stops at 0 or 2. An independent failure time is exponential with rate $1/2$. Find the probability of reaching 2 before failure and before 0, and the mean duration conditional on success. Compare with the absence of a failure clock.

3

Statistics and Linear Algebra

3.1 Regression Models

Key Ideas and Methods. For regressions with an intercept, center the variables before working with covariances or orthogonal projections. When adding a regressor, residualize both the response and the new column against the existing design. This reduces the added direction to a one-dimensional problem. Check full rank and a nonzero residual direction before dividing by a variance or a squared norm.

Measurement error and selection change the relevant covariance structure. Specify the observed variables and the conditional population before comparing linear prediction with prediction by a conditional mean. An instrument must correlate with the latent signal and be uncorrelated with the relevant errors. Projection identities are algebraic; an exact finite-sample t test additionally requires a correctly specified model with independent, homoscedastic normal errors.

Example A01. Measurement Error and Instruments

Let T, U, V, ε be mutually independent, centered random variables with finite variances, where $s = \mathrm{Var}(T) > 0$ and $u = \mathrm{Var}(U) > 0$. The observed variables are

$$W = T + U, \qquad Z = T + V, \qquad Y = \beta T + \varepsilon.$$

Find the population regression slope of Y on W. Can β be identified from the covariances of (W, Z, Y) alone? If k readings of the same T, with independent identically distributed measurement errors, are averaged, how does the ordinary regression bias change? All regressions include an intercept.

Solution. Independence eliminates the cross-covariances, giving

$$\mathrm{Var}(W) = s + u, \quad \mathrm{Cov}(W, Y) = \beta s, \quad \mathrm{Cov}(Z, W) = s, \quad \mathrm{Cov}(Z, Y) = \beta s.$$

The ordinary slope is $\beta s/(s + u)$, which is attenuated toward zero. Unless $\beta = 0$, it differs from the coefficient of the latent variable. Since $s > 0$, the instrumental-variable ratio is well defined and identifies

$$\boxed{\beta = \frac{\mathrm{Cov}(Z, Y)}{\mathrm{Cov}(Z, W)}}.$$

The instrument must correlate with the latent regressor and be uncorrelated with both the measurement error and the response noise. For readings $W_j = T +$

U_j, suppose the U_j are mutually independent, independent of (T, ε), and have variance u. Then $\bar{W} = T + \bar{U}$ has measurement-error variance u/k, so

$$\beta_k = \frac{\beta s}{s + u/k}, \qquad \beta_k - \beta = -\frac{\beta u}{ks + u} \longrightarrow 0.$$

Averaging reduces measurement noise; it does not create k independent realizations of the latent variable.

Method. Computing a projection coefficient and identifying a structural coefficient are different tasks. List the observable covariances and look for a ratio that eliminates the unknown variance.

Example A02. The Gain from an Extra Regressor

Let the fixed design $X \in \mathbb{R}^{n\times k}$ have full column rank and contain an intercept. Add a column $z \notin \operatorname{col}(X)$, with $n > k + 1$. Let S_0 be the residual sum of squares using X, and $S_1 > 0$ that using (X, z). Write $M = I - X(X^\top X)^{-1}X^\top$ and $r = Mz$. Find the added coefficient and $S_0 - S_1$, and relate the partial coefficient of determination to the usual t statistic. State assumptions for an exact t test.

Solution. For a fixed added coefficient b, minimizing over the old coefficients leaves $\|My - br\|^2$. Since $r \neq 0$, completing the square gives

$$\widehat{b} = \frac{r^\top y}{r^\top r}, \qquad S_0 - S_1 = \frac{(r^\top y)^2}{r^\top r} =: D.$$

The decrease is nonnegative and is zero exactly when $r^\top y = 0$. It measures alignment with the old residual, rather than correlation with the raw response. Put $\nu = n - k - 1$. With $\widehat{\sigma}^2 = S_1/\nu$, the standard error and squared statistic are

$$\operatorname{se}(\widehat{b}) = \sqrt{\frac{S_1}{\nu r^\top r}}, \qquad t^2 = \frac{\nu D}{S_1}.$$

Since $S_0 = S_1 + D$,

$$\boxed{R^2_{\text{partial}} := \frac{D}{S_0} = \frac{t^2}{t^2 + \nu}}.$$

These algebraic identities require no random model. Under the correctly specified model $y = X\alpha + bz + \varepsilon$, conditional normal errors $\varepsilon \sim N(0, \sigma^2 I)$ with $\sigma^2 > 0$ yield $t \sim t_\nu$ under $b = 0$: the normal coefficient component is independent of the orthogonal residual, whose scaled squared norm is χ^2_ν. Without normality, the finite-sample t law does not follow from these identities.

Method. One extra regressor adds one orthogonal direction. The fitting gain, significance statistic, and partial explained fraction all derive from that projection.

Example A03. Prediction after Sign Selection

Let X, Y be independent standard normal variables, and retain only $S = \{XY > 0\}$. In this conditional population, find the best affine predictor of Y from X and the best predictor under squared loss. Compare their mean squared errors.

Solution. For a standard normal variable, sign and magnitude are independent, and $\mathbb{P}(S) = 1/2$. After selection the signs agree, with each common sign equally likely; the magnitudes remain independent half-normal variables. Thus the conditional means are zero, the variances are one, and

$$\mathbb{E}[XY \mid S] = \mathbb{E}|X|\,\mathbb{E}|Y| = \frac{2}{\pi}.$$

The optimal affine predictor is $\widehat{Y}_{\text{lin}} = 2X/\pi$, with zero intercept and risk $1 - 4/\pi^2$. Given $X = x \neq 0$ and S, only the sign of Y is determined. Hence

$$m(x) = \mathbb{E}[Y \mid X = x, S] = \operatorname{sign}(x)\sqrt{2/\pi}.$$

This minimizes squared loss among all measurable predictors: for any g, conditional squared error decomposes into the conditional variance plus $(m(X) - g(X))^2$. The half-normal variance is $1 - 2/\pi$, so the optimal risk is $1 - 2/\pi$, improving on the linear risk by

$$\frac{2}{\pi} - \frac{4}{\pi^2} > 0.$$

The value assigned at $x = 0$ is immaterial because that event has probability zero.

Method. Selection can create correlation and a nonlinear conditional mean. Determine the conditional distribution before comparing linear and unrestricted prediction.

Section Exercises

OA02. Regression on a Residual

Let Y, X_1, X_2 have finite second moments. Assume that both predictors have positive variance and correlation ρ with $|\rho| < 1$. Every regression below is a population least-squares regression with an intercept. Regress Y on X_1, and denote the residual by e. Let β_1 be the slope from regressing e on X_2, and let β_2 be the coefficient of X_2 in the joint regression of Y on (X_1, X_2). Find their relationship.

OA03. Regression after Truncation

Let X and Y be independent $U[0, 1]$ variables. In the conditional population defined by $X + Y > 1$, find the slope in the linear regression of Y on X with an intercept.

3.2 Estimation

Key Ideas and Methods. Specify the comparison class before optimizing: linear unbiased estimation, constrained least squares, and posterior estimation use different criteria. With a positive definite covariance matrix, multiplier equations determine optimal linear weights. Completing a quadratic loss expresses the cost of a constraint as a distance in the appropriate metric. Correlated readings cannot be counted as independent information.

An absolute-value penalty requires sign cases or a subgradient at zero. For collinear regressors, separate their combined fitted contribution from its allocation among coefficients, and check uniqueness separately. Distinguish a hard constraint from a finite penalty. For Gaussian observations, separate signal-bearing directions from noise-only directions before conditioning; joint normality justifies the step from zero covariance to independence.

Example A04. A Mean from Correlated Observations

Observe $Y_i = \mu + \varepsilon_i$, $i = 1, \dots, n$, where $n \geq 2$, $\mathbb{E}\varepsilon_i = 0$, and

$$\operatorname{Cov}(\varepsilon_i, \varepsilon_j) = \sigma^2 \rho^{|i-j|}, \qquad \sigma^2 > 0, \quad 0 \leq \rho < 1.$$

Find the minimum-variance linear unbiased estimator and its variance. Compare it with the arithmetic mean and determine the limit as $\rho \uparrow 1$.

Solution. Write the correlation matrix as R. Direct multiplication verifies $R^{-1} = T/(1-\rho^2)$, where T is tridiagonal: its two endpoint diagonal entries are 1, its interior diagonal entries are $1+\rho^2$, and adjacent off-diagonal entries are $-\rho$. There are no interior entries when $n = 2$. An estimator $w^\top Y$ is unbiased exactly when $\mathbf{1}^\top w = 1$. Lagrange multipliers minimize $w^\top R w$ at $w = R^{-1}\mathbf{1}/(\mathbf{1}^\top R^{-1}\mathbf{1})$. The endpoint row sums are $1/(1+\rho)$; interior row sums are $(1-\rho)/(1+\rho)$. Thus, with $D = 2 + (n-2)(1-\rho)$,

$$\widehat{\mu} = \frac{Y_1 + Y_n + (1-\rho)\sum_{i=2}^{n-1} Y_i}{D}, \qquad \boxed{\operatorname{Var}(\widehat{\mu}) = \frac{\sigma^2(1+\rho)}{D}}.$$

Positive definiteness makes the solution unique. Alternatively, any other unbiased weights are $w + h$, where $\mathbf{1}^\top h = 0$. The cross term vanishes, and the variance increases by $\sigma^2 h^\top R h \geq 0$. For $\rho = 0$ this is the arithmetic mean. For $n > 2$ and $\rho > 0$, endpoints receive more weight because interior observations correlate with neighbors on both sides. For $n = 2$ the weights remain equal. With n fixed and $\rho \uparrow 1$, the endpoint weights approach $1/2$ each and the minimum variance approaches σ^2, rather than zero.

Method. More readings do not necessarily mean more independent information. Optimal linear weights depend on the inverse covariance matrix; normality is unnecessary.

Example A05. Penalizing Identical Regressors

Let $\|x\|^2 = 1$ and $z = x^\top y$, and suppose both regressors equal x. For $\lambda > 0$ and $\eta \geq 0$, find all minimizers of

$$\min_{b_1, b_2} \quad \frac{1}{2}\|y - x(b_1 + b_2)\|^2 + \lambda(|b_1| + |b_2|) + \frac{\eta}{2}(b_1^2 + b_2^2).$$

Explain why an arbitrarily small positive quadratic penalty changes uniqueness of the coefficient allocation.

Solution. Set $s = b_1 + b_2$. For fixed s, the triangle inequality gives $|b_1| + |b_2| \geq |s|$, while $b_1^2 + b_2^2 \geq s^2/2$. Both lower bounds hold with equality at $b_1 = b_2 = s/2$. Ignoring a constant, the remaining problem is

$$\min_s \frac{1}{2}(s - z)^2 + \lambda|s| + \frac{\eta}{4}s^2.$$

Differentiating on each side of zero and checking the subgradient at zero yields

$$s^* = \frac{\operatorname{sign}(z)(|z| - \lambda)_+}{1 + \eta/2}.$$

If $\eta > 0$, equality in the squared-norm bound forces the unique allocation $b_1 = b_2 = s^*/2$. If $\eta = 0$, the sum remains unique. For $s^* > 0$, all solutions are $(t, s^* - t)$ with $0 \leq t \leq s^*$; for $s^* < 0$, they have $s^* \leq t \leq 0$. If $s^* = 0$, the assumption $\lambda > 0$ forces both coefficients to be zero. Thus a nonzero lasso fit admits multiple allocations between identical regressors. Any $\eta > 0$ selects the equal allocation and also shrinks their total coefficient. The boundary $|z| = \lambda$ belongs to the zero-solution case.

Method. First fix the combined fitted contribution of collinear variables, then optimize how that contribution is allocated.

Example A06. Least Squares under a Constraint

Let X have full column rank, $A = X^\top X$, and $\widehat{\beta} = A^{-1}X^\top y$. For a nonzero vector c and a scalar d, minimize residual sum of squares subject to $c^\top \beta = d$, and find its increase over the unconstrained minimum. For $\lambda \geq 0$, also solve the penalized problem with objective $\|y - X\beta\|^2 + \lambda(c^\top \beta - d)^2$ and analyze $\lambda \to \infty$.

Solution. The normal equations eliminate the cross term, so

$$\|y - X\beta\|^2 = \|y - X\widehat{\beta}\|^2 + (\beta - \widehat{\beta})^\top A(\beta - \widehat{\beta}).$$

Write $q = c^\top A^{-1} c > 0$ and $e = c^\top \widehat{\beta} - d$. The multiplier equation puts the correction in direction $A^{-1}c$. Enforcing the constraint gives

$$\boxed{\widehat{\beta}_c = \widehat{\beta} - \frac{e}{q}A^{-1}c}, \qquad \Delta\mathrm{RSS} = \frac{e^2}{q}.$$

Positive definiteness guarantees uniqueness. If $e = 0$, the constraint costs nothing. For the penalized minimizer β_λ, the first-order condition is

$$A(\beta_\lambda - \widehat{\beta}) + \lambda c(c^\top \beta_\lambda - d) = 0.$$

Multiplication by $c^\top A^{-1}$ shows that the constraint error is $e/(1 + \lambda q)$. Therefore

$$\beta_\lambda = \widehat{\beta} - \frac{\lambda e}{1 + \lambda q}A^{-1}c \longrightarrow \widehat{\beta}_c.$$

A finite penalty generally does not enforce the constraint exactly. Unless $e = 0$, the discrepancy vanishes only at order $1/\lambda$.

Method. This projection uses the metric defined by A, not the ordinary Euclidean metric. The initial constraint error and the cost of its correction are measured by e and q.

Example P34. Combining Correlated Sensors

Let $X \sim N(\mu, s^2)$ with $s^2 > 0$. Two sensors report $Z_1 = X+\varepsilon_1$ and $Z_2 = X+\varepsilon_2$. The noise pair is independent of X, jointly normal and centered, with common variance $\tau^2 > 0$ and correlation $-1 < \rho < 1$. Find the posterior law of X, show that the difference of the readings adds no information, and examine both endpoint limits of ρ.

Solution. Set $M = (Z_1 + Z_2)/2$ and $D = Z_1 - Z_2$. Then $M = X + \bar{\varepsilon}$, with

$$v := \mathrm{Var}(\bar{\varepsilon}) = \frac{\tau^2(1+\rho)}{2}, \quad \mathrm{Var}(D) = 2\tau^2(1-\rho).$$

Equal noise variances give $\mathrm{Cov}(\bar{\varepsilon}, D) = 0$, and $\mathrm{Cov}(X, D) = 0$. Joint normality therefore makes D independent of (X, M). Since (M, D) and (Z_1, Z_2) determine each other, the posterior depends only on the average M. Multiplying the prior density by the likelihood of M gives the following part of the exponent involving x:

$$-\frac{1}{2}\left\{\frac{(x-\mu)^2}{s^2} + \frac{(M-x)^2}{v}\right\}.$$

Completing the square yields

$$X \mid Z_1, Z_2 \sim N(m_*, V_*), \qquad V_* = (s^{-2} + v^{-1})^{-1}, \quad m_* = V_*\left(\frac{\mu}{s^2} + \frac{M}{v}\right).$$

As $\rho \uparrow 1$, $v \to \tau^2$: the readings become duplicates and carry the information of one sensor. As $\rho \downarrow -1$, $v \to 0$, so the average noise cancels and posterior variance tends to zero. In the endpoint model, $M = X$ almost surely. For fixed s^2, τ^2, V_* is strictly increasing in ρ.

Method. Decompose multiple measurements into a signal-bearing direction and a pure-noise direction before conditioning. Correlation determines whether the measurements complement or duplicate one another.

Section Exercises

OA05. When Lasso Selects Zero

For $X \in \mathbb{R}^{n\times p}$ and $y \in \mathbb{R}^n$, with $n \geq 1$, consider

$$L(\beta) = \frac{1}{2n}\|y - X\beta\|^2 + \lambda\|\beta\|_1, \qquad \lambda \geq 0.$$

There is no separate unpenalized intercept. Find a necessary and sufficient condition for $\beta = 0$ to be a minimizer.

3.3 Matrices and Quadratic Forms

Key Ideas and Methods. Correlation and Gram matrices are positive semidefinite. Trace, rank, and Rayleigh quotients connect correlations, vector geometry, and eigenvalues. Valid individual entries need not form a compatible matrix. After proving a bound, construct an equality case and allow singular boundary matrices.

With only moment information, optimize a shifted squared-moment bound and check its equality distribution against the event being studied. A strict event may have only a supremum. Conditioning on a sufficient statistic does not increase variance; completeness can then establish optimality within the unbiased class.

For testing simple hypotheses, the likelihood ratio orders outcomes, while the null distribution calibrates size. Normal shrinkage compares total squared-error risk. Check its dimension and covariance assumptions carefully: improvement in total risk does not imply improvement in every coordinate.

Example P07. A Missing Correlation

Let X, Y, Z be standardized random variables with $\mathrm{Corr}(X, Y) = a$ and $\mathrm{Corr}(X, Z) = b$, where $|a|, |b| < 1$. Find all feasible values of $c = \mathrm{Corr}(Y, Z)$ and construct a joint distribution for each. Specialize to $a = 3/5, b = 4/5$ and explain the endpoint degeneracies.

Solution. Put $U = Y - aX$ and $V = Z - bX$. These residuals are uncorrelated with X, and

$$\mathrm{Var}(U) = 1 - a^2, \quad \mathrm{Var}(V) = 1 - b^2, \quad \mathrm{Cov}(U, V) = c - ab.$$

Cauchy–Schwarz gives the necessary condition

$$\boxed{|c - ab| \le \sqrt{(1 - a^2)(1 - b^2)}}.$$

Conversely, set $q = (c - ab)/\sqrt{(1 - a^2)(1 - b^2)} \in [-1, 1]$. For independent standard normals G_0, G_1, G_2, define

$$\begin{aligned} X &= G_0, \\ Y &= aG_0 + \sqrt{1 - a^2}\,G_1, \\ Z &= bG_0 + \sqrt{1 - b^2}\big(qG_1 + \sqrt{1 - q^2}\,G_2\big). \end{aligned}$$

Direct calculation gives unit variances and the required correlations, proving sufficiency. For $|q| < 1$ the transformation is invertible and the correlation matrix is positive definite. For $q = \pm 1$, the two residuals are linearly dependent and the matrix is singular. In the numerical case, both ab and the square-root term equal $12/25$, so $0 \le c \le 24/25$. At the lower endpoint $V = -3U/4$; at the upper endpoint $V = 3U/4$. These are valid correlations: an endpoint introduces an exact linear relation among the variables.

Method. Removing the common variable turns the missing correlation into a residual covariance. The bound and the attaining construction then use the same coordinates.

Example A25. A Sharp One-Sided Tail Bound

Suppose $\mathbb{E}X = 0$ and $\mathrm{Var}(X) = \sigma^2 > 0$. For $t > 0$, maximize $\mathbb{P}(X \geq t)$ over all such distributions and characterize equality. Is the same bound attained for the strict event $X > t$?

Solution. For every $a \geq 0$, $X \geq t$ implies $(X + a)^2 \geq (t + a)^2$. Therefore

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}(X + a)^2}{(t + a)^2} = \frac{\sigma^2 + a^2}{(t + a)^2}.$$

The derivative with respect to a has the sign of $at - \sigma^2$, so the sharpest bound uses $a = \sigma^2/t$:

$$\boxed{\mathbb{P}(X \geq t) \leq \frac{\sigma^2}{\sigma^2 + t^2}}.$$

For equality, the squared term must vanish off the event and equal the threshold on it. Thus X must take only the values t and $-\sigma^2/t$. The mean condition forces $\mathbb{P}(X = t) = \sigma^2/(\sigma^2 + t^2)$, with the remaining probability at the lower point. Its second moment is σ^2, so this is the unique equality distribution. The strict event has the same upper bound. If it attained the bound, the non-strict bound would also be an equality, forcing the distribution just found. But its mass at t does not belong to the strict event, a contradiction. There is consequently no maximum. Using upper point $t + \varepsilon$ and lower point $-\sigma^2/(t + \varepsilon)$ gives strict-tail probabilities approaching the bound, so it is the supremum.

Method. Shift before applying a squared-moment bound to retain one-sided information; optimize the shift as well.

Example A29. Coherence of Unit Vectors

Let $m > d \geq 1$, and let $u_1, \ldots, u_m$ be unit vectors in $\mathbb{R}^d$. Prove

$$\max_{i \neq j} |u_i^\top u_j|^2 \geq \frac{m - d}{d(m - 1)}.$$

Give two necessary and sufficient conditions for equality, and construct an equality example for $d = 2, m = 3$.

Solution. The Gram matrix G is positive semidefinite, has rank at most d, and satisfies $\mathrm{tr}\, G = m$. Take its nonzero eigenvalues and pad them with zeros to obtain d entries. Cauchy–Schwarz yields

$$\mathrm{tr}(G^2) \geq \frac{m^2}{d}.$$

If $M = \max_{i \neq j} |u_i^\top u_j|^2$, then

$$\operatorname{tr}(G^2) = m + \sum_{i \neq j} |u_i^\top u_j|^2 \leq m + m(m-1)M.$$

Combining the inequalities proves the bound. Equality holds exactly when G has d nonzero eigenvalues, all equal to m/d, and all off-diagonal inner products have the same absolute value. The first condition is equivalent to

$$\sum_{i=1}^{m} u_i u_i^\top = \frac{m}{d} I_d.$$

In the plane, take unit vectors with angles $0, \pi/3, 2\pi/3$. Their pairwise inner products have absolute value $1/2$, whose square is $1/4 = (3-2)/(2 \cdot 2)$, attaining the bound.

Method. The spectrum controls the sum of squared inner products, and averaging forces at least one large pair. Absolute size, rather than sign, is the relevant quantity.

Example A30. Eigenvalues after a Rank-One Update

A real symmetric matrix A has eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$. Let $B = A + tuu^\top$, where $t \geq 0$ and $u \in \mathbb{R}^n$, and write its eigenvalues as $\mu_1 \leq \cdots \leq \mu_n$. Prove, for $i < n$,

$$\lambda_i \leq \mu_i \leq \lambda_{i+1}.$$

Also show that $\mu_n \geq \lambda_n$, that each eigenvalue increase is at most $t\|u\|^2$, and that their sum is exactly $t\|u\|^2$.

Solution. Every Rayleigh quotient increases by $t(u^\top x)^2/\|x\|^2 \geq 0$, so the minimax principle gives $\mu_i \geq \lambda_i$. The increase in any quotient is at most $t\|u\|^2$; applying the same principle gives $\mu_i \leq \lambda_i + t\|u\|^2$. For the sharper interlacing upper bound, let E span the first $i+1$ eigenvectors of A. If $u \neq 0$, the space $E \cap u^\perp$ has dimension at least i. The update vanishes there, and the Rayleigh quotient of A is at most λ_{i+1}. Insert an i-dimensional subspace of this intersection into the minimax formula for B to obtain $\mu_i \leq \lambda_{i+1}$. If $u = 0$ or $t = 0$, the assertions are immediate. Finally, taking traces gives

$$\sum_{i=1}^{n} (\mu_i - \lambda_i) = \operatorname{tr}(tuu^\top) = t\|u\|^2.$$

Thus the total spectral displacement is fixed, and no ordered eigenvalue decreases. When $n = 1$, the interlacing assertion is empty but the other statements still hold.

Method. A rank-one perturbation vanishes on a subspace of codimension one. This yields finer location control than a norm bound alone.

Example CT01. Unbiased Estimation of a Squared Probability

Let $X_1, \dots, X_n$ be independent Bernoulli(p) variables, where $n \geq 2$ and $0 < p < 1$. Starting with the unbiased estimator X_1X_2, construct an unbiased estimator of p^2 with no larger variance. Find its exact variance and prove minimum variance among all unbiased estimators.

Solution. Put $S = \sum_i X_i$. Given $S = s$, the success locations are uniform over subsets of size s, so

$$U := \mathbb{E}[X_1X_2 \mid S] = \frac{S(S-1)}{n(n-1)}.$$

This conditional law does not involve p, making S sufficient. Total variance gives $\mathrm{Var}(X_1X_2) = \mathrm{Var}(U) + \mathbb{E}\,\mathrm{Var}(X_1X_2 \mid S)$, so conditioning preserves unbiasedness and cannot increase variance. Also, U is the average of all $\binom{n}{2}$ products X_iX_j. Each has variance $p^2(1-p^2)$; two products sharing one index have covariance $p^3(1-p)$; disjoint products are independent. There are $n\binom{n-1}{2}$ unordered pairs sharing an index. Hence

$$\mathrm{Var}(U) = \frac{2p^2(1-p^2) + 4(n-2)p^3(1-p)}{n(n-1)}.$$

To prove minimum variance, first establish completeness. If $\mathbb{E}_p h(S) = 0$ for every $p \in (0,1)$, divide by $(1-p)^n$. The resulting polynomial in $p/(1-p)$ vanishes identically, so every coefficient $\binom{n}{s}h(s)$ is zero. For any other unbiased estimator W, its conditional mean given S is a parameter-free function with the same expectation as U. Completeness forces equality. Total variance then gives $\mathrm{Var}(W) \geq \mathrm{Var}(U)$; an estimator without finite variance cannot improve the bound either. For $n = 2$, $U = X_1X_2$ and the variance formula still applies.

Method. Conditioning removes parameter-free randomness. Completeness rules out any different unbiased function of the sufficient statistic that could improve the result.

Example CT02. A Most Powerful Exponential Test

Let $X_1, \dots, X_n$ be independent exponential variables with rate θ. Test $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1 > \theta_0$. Find the most powerful test among those with type I error at most $\alpha \in (0,1)$, and calculate its power. Give the explicit cutoff and power for $n = 2, \theta_0 = 1, \theta_1 = 2, \alpha = 1 - 2/e$.

Solution. Set $T = \sum_i X_i$. The joint likelihood ratio of alternative to null is

$$L = \left(\frac{\theta_1}{\theta_0}\right)^n e^{-(\theta_1 - \theta_0)T}.$$

It decreases strictly in T, suggesting $\phi_* = \mathbf{1}_{\{T<c\}}$, where $c > 0$ uniquely satisfies $\mathbb{P}_{\theta_0}(T < c) = \alpha$. Continuity makes boundary randomization unnecessary. Let k be the corresponding likelihood-ratio threshold. Every competing test $0 \leq \phi \leq 1$ satisfies $(\phi_* - \phi)(L - k) \geq 0$. If its size is at most α, then

$$\mathbb{E}_{\theta_1}(\phi_* - \phi) \geq k\mathbb{E}_{\theta_0}(\phi_* - \phi) \geq 0.$$

Thus the candidate maximizes power, proving the likelihood-ratio principle in this setting. The sum has density $\theta^n t^{n-1} e^{-\theta t}/(n-1)!$, obtainable by induction through convolution. Repeated integration by parts gives

$$\mathbb{P}_\theta(T < c) = 1 - e^{-\theta c} \sum_{j=0}^{n-1} \frac{(\theta c)^j}{j!}.$$

Use θ_0 to solve for the cutoff and θ_1 to obtain power. In the specified case, $c = 1$ and power is $1 - 3e^{-2}$. A larger rate makes total waiting time shorter, so rejection lies in the left tail.

Method. The likelihood ratio determines which tail to use; the null distribution calibrates the size. The rejection direction should not be guessed from the parameter's name.

Example CT03. Risk Reduction by Normal Shrinkage

Observe $X \sim N_d(\theta, I_d)$, where $d \geq 3$ and $\theta \in \mathbb{R}^d$ is unknown. Within the family

$$\delta_a(X) = \left(1 - \frac{a}{\|X\|^2}\right) X,$$

find the constant a minimizing squared-error risk and show that it improves on X for every θ. Define the estimator arbitrarily at $X = 0$, a null event. How does the risk change if the same estimator predicts an independent $Y \sim N_d(\theta, I_d)$?

Solution. Write $\delta_a = X + g(X)$, where $g(x) = -ax/\|x\|^2$. Integration by parts against the normal density gives

$$\mathbb{E}[(X_i - \theta_i) g_i(X)] = \mathbb{E}[\partial_i g_i(X)].$$

The singularity at zero is harmless here. First integrate over $\|x\| > \varepsilon$: the spherical boundary term has order $\varepsilon^{d-2} \to 0$. The derivative has local order $\|x\|^{-2}$, integrable when $d > 2$; Gaussian decay removes the boundary at infinity. Direct differentiation gives

$$\|g(x)\|^2 = \frac{a^2}{\|x\|^2}, \qquad \sum_i \partial_i g_i(x) = -\frac{a(d-2)}{\|x\|^2}.$$

Expanding risk and using the identity yields

$$\begin{aligned} R_\theta(a) &= \mathbb{E}_\theta \|\delta_a(X) - \theta\|^2 \\ &= d + \big(a^2 - 2a(d-2)\big) \mathbb{E}_\theta \frac{1}{\|X\|^2}. \end{aligned}$$

The final expectation is finite and strictly positive, so the same constant $a = d - 2$ minimizes risk for every θ. Its risk is

$$d - (d-2)^2 \mathbb{E}_\theta \frac{1}{\|X\|^2} < d,$$

whereas X has risk d. This compares total squared error across coordinates, not the risk of each coordinate separately. Independence and $\mathbb{E}(Y - \theta) = 0$ eliminate

the predictive cross term, giving $\mathbb{E}\|Y - \delta_a(X)\|^2 = d + R_\theta(a)$. The optimal shrinkage constant is unchanged, and its prediction risk is strictly below the direct predictor's risk $2d$.

Method. A suitable bias can reduce total squared risk. Gaussian integration by parts converts the difficult error cross term into the divergence of the shrinkage adjustment.

Section Exercises

OA30. Eigenvalues of a Principal Submatrix

Let A be a real symmetric $n \times n$ matrix, with $n \geq 2$, and let B be obtained by deleting row k and column k. Write the eigenvalues in increasing order as

$$\lambda_1 \leq \cdots \leq \lambda_n, \qquad \mu_1 \leq \cdots \leq \mu_{n-1}.$$

Prove the interlacing relation between these eigenvalues.

3.4 Exercises

Key Ideas and Methods. Translate each model into equations and inequalities that can be checked together. For observed moments, test whether one latent model satisfies all covariance restrictions. For probability bounds, separate the upper-bound proof from an attaining construction; mass at the boundary matters for strict events.

For constrained estimation, state feasibility, multiplier signs, complementary slackness, and uniqueness. A shared budget couples the coordinates, so separate thresholding is insufficient. Check the candidate independently by substitution, elimination, or its increase in the objective, and state the parameter range in which the argument applies.

AG1-1. Compatibility of Observed Moments

Use the independent centered model in A01. The observed moments are

$$\operatorname{Var}(W) = 5, \quad \operatorname{Var}(Z) = 4, \quad \operatorname{Cov}(W, Z) = 3,$$
$$\operatorname{Cov}(W, Y) = \operatorname{Cov}(Z, Y) = 6.$$

Identify β and the variances of T, U, V, and find the regression slope of Y on W. If only $\operatorname{Cov}(Z, Y)$ changes to 5, can the same model still hold?

AG1-2. A Strict Tail Event

Over all distributions with $\mathbb{E}X = 0$ and $\operatorname{Var}(X) = 1$, maximize $\mathbb{P}(X \geq 2)$ and find the supremum of $\mathbb{P}(X > 2)$. Give the first equality distribution and a sequence approaching the second supremum.

AG1-3. Sparse Estimation with a Budget

A centered design satisfies $X^\top X/n = I_3$ and $X^\top y/n = (3, 1, -1)^\top$. Solve

$$\min_{b_j \geq 0,\ \sum_j b_j \leq 2} \frac{\|y - Xb\|^2}{2n} + \frac{1}{2} \sum_{j=1}^{3} b_j.$$

Find the budget multiplier and explain why independent coordinate thresholding alone is insufficient.

AG1-4. The Cost of a Coefficient Constraint

A full-rank least-squares problem has $X^\top X = \operatorname{diag}(1, 4)$ and unconstrained solution $(2, -1)^\top$. Impose that the two coefficients sum to zero. Find the constrained solution and the increase in residual sum of squares, and check by direct one-variable minimization.

4

Mathematical Reasoning

4.1 Mathematical Reasoning

Key Ideas and Methods. For discrete structures, look for congruences, parity, conserved totals, or incidence counts. Separate necessary conditions from an explicit construction. Double counting links local overlaps to global size; factorial divisibility often reduces to prime valuations or the possible locations of prime factors.

In geometric optimization, choose coordinates, write the objective, and check boundaries and small cases. On a convex feasible set, strict convexity can establish uniqueness. In geometric probability, retain the transformed density and integration region. Equidistribution alone does not make events at a fixed lag independent.

For analysis problems, state the domain and regularity assumptions. Reciprocals or subtraction of a polynomial may simplify an equation. Carefully chosen difference-quotient sequences disprove differentiability, while a finite prefix and a uniform tail bound handle dense jumps. Probability integral transforms require special treatment of atoms.

Example A14. Queens on a Torus

Identify opposite edges of an $n \times n$ chessboard, indexing rows and columns modulo n. Place n queens with one in each row and column, and require both diagonal labels $i + j$ and $i - j$ of their occupied cells to be distinct. Determine all positive integers n for which this is possible and give a construction.

Solution. Let the chosen columns form a permutation $\pi(i)$, $i = 0, \ldots, n-1$. If $i + \pi(i)$ also forms a permutation, summing modulo n gives

$$\frac{n(n-1)}{2} \equiv n(n-1) \equiv 0 \pmod{n},$$

so n must be odd. Put $S = \sum_{i=0}^{n-1} i^2$. Since both diagonal label lists are permutations,

$$2S \equiv \sum_i \big((i + \pi(i))^2 + (i - \pi(i))^2\big) = 4S \pmod{n}.$$

Consequently $n \mid 2S = n(n-1)(2n-1)/3$. If $3 \mid n$, the product $(n-1)(2n-1)$ is not divisible by 3, contradicting this divisibility. Hence $\gcd(n, 6) = 1$ is necessary. Conversely, under that condition take $\pi(i) = 2i \pmod{n}$. The column and

two diagonal labels are $2i, 3i, -i$, respectively. All three multipliers are invertible modulo n, so all three lists are permutations. This also handles $n = 1$.

Method. First moments rule out even sizes; second moments rule out multiples of three. The two diagonal constraints require more than one invariant.

Example A17. A Seven-Person Sharing Design

Seven people each hold three types of item. Each pair of people shares exactly one type, and each type is held by at least one and at most three people. Prove that there are exactly seven types and that each is held by three people. Construct an arrangement satisfying every condition.

Solution. Let n_j count types held by exactly j people, for $j = 1, 2, 3$. Counting person–item incidences gives

$$n_1 + 2n_2 + 3n_3 = 21.$$

Counting the $\binom{7}{2} = 21$ pairwise sharing relations by item gives $n_2 + 3n_3 = 21$. Subtraction yields $n_1 + n_2 = 0$, hence $n_1 = n_2 = 0$ and $n_3 = 7$. Label the people $0, \ldots, 6$. For each i, introduce an item shared by

$$\{i, i + 1, i + 3\} \pmod 7.$$

Each person occurs in three translated sets and therefore holds three types. The six directed differences within $\{0, 1, 3\}$ are exactly the six nonzero residues modulo 7. Thus each ordered pair of distinct people occurs in exactly one translate, proving the pairwise sharing condition.

Method. Count both person–item incidences and pairwise sharing incidences. Together they can force the size of every item group; uniqueness of differences verifies the construction.

Example A19. Recovering Edge Bits

Each edge of a finite undirected simple graph carries a bit. The parity of all incident edge bits is known at every vertex, as are some edge values. Let H contain the unknown edges and all v vertices; let it have q edges and c connected components, including isolated vertices. Give necessary and sufficient consistency conditions, count the solutions, characterize unique recovery, and construct all solutions.

Solution. XOR the known incident bits out of each vertex check, leaving residual requirements b_x. Within any component of H, each unknown edge is counted twice. Thus the XOR of the b_x in each component must be zero. For sufficiency, choose a spanning tree in each nontrivial component. Assign arbitrary values to all non-tree edges and remove their contributions from the vertex requirements. On a tree, process leaves: the unique remaining edge at a leaf must equal its residual check. Fix it, delete the leaf, and update its neighbor. The final root check holds automatically by the zero component parity. At an isolated vertex, the condition is simply $b_x = 0$. The spanning forest has $v - c$ edges. There are therefore $q - v + c$

freely assignable non-tree bits, with all tree bits uniquely determined afterward. The number of solutions is

$$\boxed{2^{q-v+c}}.$$

For consistent data, recovery is unique exactly when $q = v - c$, that is, when the unknown-edge graph is a forest. If it contains a cycle, flipping every bit on that cycle preserves all checks and directly proves nonuniqueness.

Method. Uniqueness depends on the unknown subgraph, not on whether the original graph has cycles. Observed edges may have already broken every remaining cycle.

Example A20. Balancing Three Accounts

Three accounts start with integer balances a_1, a_2, a_3. Each step subtracts 2 from one account and adds 1 to each of the others. Negative intermediate balances are allowed. When can the balances be made equal? Find the minimum number of steps and construct a sequence attaining it.

Solution. The total is invariant, so the common final value must be $t = (a_1 + a_2 + a_3)/3$. Each step changes any pairwise difference by 0 or ± 3. Consequently the three initial balances must be congruent modulo 3; this also makes t an integer. Suppose there are N steps and account i is chosen for subtraction n_i times. Then $\sum_i n_i = N$, and its final balance is

$$a_i + N - 3n_i = t.$$

Thus $n_i = (a_i + N - t)/3 \geq 0$, implying $N \geq t - \min_i a_i$. Let $m = \min_i a_i$ and choose

$$N = t - m, \qquad n_i = \frac{a_i - m}{3}.$$

Under the congruence condition these are nonnegative integers summing to N. Choose account i for subtraction exactly n_i times, in any order, to reach the common balance t. This proves sufficiency and optimality. Initially equal balances require zero steps, as the formula states.

Method. Invariants reveal reachability obstacles; counting each operation type then supplies both a construction and a shortest-length proof. Allowing negative intermediate balances makes arbitrary ordering valid.

Example A22. Powers of Two in a Binomial Coefficient

For a positive integer n, find the exact exponent of 2 dividing $\binom{2n}{n}$. Determine when it is even but not divisible by 4, and interpret the result in binary notation.

Solution. Each multiple of 2 contributes one factor of 2 to a factorial, each multiple of 4 contributes another, and so on. Hence

$$v_2(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor.$$

Write $n = \sum_{k\ge 0} b_k 2^k$, with $b_k \in \{0,1\}$. Interchanging finite sums gives

$$v_2(n!) = \sum_k b_k(2^k - 1) = n - s_2(n), \qquad s_2(n) = \sum_k b_k.$$

Consequently,

$$v_2\binom{2n}{n} = 2n - s_2(2n) - 2(n - s_2(n)) = s_2(n),$$

since multiplication by two appends a zero and leaves the binary digit sum unchanged. A positive integer has at least one nonzero binary digit, so the central binomial coefficient is always even. It is not divisible by 4 exactly when $s_2(n) = 1$, equivalently when n is a nonnegative integral power of 2. For example, $n = 13 = (1101)_2$ gives exact exponent 3, without expanding the large binomial coefficient.

Method. Factorial divisibility can reduce to digit counting. An exact valuation provides more information than parity alone.

Example A15. Minimum Travel Time across an Interface

The line $x = 0$ divides the plane into regions with speeds $v_1, v_2 > 0$ on the left and right. Travel from $A = (-a, 0)$ to $B = (b, h)$, where $a, b, h > 0$, using two straight segments that cross the interface once at $C = (0, y)$. Derive the equation for the optimal crossing, prove uniqueness, and find the equal-speed solution and the limit as $v_2 \to \infty$.

Solution. Total time is

$$T(y) = \frac{\sqrt{a^2 + y^2}}{v_1} + \frac{\sqrt{b^2 + (h-y)^2}}{v_2}, \qquad y \in \mathbb{R}.$$

It tends to infinity as $|y| \to \infty$, and

$$T''(y) = \frac{a^2}{v_1(a^2 + y^2)^{3/2}} + \frac{b^2}{v_2(b^2 + (h-y)^2)^{3/2}} > 0.$$

Thus the minimizer is unique. Since $T'(0) < 0$ and $T'(h) > 0$, it lies in $(0, h)$ and solves

$$\boxed{\frac{y_*}{v_1\sqrt{a^2 + y_*^2}} = \frac{h - y_*}{v_2\sqrt{b^2 + (h-y_*)^2}}}.$$

If θ_1, θ_2 are the angles of the two segments to the interface normal, this is $\sin\theta_1/v_1 = \sin\theta_2/v_2$. When $v_1 = v_2$, both sides are positive. Squaring and simplifying gives $by_* = a(h - y_*)$, hence $y_* = ah/(a+b)$: the path is the straight segment AB. With the other parameters fixed and $v_2 \to \infty$, the right-hand side is at most $1/v_2 \to 0$. Since $0 < y_* < h$, this forces $y_* \to 0$, minimizing distance spent in the slower region.

Method. Strict convexity proves that the stationary point is the unique global optimum before any equation is solved. A complicated radical expression for a general speed ratio is unnecessary.

Example A16. Choosing a Cut in a Folded Rope

A rope of length $L > 0$ is folded in half $n \geq 1$ times, each time folding the current right half onto the left. The final segment has length $h = L/2^n$. Cut through every layer at distance th from its left end, where $0 < t < 1$. Choose t to minimize the longest unfolded piece. Treat $n = 1$ and $n \geq 2$ separately, and decide whether all pieces can have equal length.

Solution. Put $N = 2^n$. The N original intervals of length h cover the folded segment in alternating orientations. The unfolded cut locations are

$$th,\ (2-t)h,\ (2+t)h,\ (4-t)h, \ldots, (N-t)h.$$

Both endpoint pieces have length th, and interior gaps alternate between $2(1-t)h$ and $2th$. For $n = 1$, there is only one interior gap. The longest piece has length $h \max\{t, 2(1-t)\}$, uniquely minimized by equating the two terms: $t = 2/3$. All three pieces then have length $2h/3 = L/3$. For $n \geq 2$, both interior gap types occur. The longest length is therefore

$$2h \max\{t, 1-t\} \geq h.$$

The unique optimum is $t = 1/2$, with longest length $h = L/2^n$. The two endpoint pieces have length $h/2$, while the other $N - 1$ pieces have length h. For every $t > 0$, an endpoint length th differs from an interior length $2th$, so equal lengths for all pieces are impossible.

Method. List every type of unfolded gap, not just the number of cuts. At small fold counts, one interior gap type is absent and changes the optimum.

Example A18. Manhattan Distance between Random Points

Choose two independent uniform points (X_1, Y_1) and (X_2, Y_2) in the unit square. Let $D = |X_1 - X_2| + |Y_1 - Y_2|$. Find $\mathbb{P}(D \leq t)$ for $0 \leq t \leq 1$, as well as $\mathbb{P}(D > 1)$ and $\mathbb{E}D$.

Solution. For $U = |X_1 - X_2|$, the two corner triangles of the unit square give $\mathbb{P}(U > u) = (1-u)^2$. Thus $f_U(u) = 2(1-u)$ for $0 < u < 1$. The variable $V = |Y_1 - Y_2|$ is independent of U and has the same law. For $0 \leq t \leq 1$, the integration region is $u, v \geq 0, u + v \leq t$, so

$$\begin{aligned}\mathbb{P}(D \leq t) &= 4 \int_0^t (1-u) \int_0^{t-u} (1-v)\ \mathrm{d}v\ \mathrm{d}u \\ &= 2t^2 - \frac{4}{3}t^3 + \frac{1}{6}t^4.\end{aligned}$$

At $t = 1$ this gives $\mathbb{P}(D > 1) = 1 - 5/6 = 1/6$. Also,

$$\mathbb{E}U = 2 \int_0^1 u(1-u)\ \mathrm{d}u = \frac{1}{3}, \qquad \mathbb{E}D = \frac{2}{3}.$$

The distribution is continuous, so the boundary $D = 1$ has zero probability. For $t > 1$, the region is truncated by $u, v \leq 1$ and the displayed polynomial no longer applies.

Method. Find each coordinate difference first and then convolve. The expected distance follows directly from linearity, without requiring the full distribution.

Example A21. Leading Digits Three Terms Apart

Choose n uniformly from $1, \ldots, N$. As $N \to \infty$, find the probability limit that both 2^n and 2^{n+3} have leading decimal digit 1. Also find the limiting conditional probability for the second event given the first. Are the events asymptotically independent?

Solution. Put $\alpha = \log_{10} 2$. If it were rational, an equality $2^q = 10^p$ would contradict unique prime factorization, so α is irrational. The sequence $\{n\alpha\}$ is equidistributed in $[0, 1)$: for each nonzero integer h, the corresponding exponential sum is a bounded geometric sum, whose average tends to zero. Weyl's criterion applies. Leading digit 1 means that the fractional part lies in $I = [0, \alpha)$. A gap of three terms is a circle rotation by $3\alpha = \log_{10} 8$. For $x \in I$, the point $x + 3\alpha$ cannot return to I without crossing 1. After crossing, the required overlap is

$$1 - 3\alpha \le x < \alpha.$$

The additional upper condition $x < 1 - 2\alpha$ is automatic because $3\alpha < 1$. The joint limit is therefore

$$4\alpha - 1 = \log_{10}(8/5),$$

and the conditional limit is $(4\alpha - 1)/\alpha$, approximately 0.6781. The marginal limit is $\alpha \approx 0.3010$. The joint limit, approximately 0.2041, differs from $\alpha^2 \approx 0.0906$, so there is no asymptotic independence.

Method. Equidistribution makes a single phase uniform; it does not make phases at a fixed lag independent. Joint frequencies require intersecting rotated intervals.

Example A23. A Differential Blow-Up Bound

Let $a > 0$ and let $f : [0, T) \to (0, \infty)$ be differentiable, with a right derivative at zero, satisfying $f(0) = a$ and $f'(x) \ge f(x)^2$. Prove that $T \le 1/a$ and exhibit a function on the longest possible interval. Also obtain a lower bound for $f(x)$ when $x < \min\{T, 1/a\}$.

Solution. Set $g = 1/f$. Positivity makes this well defined, and

$$g'(x) = -\frac{f'(x)}{f(x)^2} \le -1.$$

The mean value theorem gives $g(x) - g(0) \le -x$, hence

$$0 < \frac{1}{f(x)} \le \frac{1}{a} - x.$$

If $T > 1/a$, choosing $x = 1/a$ is a contradiction. Thus $T \le 1/a$. Where the right-hand side is positive, taking reciprocals gives

$$f(x) \ge \frac{a}{1 - ax}.$$

The function $f(x) = a/(1 - ax)$ is positive on $[0, 1/a)$ and satisfies $f' = f^2$, so the interval-length bound is sharp. It diverges at the endpoint and admits no differentiable extension there with a finite positive value. A general function satisfying the inequality may blow up earlier, or simply be specified on a shorter interval. The assumptions do not force explosion exactly at $1/a$ or force the function to have the displayed equality form.

Method. Taking a reciprocal turns a quadratic growth lower bound into a linear decrease and directly controls the possible existence time.

Example A24. A Quadratic Perturbation of Additivity

Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f(x + y) = f(x) + f(y) + 2xy \quad (x, y \in \mathbb{R}), \qquad f(1) = 0,$$

and bounded on $[0, 1]$. Explain where boundedness is needed.

Solution. Put $g(x) = f(x) - x^2$. Then $g(x + y) = g(x) + g(y)$ and $g(1) = -1$. Additivity gives $g(0) = 0$, $g(-x) = -g(x)$, and the integer-multiple identities imply $g(q) = -q$ for every rational q. Boundedness of f on $[0, 1]$ implies the same for g. Oddness extends this to $|g(x)| \le M$ for $|x| \le 1$, with some $M > 0$. Given $\varepsilon > 0$, choose an integer $k > M/\varepsilon$. If $|h| < 1/k$, then

$$|g(h)| = \frac{|g(kh)|}{k} \le \frac{M}{k} < \varepsilon.$$

Thus g is continuous at zero, and additivity makes it continuous everywhere. For rationals $q_j \to x$, we obtain $g(x) = \lim_j g(q_j) = -x$. The unique answer is

$$\boxed{f(x) = x^2 - x}.$$

Direct substitution checks the equation, normalization, and boundedness. Boundedness extends a rational identity to the reals; without any regularity condition, additive functions can be discontinuous, so a calculation on rational arguments is insufficient.

Method. Subtract the quadratic term that produces the cross term, then solve the remaining additive equation. Local regularity is needed only for the passage from rational to real arguments.

Example A26. Local Regularity of an Oscillation

For $\alpha, \beta > 0$, define $f(0) = 0$ and, for $x \neq 0$,

$$f(x) = |x|^{\alpha} \sin(|x|^{-\beta}).$$

Determine continuity and differentiability at zero, and determine when the derivative is continuous there. Prove necessity using explicit sequences.

Solution. Since $|f(x)| \le |x|^{\alpha}$, continuity at zero holds for all the stated parameters. The absolute difference quotient is at most $|x|^{\alpha-1}$, so $\alpha > 1$ gives $f'(0) = 0$. Conversely, choose positive sequences satisfying $x_k^{-\beta} = 2\pi k + \pi/2$

and $y_k^{-\beta} = 2\pi k + 3\pi/2$. Their difference quotients are $x_k^{\alpha-1}$ and $-y_k^{\alpha-1}$. They have distinct limits for $\alpha = 1$ and diverge with opposite signs for $\alpha < 1$. Thus differentiability holds exactly when $\alpha > 1$. For $x > 0$,

$$f'(x) = \alpha x^{\alpha-1}\sin(x^{-\beta}) - \beta x^{\alpha-\beta-1}\cos(x^{-\beta}).$$

On the negative half-line use the oddness of the derivative of this even function. If $\alpha > \beta+1$, both terms tend to zero, so the derivative is continuous. If $1 < \alpha \le \beta+1$, take $x_k^{-\beta} = 2\pi k$. Then $f'(x_k) = -\beta x_k^{\alpha-\beta-1}$, which is constantly $-\beta$ in the equality case and diverges in the strict case. Neither converges to $f'(0) = 0$. Therefore the derivative is continuous at zero exactly when $\alpha > \beta + 1$.

Method. Differentiability is controlled by amplitude divided by distance. Continuity of the derivative also depends on oscillation frequency, producing a different threshold.

Example A27. One-Sided Derivatives and Convexity

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous, with a finite right derivative $g(x) = f'_+(x)$ at every point. Suppose g is nondecreasing. Prove that f is convex, that finite left derivatives exist everywhere, and that the set where f is not differentiable is at most countable. Do not assume continuity of g.

Solution. First prove a comparison lemma: if a continuous h has nonnegative right derivative on $[a, b)$, then $h(b) \ge h(a)$. Otherwise choose a sufficiently small $\varepsilon > 0$ so that $q(x) = h(x) + \varepsilon x$ still satisfies $q(b) < q(a)$. Its maximum on $[a, b]$ occurs at some $t < b$, where its right difference quotients are nonpositive. This contradicts $q'_+(t) > 0$. Apply the lemma to $f(x) - g(a)x$ and $g(b)x - f(x)$ to obtain

$$g(a) \le \frac{f(b) - f(a)}{b - a} \le g(b), \qquad a < b.$$

For $a < b < c$, the left secant slope is therefore no greater than the right one. Rearranging this inequality gives convexity. Fix x. Monotonicity gives a finite left limit $g(x-)$, bounded by values at nearby points. For $a < x$, the comparison lemma also permits upper bound $g(x-)$ because $g(t) \le g(x-)$ for $t < x$. Hence

$$g(a) \le \frac{f(x) - f(a)}{x - a} \le g(x-).$$

Letting $a \uparrow x$ yields $f'_-(x) = g(x-)$. Nondifferentiability thus occurs exactly when $g(x-) < g(x)$. Select a rational number in each nonempty interval $(g(x-), g(x))$. These intervals are disjoint for distinct points, so this choice injects the exceptional set into the rationals. It is at most countable.

Method. A monotone one-sided derivative controls secant slopes and yields convexity. Corners correspond to positive jumps of the derivative; continuity of that derivative is unnecessary.

Example A28. Randomizing a Discrete Distribution

Enumerate the rationals without repetition as $r_1, r_2, \ldots$. Let $\mathbb{P}(X = r_j) = w_j > 0$, with $\sum_j w_j = 1$, and let F be the distribution function. Independently take $V \sim \mathrm{Unif}(0,1)$ and define

$$U = F(X-) + V\big(F(X) - F(X-)\big).$$

Prove that U is uniform on $(0,1)$ and that the generalized inverse $Q(u) = \inf\{x : F(x) \geq u\}$ recovers $Q(U) = X$ almost surely. Why does $F(X)$ alone fail to be uniform?

Solution. Conditional on $X = r_j$, the variable U is uniform on

$$I_j = (F(r_j-), F(r_j)),$$

whose length is w_j. These intervals are disjoint and their lengths sum to one, so their union covers $(0,1)$ except for a set of Lebesgue measure zero. For every Borel set $B \subset (0,1)$,

$$\mathbb{P}(U \in B) = \sum_j w_j \frac{|B \cap I_j|}{w_j} = |B|,$$

where $|\cdot|$ denotes Lebesgue length. Thus U is uniform. If $u \in I_j$, then every $x < r_j$ satisfies $F(x) \leq F(r_j-) < u$, whereas $F(r_j) > u$. Hence $Q(u) = r_j$, and the conditional probability-one interval events imply $Q(U) = X$ almost surely. In contrast, $F(X)$ takes values in the countable set $\{F(r_j) : j \geq 1\}$ and cannot have a continuous uniform law. The auxiliary variable spreads each atomic probability uniformly across its jump interval.

Method. The probability integral transform needs randomization at atoms. The inverse quantile transform can then recover the original atom from the same uniform variable.

Section Exercises

OA17. Four Postcard Collections

Four people each have 10 distinct types of postcard. Any two collections have at most 6 types in common. What is the smallest possible number of distinct types in their combined collections? Prove a lower bound and construct collections attaining it.

OA22. A Factorial Divisibility Inequality

Let x, y be positive integers such that

$$\frac{x!\,y!}{x! + y!}$$

is an integer. Prove that $3x - 2y \geq 2$.

OA28. A Function with Dense Jumps

Enumerate all rational numbers exactly once as $r_0, r_1, r_2, \ldots$, and define

$$f(x) = \sum_{\{n \geq 0 : r_n < x\}} 2^{-n}, \qquad x \in \mathbb{R}.$$

Determine every continuity and discontinuity point of f, and find the one-sided limits at each rational point.

4.2 Games

Key Ideas and Methods. Before solving a game, specify timing, information, payments, and tie rules. Evaluate the actual payoff after a unilateral deviation. Private information and selective acceptance change conditional distributions, so identify the relevant population before optimizing a bid. Compute individual equilibrium and minimum total cost separately.

In zero-sum games, matching guarantees for both players establish optimality. Equal payoffs within a proposed support only produce candidates; actions outside it must also be checked. For finite removal games, classify positions by strong induction or construct responses whose legality is preserved. A short table does not prove periodicity.

Constructions must also obey implementation constraints. Matchings require neighbor-set conditions, simultaneous congruences require compatibility, and weighing codes must distinguish states while keeping the two pans balanced in every weighing.

Example P01. The Value of a Quality Test

A seller knows an item's value $X \sim \mathrm{Unif}[0, 1]$; the buyer values it at $3X/2$. The buyer offers a single price $p \geq 0$, accepted exactly when $X \leq p$. Before bidding, the buyer may pay $c \geq 0$ for a test: he chooses a threshold $t \in (0, 1)$ in advance and learns only whether $X < t$ or $X \geq t$, then makes a contingent offer. Find the best threshold and offers, and decide when to purchase the test.

Solution. Without testing, an offer $0 \leq p \leq 1$ gives expected trading profit

$$\int_0^p (3x/2 - p)\, \mathrm{d}x = -p^2/4 \leq 0.$$

For $p > 1$ the profit is $3/4 - p < 0$. Thus not trading attains the optimum zero. After a low result, X is uniform on $[0, t)$. Offers $p \leq t$ have nonpositive profit; offers $p \geq t$ have conditional profit $3t/4 - p < 0$. The buyer should not trade. After a high result, consider only $p \in [t, 1]$: lower prices do not trade, and higher prices only increase the payment. Multiplying conditional profit by the high-result probability $1 - t$ gives the unconditional contribution

$$H_t(p) = \int_t^p (3x/2 - p)\, \mathrm{d}x = -\frac{p^2}{4} + tp - \frac{3t^2}{4}.$$

This concave quadratic is maximized at $p_t = \min\{2t, 1\}$. The gross benefit of testing is therefore

$$W(t) = \begin{cases} t^2/4, & 0 < t \le 1/2, \\ t - 1/4 - 3t^2/4, & 1/2 \le t < 1. \end{cases}$$

The first branch increases; the second has its unique maximum at $t = 2/3$. The optimal test uses threshold $2/3$, no trade after a low result, and offer 1 after a high result. Gross profit is $1/12$. Net improvement over no test is $1/12 - c$: test below this cost, decline above it, and be indifferent at equality.

Method. An offer selects which goods trade, while a test changes the information available before offering. Optimize the offer within each information branch before optimizing the test itself.

Example A07. Searching Two Locations

A hides a target in one of three locations while B simultaneously chooses two distinct locations to search. If location i is found, A pays B w_i, where $(w_1, w_2, w_3) = (2, 3, 4)$; otherwise the payment is zero. Both players may randomize. Find all optimal strategies and the value.

Solution. If A uses $p = (6, 4, 3)/13$, each location contributes $w_i p_i = 12/13$ to expected payment. Every two-location search therefore earns $24/13$, giving A an upper guarantee. Let B omit locations $1, 2, 3$ with probabilities $(1, 5, 7)/13$, respectively. Their search probabilities are $(12, 8, 6)/13$, and multiplication by their respective payments gives $24/13$ in each case. Thus B guarantees that amount against every hiding location. The matching guarantees establish

$$\boxed{V = 24/13}.$$

To prove uniqueness, note that all three search actions in B's displayed strategy have positive probability. Against any other optimal hiding distribution, their expected payoffs must all equal V: their weighted average is V, and none can exceed it. Equality of the three pair sums forces all $w_i p_i$ to be equal, which uniquely determines the displayed p after normalization. Similarly, A assigns positive probability to every location. Any optimal search strategy must therefore pay V against each one, making its inclusion probabilities V/w_i and its omission probabilities unique. Both optimal strategies are thus unique.

Method. Here a search action is a subset. Equalize subset payoffs, then obtain an independent optimality certificate from the opponent's coverage probabilities.

Example A08. Adding a Spot-Check Option

An inspector chooses full inspection I or no inspection N, while the other player simultaneously chooses violation C or compliance H. The inspector's payoff matrix, with rows I, N and columns C, H, is

$$\begin{pmatrix} 1 & -1 \\ -2 & 0 \end{pmatrix}.$$

The other player's payoff is its negative. Find the mixed equilibrium. Then add an inspector action S whose expected payoff is $-1/4$ against either column. Find all optimal strategies and the new value.

Solution. If violation has probability p, the inspector's original actions earn $2p - 1$ and $-2p$. Equality gives $p = 1/4$ and value $-1/2$. If full inspection has probability q, the payoffs against violation and compliance are $3q - 2$ and $-q$. Equality gives $q = 1/2$, again with value $-1/2$. These checks cover all pure actions and establish the unique original mixed equilibrium. With S available, the inspector guarantees $-1/4$ by choosing it. The opponent holds every inspection payoff to at most $-1/4$ exactly when

$$2p - 1 \le -1/4, \qquad -2p \le -1/4,$$

that is, $p \in [1/8, 3/8]$. The new value is therefore $-1/4$, and this interval gives all optimal violation probabilities. The inspector's unique optimal strategy is pure S: against $p = 1/4$, each old action earns only $-1/2$, so any positive weight on them lowers the payoff strictly below $-1/4$. The new action leaves the old indifference intersection unchanged but changes optimal support.

Method. Indifference equations are valid only for a proposed support. A new action requires checking the entire payoff table again.

Example A09. A Fixed Subtraction Game

A pile contains $n \ge 0$ stones. Players alternate removing 1, 3, or 4 stones, never more than remain. Taking the last stone wins; a player with no legal move loses. Classify all losing positions and give a legal winning first move from every winning position.

Solution. Write P for losing and N for winning. For $n = 0, \dots, 6$, the labels are

$$\mathrm{P}, \mathrm{N}, \mathrm{P}, \mathrm{N}, \mathrm{N}, \mathrm{N}, \mathrm{N}.$$

For example, $n = 2$ can only leave winning position 1; positions $3, 4$ can be emptied; positions $5, 6$ can leave 2 by removing $3, 4$. We claim that all losing positions have $n \bmod 7 \in \{0, 2\}$. From residue 0, removing $1, 3, 4$ gives residues $6, 4, 3$; from residue 2, it gives $1, 6, 5$. None is a losing residue. Conversely, from residues $1, 3, 4, 5, 6$, remove respectively

$$1,\ 1,\ 4,\ 3,\ 4$$

stones to reach residue 0 or 2. These moves are already legal at the smallest representatives $1, 3, 4, 5, 6$, hence at all larger ones. Strong induction on the remaining pile proves both the classification and the strategy.

Method. A finite table suggests periodicity but does not prove it. Check both that losing positions cannot reach one another and that every other position can legally reach one.

Example A10. Congestion with Unequal Servers

There are $n \geq 2$ users choosing simultaneously between a slow and a fast server. If k choose the slow server, each pays cost k; the other $n - k$ users each pay $(n - k)/2$. Users minimize their own costs and are indifferent at ties. Find every pure Nash equilibrium occupancy. For $n = 8$, compare equilibrium and minimum total cost.

Solution. When both servers are occupied, a slow-server user does not move exactly when

$$k \leq \frac{n - k + 1}{2}.$$

A fast-server user does not move exactly when $(n - k)/2 \leq k + 1$. Together these give

$$\boxed{n - 2 \leq 3k \leq n + 1}.$$

The case $k = n$ is not an equilibrium because moving to the empty fast server lowers cost. For $k = 0$, only fast-server users need checking: no move is profitable exactly when $n/2 \leq 1$, or $n = 2$, which also satisfies the displayed condition. Thus that condition, with integer $0 \leq k \leq n$, describes every equilibrium. Any assignment of users with such occupancy works. For $n = 8$, equilibria have $k = 2$ or 3. Total cost is

$$C(k) = k^2 + \frac{(8 - k)^2}{2} = \frac{3}{2}\left(k - \frac{8}{3}\right)^2 + \frac{64}{3}.$$

The integer optimum is $k = 3$, with cost $43/2$; the other equilibrium, $k = 2$, costs 22. Individual stability therefore does not imply minimum social cost. The worst equilibrium-to-optimum ratio is $44/43$.

Method. A unilateral switch adds one user to the destination. Equilibrium and total-cost minimization use different objectives and must be analyzed separately.

Example A11. Location with Uneven Demand

Two providers simultaneously choose locations in $[0, 1]$ and may coincide. Customer location T has density $2t$. Customers choose the nearer provider; ties split demand equally. Each provider maximizes its demand share, with no prices. Find all pure Nash equilibria and the mean customer travel distance at equilibrium.

Solution. The customer distribution function is $F(t) = t^2$. Suppose locations differ, with $a < b$. The left provider receives $F((a+b)/2)$. Moving to any $a' \in (a, b)$ shifts the boundary strictly right. Every nonempty interval has positive demand, so this strictly improves its share. Distinct locations cannot be an equilibrium. If both providers locate at x, each gets $1/2$. A unilateral move to $z < x$ gives $F((z + x)/2)$, while a move to $z > x$ gives $1 - F((z + x)/2)$. If $F(x) > 1/2$, a sufficiently close move to the left earns more than half; if $F(x) < 1/2$, a close move to the right improves the share. Thus the only candidate is the median

$$m = 1/\sqrt{2}.$$

At this point every left deviation earns less than $F(m) = 1/2$, and every right deviation less than $1 - F(m) = 1/2$. Staying earns half. Hence both at m is the unique pure equilibrium. Travel distance is then $|T - m|$, so

$$\begin{aligned}\mathbb{E}|T - m| &= \int_0^m 2t(m - t)\ \mathrm{d}t + \int_m^1 2t(t - m)\ \mathrm{d}t \\ &= \frac{2}{3} - m + \frac{2}{3}m^3 = \frac{2 - \sqrt{2}}{3}.\end{aligned}$$

Method. First rule out separated locations, then test deviations from coincidence. Competition selects a demand median, not the mean customer location.

Example A12. A Reserve in a Sealed-Bid Auction

Two risk-neutral bidders have independent private values uniform on $[0, 1]$. The seller first sets a reserve $r \in [0, 1]$. Bidders then submit simultaneous sealed bids or abstain. The highest qualifying bid, at least r, wins and pays its own bid; everyone else pays zero. Ties are broken randomly. Construct a symmetric equilibrium, calculate expected seller revenue, and choose the best reserve.

Solution. For $0 < r < 1$, consider abstention for $v < r$ and, for $v \geq r$,

$$b(v) = \frac{v^2 + r^2}{2v}.$$

We have $b(r) = r$, and b is strictly increasing for $v > r$. If a bidder with value v imitates type $t \in [r, 1]$, he wins exactly when the opponent's value is below t, up to zero-probability ties. His expected payoff is

$$(v - b(t))t = vt - \frac{t^2 + r^2}{2}.$$

This is strictly concave in t, with derivative $v - t$. For $v \geq r$, the best choice is $t = v$, giving nonnegative payoff $(v^2 - r^2)/2$. For $v < r$, the best point in the interval is $t = r$, giving $r(v - r) < 0$, so abstention is preferable. Bids below the reserve are equivalent to abstention; bids above $b(1)$ cannot raise the winning probability and only raise payment. All deviations are covered. The maximum value has density $2v$, so seller revenue is

$$R(r) = \int_r^1 2v\, b(v)\ \mathrm{d}v = \frac{1}{3} + r^2 - \frac{4}{3}r^3.$$

At $r = 0$, use $b(v) = v/2$ with $b(0) = 0$. At $r = 1$, the item is almost surely unsold. The revenue formula remains valid at both endpoints. Since $R'(r) = 2r(1 - 2r)$, the unique revenue-maximizing reserve is $r = 1/2$, yielding $5/12$.

Method. Checking deviations by imitation of another type both derives payoffs and verifies equilibrium. Revenue must include the possibility that no bidder meets the reserve.

Example A13. A Mirror Strategy for a Row

A row of $n \geq 1$ numbered positions contains one token per position. Players alternate removing one token, or two tokens whose original positions are adjacent and still occupied. Gaps do not make originally nonadjacent positions adjacent. The player taking the last token wins. Give a first-player winning strategy for every n and prove that every response is legal.

Solution. For odd n, remove the middle token first; for even n, remove the two adjacent middle tokens. The remaining position is symmetric under reflection about the original midpoint, with a central gap. After every opposing move, remove its reflected image. The opponent's legal move cannot cross the central gap, so it is disjoint from its image. The position was symmetric before that move, ensuring that the reflected tokens remain present. Reflection preserves original adjacency, so the response is legal. It restores symmetry. Each pair of moves reduces the number of tokens, so play terminates. Whenever the opponent has a legal move, the first player has a legal response. The opponent therefore cannot take the final token, and the first player wins. For $n = 1, 2$, the opening move already takes everything.

Method. A pairing argument must ensure that a move and its reflected response do not overlap. Creating a central gap supplies that missing guarantee.

Example CM01. Scheduling a Regular Bipartite Graph

There are n workers and n tasks. Each worker can perform exactly d tasks, and each task can be performed by exactly d workers, where $1 \leq d \leq n$. Prove that d rounds can be scheduled so that each worker performs a distinct task in every round and every allowed worker–task pair is used exactly once. Give the key proof of the matching theorem used.

Solution. Draw the allowed pairs as a bipartite graph. For any worker set S, there are $d|S|$ incident edges, all ending in its neighbor set $N(S)$. Each task receives at most d of these edges, so $d|S| \leq d|N(S)|$, giving $|N(S)| \geq |S|$. Hall's theorem states that this condition guarantees a matching covering all workers. Its proof is by induction on the left-side size. If a nonempty proper set S has equality, match S to $N(S)$ by induction. In the remaining graph, the original condition applied to $S \cup T$ gives $|N(T) \setminus N(S)| \geq |T|$ for every remaining worker set T, so induction also matches the complement. If every nonempty proper set has strict inequality, choose an edge and remove its endpoints. Every remaining worker set loses at most one neighbor and still satisfies Hall's condition. Apply induction, then restore the chosen edge. A single vertex is the immediate base case. The original graph thus has a perfect matching for the first round. Deleting its edges lowers every degree to $d - 1$. Repeat on the remaining regular graph until degree zero, obtaining d disjoint perfect matchings. Each round covers all workers and tasks, and all nd allowed edges are used once.

Method. Degree counts turn a local regularity assumption into Hall's condition. Regularity survives removal of a matching, allowing a complete round-by-round decomposition.

Example CM02. Congruences with Noncoprime Moduli

For positive integers m, n and integers a, b, determine when $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$ are simultaneously solvable. Give an effective construction and the solution period. Then solve

$$x \equiv 2 \pmod{6}, \qquad x \equiv 5 \pmod{9}, \qquad x \equiv 4 \pmod{5}$$

over the integers.

Solution. Let $g = \gcd(m, n)$. Subtracting the congruences shows the necessary condition $g \mid b - a$. If it holds, write $x = a + mt$. The second congruence becomes

$$\frac{m}{g}t \equiv \frac{b-a}{g} \pmod{n/g}.$$

The reduced coefficient and modulus are coprime. Extended Euclid computes the inverse of m/g modulo n/g, and hence t. If $n/g = 1$, there is no remaining restriction and one may take $t = 0$. This proves sufficiency. The difference of any two solutions is divisible by both m and n, hence by $\operatorname{lcm}(m, n) = mn/g$. Conversely, adding any multiple of this number preserves both congruences. Thus the solutions form one residue class modulo mn/g. In the example, put $x = 2 + 6t$. Then $6t \equiv 3 \pmod{9}$, or $2t \equiv 1 \pmod{3}$, so $t \equiv 2 \pmod{3}$ and $x \equiv 14 \pmod{18}$. Write $x = 14 + 18k$; the third congruence gives $3k \equiv 0 \pmod{5}$, hence $k \equiv 0 \pmod{5}$. All solutions are

$$\boxed{x = 14 + 90j, \qquad j \in \mathbb{Z}}.$$

Method. Noncoprime moduli require a compatibility check before inversion. Cancel the common divisor before seeking an inverse; do not divide by a noninvertible residue.

Example CM03. Twelve Balls with Fixed Weighings

Exactly one of twelve identical-looking balls has abnormal mass, either heavier or lighter; all normal balls have equal mass. Using a balance without reference weights, with equal numbers of balls on the pans each time, design three weighings fixed in advance to identify the abnormal ball and whether it is heavy or light. Prove that two weighings are insufficient and that thirteen balls cannot be handled by this class of three predetermined weighings.

Solution. Encode left-heavy, balance, and right-heavy as $1, 0, -1$. There are 24 ball-and-sign states, but only $3^2 = 9$ outcomes in two weighings, so at least three

are required. In the matrix below, columns represent balls and rows represent weighings: 1 means left pan, −1 right pan, and 0 omission.

	1	2	3	4	5	6	7	8	9	10	11	12
1	0	0	0	0	−1	−1	−1	1	1	1	−1	1
2	0	−1	−1	−1	1	1	1	0	0	0	−1	1
3	−1	1	0	−1	1	0	−1	−1	0	1	1	0

Each row has four positive and four negative entries, so all weighings are legal. Every column is nonzero, and no two are equal or opposite. Heavy ball j produces column j as its outcome vector; a light ball produces its negative. The 24 possible outcomes are distinct, allowing unique decoding. For thirteen balls, all $3^3 - 1 = 26$ nonzero outcomes would be needed. A column and its negative form a pair, so the thirteen columns would represent all such pairs. In any fixed coordinate, $2 \cdot 3^2 = 18$ of the 26 nonzero vectors have a nonzero entry. Thus that row of the thirteen representative columns would contain exactly nine nonzero entries. Nine balls cannot be split equally between the pans, contradicting the balance constraint. Thirteen are impossible, and twelve achieves the maximum for this class of predetermined weighings.

Method. Counting outcomes gives only a necessary condition; a code must also satisfy physical balance constraints. Pairing a column with its negative handles the unknown heavy-or-light sign.

Section Exercises

OA09. A Decreasing-Limit Stone Game

There are $n \geq 2$ stones. Two players alternate moves, and the player taking the last stone wins. On the first move, the first player may take between 1 and $n - 1$ stones. Each later move must take at least one stone and no more than the preceding move; taking all remaining stones is allowed if that number satisfies the limit. For which n can the first player force a win? Give a winning strategy when one exists.

OA13. A Coprime Selection Game

Two players alternately choose an unused integer from $2, 3, \ldots, 30$. Each newly chosen number must be coprime to every number chosen earlier. A player with no legal move loses. Determine who can force a win, and prove a winning strategy.

4.3 Exercises

Key Ideas and Methods. Identify the remaining degrees of freedom. Unknown graph edges are constrained by connectivity, affine arrangements by invertibility modulo the board size, and sharing designs by incidence counts. For a randomized transform, first condition on one atom, then mix over atomic probabilities.

Recheck every changed game rule. Losing on the last move changes terminal positions; fees change deviation payoffs without necessarily changing service costs. When asked for all optimal strategies, describe the full probability or parameter range, including endpoints. Verify cooperative and competitive objectives separately.

AG2-1. Parity Recovery on a Cycle

A graph has vertices $1, 2, 3, 4$, cycle edges $12, 23, 34, 41$, and diagonal 13. The diagonal bit is known to be 1; vertex parity checks are $(1, 0, 1, 0)$. List all unknown cycle-edge solutions. If those four bits were independent and fair, what is the probability of satisfying the checks? Does observing $12 = 0$ make recovery unique?

AG2-2. Affine Queen Arrangements

On a 5×5 toroidal board, choose column $j = ai + b \pmod 5$ in row i, with $a, b \in \{0, 1, 2, 3, 4\}$. Find all pairs satisfying every condition in A14. Then decide whether any arrangement, affine or otherwise, is possible for a 9×9 board.

AG2-3. Spreading an Atom Uniformly

In A28, an atom x_0 satisfies $F(x_0-) = 3/8$ and $\mathbb{P}(X = x_0) = 1/4$. Find the law of $U \mid X = x_0$ and $\mathbb{P}(X = x_0, U > 1/2)$. Prove that no deterministic function h can make $h(X)$ continuously uniform.

AG2-4. Deleting One Member

Delete one person from the seven-person design in A17, retaining every item with its remaining holders. Count items held by two and by three people. Check that each remaining person still holds three types and each pair still shares exactly one. Independently verify the counts through incidence equations.

AG3-1. All Optimal Cyclic Searches

Four hiding locations form the cycle $1, 2, 3, 4$. A chooses one location; B simultaneously searches one adjacent pair from $12, 23, 34, 41$. B receives 1 upon finding the target and zero otherwise, with opposite payoffs for A. Find the value and every optimal mixed strategy.

AG3-2. Losing by Taking the Last Stone

Keep A09's subtraction set $\{1, 3, 4\}$, but make taking the last stone an immediate loss. For positive pile sizes, classify losing residues and prove the classification. Compare winning first moves from 20 stones under this rule and under A09's rule.

AG3-3. A Fee That Selects an Occupancy

In A10 take $n = 8$ and charge each fast-server user an additional fee $\delta \geq 0$. There is no fee at the slow server. The fee affects individual decisions but is excluded as a transfer when measuring total service cost. Find all fees for which every pure equilibrium has exactly three slow-server users, and explain both boundary cases.

AG3-4. Cooperative and Competitive Locations

Customers are uniform on $[0, 1]$. Two providers cooperate to choose $0 \leq a \leq b \leq 1$, minimizing average distance to the nearer provider. Find the unique ordered optimum and minimum distance. Compare with a simultaneous game in which each maximizes its own demand share and coincident providers split demand.

5

Algorithms and Programming

5.1 Summaries and Interval Scans

Key Ideas and Methods. For a data stream, first identify the summary that must be retained. Sample size, mean, and centered sum of squares suffice to update variance; merging groups also requires a correction for their different means. For interval extrema, a monotone deque retains candidates that have neither expired nor been dominated by a later element. If extending a window can only make its constraint harder to satisfy, moving its left endpoint forward can locate the longest feasible interval.

Correctness follows from the summary or deque invariant. Linear total time follows because each element enters and leaves at most once. Random simulation also requires a proof of the accepted distribution and termination; retries must use fresh independent trials.

Example A31. Merging and Separating Summary Statistics

Two disjoint data groups are available only through summaries (n_A, μ_A, M_A) and (n_B, μ_B, M_B), with $n_A, n_B \geq 1$ and $M_A = \sum_{i \in A}(x_i - \mu_A)^2$, similarly for B. Raw observations cannot be read. Recover the combined mean, centered sum of squares, and both variance conventions. Also recover B's summary from the combined summary and A's summary. Discuss cost and numerical limitations.

Solution. Put $n = n_A + n_B$ and $\delta = \mu_B - \mu_A$. Adding the two data sums gives

$$\mu = \mu_A + \frac{n_B}{n}\delta.$$

For group A, expand $x_i - \mu = (x_i - \mu_A) + (\mu_A - \mu)$. The cross term sums to zero because the within-group deviations sum to zero. Doing the same for B yields

$$M = M_A + M_B + n_A(\mu_A - \mu)^2 + n_B(\mu_B - \mu)^2 = M_A + M_B + \frac{n_A n_B}{n}\delta^2.$$

The population variance is M/n and the sample variance with a degrees-of-freedom correction is $M/(n-1)$. Unbiasedness of the latter additionally requires suitable iid sampling assumptions.

For separation, first find $n_B = n - n_A > 0$. Then

$$\mu_B = \frac{n\mu - n_A\mu_A}{n_B}, \qquad M_B = M - M_A - \frac{n_A n_B}{n}(\mu_B - \mu_A)^2.$$

Each operation uses constant time and a constant-size summary. In exact arithmetic, any binary merge order gives the same result: each merge recovers the ac-

tual combined moments. Floating-point rounding can make orders differ slightly. Separation can suffer severe cancellation, especially when B is small and the subtracted quantities are large. A substantially negative recovered M_B signals inconsistent summaries or numerical failure and should not simply be clipped to zero.

Method. Combined variation includes both within-group variation and variation between group means. Averaging the two variances alone omits the latter.

Example A32. The Longest Interval with Bounded Range

Given real values $x_0, \ldots, x_{T-1}$ and $D \geq 0$, find the longest nonempty interval satisfying

$$\max_{i \leq t \leq j} x_t - \min_{i \leq t \leq j} x_t \leq D.$$

Break ties by the earliest start. Return an empty interval for empty input. Give an $O(T)$ algorithm and prove that advancing the left endpoint cannot discard an optimum.

Solution. Maintain a left endpoint ℓ, a deque of indices with decreasing values for the maximum, and another with increasing values for the minimum. On adding right endpoint r, remove values at most x_r from the maximum deque's tail and values at least x_r from the minimum deque's tail, then append r to both.

While the two front values differ by more than D, advance ℓ and discard front indices smaller than the new ℓ. The front values then give the true window extrema. Record $r - \ell + 1$ only when it strictly exceeds the best length; increasing right endpoints thereby preserve the earliest start among ties.

If $[\ell, r]$ is invalid, every interval containing it is invalid. Discarding that left endpoint therefore loses no feasible interval ending now or later. When shrinking stops, ℓ is the earliest still-possible feasible start for r, so the current window is the longest feasible interval ending there. Considering every r proves global optimality.

A tail index removed by a later, more extreme value is dominated: it expires no later and can never be needed as an extremum. Each index enters and leaves each deque at most once, and ℓ advances at most T times. Time is $O(T)$ and worst-case space is $O(T)$. Every singleton is feasible when the input is nonempty.

Example P09. Simulating a Die with an Unknown Biased Coin

You may only toss one coin independently, with an unknown head probability $p \in (0, 1)$, and have no other randomness. Design an algorithm that does not know p and outputs each of $1, \ldots, 6$ with exactly equal probability. Prove uniformity and almost-sure termination, and find the expected number of coin tosses used by your algorithm.

Solution. First extract fair bits. Toss twice: output 0 for HT, output 1 for TH, and discard HH or TT and repeat. HT and TH each have probability $p(1 - p)$, so an accepted output is fair. Successive outputs use disjoint segments of indepen-

dent trials; after one segment ends, the next has the same fresh distribution. The extracted bits are consequently independent.

Let b be the mean toss count per fair bit. A pair is accepted with probability $2p(1-p) > 0$, giving a geometric tail, and

$$b = 2 + \big(1 - 2p(1-p)\big)b = \frac{1}{p(1-p)}.$$

Read three fair bits as $Z \in \{0, \ldots, 7\}$. If $Z \le 5$, output $Z + 1$; otherwise discard the block and repeat. All eight values are equally likely, so the accepted six remain uniform. A block is accepted with probability $3/4$.

Writing E for the total expected toss count, one block costs $3b$ in expectation and a rejection starts fresh. Hence

$$E = 3b + \frac{1}{4}E, \qquad \boxed{E = \frac{4}{p(1-p)}}.$$

Both acceptance probabilities are positive, proving almost-sure termination. This is the cost of the stated algorithm, not a claim of optimality. Fair-bit extraction becomes expensive as p approaches either endpoint.

Section Exercises

OA31. Rolling Mean and Variance

Real observations $x_1, x_2, \ldots$ arrive sequentially. For a fixed window size $N \ge 1$, update the mean and variance of the most recent N observations whenever a new value arrives, using all observations if fewer than N have arrived. Define the variance as the average squared deviation within the current window. Give an efficient algorithm, its update formulas, and its time and space complexity.

OA32. Sliding-Window Maxima

Given a sequential stream of real values $x_1, x_2, \ldots$ and a positive integer N, output the maximum of the most recent N values at each arrival, using all available values before the window fills. Design an efficient algorithm, prove its correctness, and distinguish its worst-case time per update from its amortized time.

5.2 Drawdown Intervals

Key Ideas and Methods. For positive prices, fix the endpoint j first. The drawdown $1 - p_j/p_i$ increases with the starting price p_i, so maximum drawdown reduces to finding the highest price among eligible starting points. A duration limit determines the candidate window. The positivity assumption is essential to this comparison. Finding the shortest interval that reaches a threshold is a different optimization problem: later starting points with higher prices dominate earlier, lower ones.

Before discarding a candidate, justify why it cannot improve a future answer: it has expired, is dominated, or has already attained its shortest feasible interval.

Arrange insertion and queries so that only indices $i < j$ are eligible. Rules for equal prices and equal objective values determine which tied interval is returned.

Example A33. Maximum Drawdown with a Duration Limit

Given strictly positive prices $p_0, \ldots, p_{T-1}$ and an integer $L \geq 1$, allow only drawdown intervals of duration between 1 and L:

$$D = \max\left(0, \max_{1 \leq j-i \leq L}\left(1 - \frac{p_j}{p_i}\right)\right).$$

For $T < 2$, define $D = 0$. If $D > 0$, return an optimum, breaking ties first by the earliest end and then by the earliest start. Otherwise return zero and no interval. Design a linear-time algorithm.

Solution. For a fixed end j, the drawdown increases with p_i because $p_j > 0$. Thus find the largest price in the eligible window

$$\max(0, j - L) \leq i \leq j - 1.$$

Maintain a decreasing deque of candidates. Before evaluating j, insert $j - 1$, removing only strictly smaller prices from the tail; keep equal prices to retain the earlier start. Then expire front indices smaller than $j - L$.

With front index i, evaluate $1 - p_j/p_i$. Scan $j = 1, \ldots, T - 1$ and update only on a strict improvement over the best positive drawdown. This retains the earliest end; the deque's handling of equal peaks retains the earliest start. For $T < 2$, return zero and no interval.

A removed tail price is dominated by a later, larger price, which is at least as good in every future eligible window and expires no earlier. The front is therefore always an eligible maximum. Optimizing separately for each end and then over all ends proves correctness. Each index enters and leaves the deque at most once, giving $O(T)$ time and $O(\min(T, L))$ space.

Caution. The historical maximum can be too old to be eligible. A cumulative maximum alone can return an infeasible interval.

Example A34. The Shortest Interval Reaching a Drawdown Threshold

For strictly positive prices and $d \in (0, 1)$, find the shortest interval (i, j) with $i < j$ and $p_j \leq (1 - d)p_i$. Report failure if none exists; break length ties by the earliest end. Give an $O(T)$ algorithm and explain why maximizing the drawdown ratio does not solve this problem.

Solution. Keep candidate starts in a deque with strictly decreasing prices. At end j, repeatedly inspect the front i. If

$$p_j \leq (1 - d)p_i,$$

update the shortest length with $j - i$ and remove i; otherwise stop this loop. Next remove tail indices whose prices are at most p_j, and append j. Querying before insertion enforces $i < j$.

If the front fails the threshold, every later candidate has a smaller price and also fails. Removing a successful front is safe: the current end is the earliest unprocessed end, and every future end would give that same start a longer interval. Tail removal is safe for a different reason. A later start with a price at least as high meets any future threshold at least as easily and produces a shorter duration.

Thus front removal records the best available duration for a start, whereas tail removal uses domination by a later, higher price. Each index enters and leaves at most once, giving $O(T)$ time and $O(T)$ space. Scan ends in order and update only on a strictly shorter length to retain the earliest end. Maximum drawdown optimizes the price ratio and may take a long time; it does not optimize duration.

Section Exercises

OA33. Maximum Drawdown

Let $p_1, \ldots, p_n$ be positive prices, with $n \geq 1$. Define the maximum drawdown as the largest relative decline from an earlier to a later price:

$$D = \max\left(0, \max_{1 \leq i < j \leq n} \frac{p_i - p_j}{p_i}\right),$$

with $D = 0$ when $n = 1$. Give an $O(n)$ algorithm, prove it correct, and state its space complexity. If $D > 0$, explain how to return an interval attaining it.

5.3 Algorithm Problems

Key Ideas and Methods. In dynamic programming, define what information a state retains before writing transitions and boundary values. Limits on deletions, inventory, or selected intervals must be represented in the state. Unreachable states must be distinguished from feasible states with zero reward. In counting problems, different transitions must describe disjoint sets of solutions to avoid double counting.

The choice of algorithm rests on a proved structure: binary search needs a monotone exclusion rule, greedy methods need an exchange or dominance argument, and divide-and-conquer needs a way to account for contributions across subproblems. Count all states, transitions, and data-structure operations when bounding complexity. Space compression also requires an update order that does not accidentally reuse current-round values. For randomized algorithms, distinguish distributional correctness, expected cost, and worst-case cost.

Example ALG01. Maximum Subarray Sum with One Deletion

Given a nonempty real array $a_0, \ldots, a_{T-1}$, choose a contiguous interval and delete at most one of its elements, retaining at least one. Find the maximum retained sum in $O(T)$ time and $O(1)$ extra space.

Solution. Let f_i be the best sum of an interval ending at i with no deletion, and g_i the best nonempty retained sum for such an interval with at most one deletion. In g_i, position i itself may be deleted. Initialize $f_0 = g_0 = a_0$. For $i \geq 1$,

$$f_i = \max(a_i, f_{i-1} + a_i), \qquad g_i = \max(a_i, g_{i-1} + a_i, f_{i-1}).$$

The three choices in g_i start afresh, retain a_i and extend a previous interval, or delete a_i. If a solution deletes its first element, moving the interval's start right gives the same retained sequence, so no additional case is needed. All remaining solutions are covered by whether their last element is deleted.

Return $\max_i g_i$. Preserve the old f when updating; the third candidate must not use its new value. Initialization uses actual elements rather than zero, preventing an invalid empty answer on an all-negative array. Induction on the two states proves optimality. Time is $O(T)$ and extra space is $O(1)$.

Example ALG02. The Minimum Number of Jumps

A nonempty array $a_0, \ldots, a_{T-1}$ contains nonnegative integers. From index 0, at index i you may jump right by any of $1, \ldots, a_i$ positions. Find the minimum jumps to $T - 1$, or report that it is unreachable. Use linear time and constant extra space; reachability is not guaranteed.

Solution. Return 0 if $T = 1$. Maintain the right boundary R of the current reachable layer, the furthest next-layer position F seen so far, and the number k of completed expansions, initially all zero. Scan $i = 0, \ldots, T - 2$ and update

$$F \leftarrow \max\{F, \min(T - 1, i + a_i)\}.$$

When $i = R$, the current layer has been fully scanned. If $F = R < T - 1$, no new position is reachable, so report failure. Otherwise increment k and set $R = F$. Return k as soon as $R = T - 1$.

Each jump can reach a whole forward interval. Thus positions reachable in at most k jumps form a prefix starting at 0. After scanning this prefix, the union of all one-step extensions ends exactly at F, the boundary for at most $k + 1$ jumps. Processing these layers in order proves that the first layer covering the target has minimum depth. Every index is scanned once, giving $O(T)$ time and $O(1)$ extra space.

Example ALG03. One Trade with a Holding-Period Constraint

Prices $p_0, \ldots, p_{T-1}$ are nonnegative, and integers $1 \leq h \leq L$ are given. Make at most one purchase and sale, with holding constraint $h \leq j - i \leq L$ and sale fee $c \geq 0$. Choosing no trade is allowed. Find the maximum net profit and an optimal date pair in $O(T)$ time.

Solution. For sale day j, eligible purchase days are

$$\max(0, j - L) \leq i \leq j - h.$$

An empty range gives no feasible trade that day. Maintain an increasing price deque. On day j, insert $j - h$ if nonnegative, removing strictly larger tail prices,

and then expire front indices smaller than $j - L$. Keeping equal prices permits the earlier purchase date.

The front minimizes eligible purchase price, so evaluate $p_j - p_i - c$. Initialize the answer to zero and record dates only on a strict improvement. Return no trade if the final value remains zero. If $T \le h$, no date pair is possible.

For a fixed sale day, the sale price and fee are fixed, making the cheapest eligible purchase optimal. A later, cheaper price dominates an older, dearer one and expires no earlier. Delayed insertion enforces the minimum holding period, and front expiration enforces the maximum. Each index enters and leaves at most once: time $O(T)$ and space $O(\min(T, L))$.

Example ALG04. Daily Trading with Bounded Inventory

Given nonnegative prices for T days and integer inventory capacity $K \ge 0$, each day buy one unit, sell one unit, or do nothing. Short selling is forbidden; each purchased or sold unit incurs fee $c \ge 0$. Start with zero cash and inventory, allow negative cash, and finish with zero inventory. Maximize final cash using an $O(T(K + 1))$ dynamic program.

Solution. Let $V_t(q)$ be the maximum cash after the first t days with q units, $0 \le q \le K$. Set $V_0(0) = 0$ and all other initial states to $-\infty$. On the day with price p_t,

$$V_{t+1}(q) = \max \left\{ \begin{array}{ll} V_t(q), & \\ V_t(q-1) - p_t - c & (q \ge 1), \\ V_t(q+1) + p_t - c & (q < K). \end{array} \right\}$$

These are no action, purchase, and sale. Every right-hand side must use the previous day's array. Updating repeatedly in place can incorrectly allow multiple units to be traded on one day.

Every legal strategy belongs to exactly one last-action case. Removing that action gives the corresponding prior state, and appending it to any finite prior state is feasible. Induction proves both the upper bound and attainability. Return $V_T(0) \ge 0$, since no action is always feasible. Time is $O(T(K + 1))$ and rolling-array space is $O(K + 1)$; stored choices recover a schedule. If $K = 0$ or $T = 0$, return zero. Allowing negative cash is a material assumption; a cash constraint would require further feasibility checks.

Example ALG05. Counting Paths through at Most One Obstacle

An $R \times C$ binary grid, with $R, C \ge 1$, marks obstacles by 1. Move only right or down from the upper-left to the lower-right cell, visiting at most one obstacle, with endpoints counted by the same rule. Find the number of paths modulo an integer $M \ge 2$ in $O(RC)$ time and $O(C)$ space.

Solution. Let $A_{i,j}$ and $B_{i,j}$ count paths reaching (i, j) after visiting zero and exactly one obstacle. Sum the corresponding counts from above and from the left, obtaining a, b. At a zero cell, set $(A, B) = (a, b)$; at an obstacle, set $(A, B) = (0, a)$. Reduce every addition modulo M.

Initialize the start separately as $(1, 0)$ if it is zero and $(0, 1)$ otherwise. Other out-of-grid predecessors contribute zero. Return $A_{R-1,C-1} + B_{R-1,C-1} \pmod{M}$.

Every path's penultimate cell is either above or left, disjoint cases. Its obstacle count then changes exactly according to the current cell. Thus the recurrence neither misses nor double-counts a path. Scan each row left to right with two length-C arrays: the old current column stores the upper predecessor, and the updated previous column stores the left predecessor. Read both before overwriting. For $R, C \geq 1$, the same initialization covers a one-cell grid.

Example ALG06. A Shortest Common Supersequence

Given strings a, b, possibly empty, construct a shortest string s containing each as a subsequence. A character of s may serve one character from each input simultaneously. Find the minimum length, recover a solution, and prove correctness.

Solution. Let $D_{i,j}$ be the shortest common supersequence length for prefixes of lengths i, j. Initialize $D_{i,0} = i$ and $D_{0,j} = j$. Index characters from 1. If the last characters agree, share one final character:

$$D_{i,j} = D_{i-1,j-1} + 1 \quad (a_i = b_j).$$

If they differ, the final character removes the last character of only one input, giving

$$D_{i,j} = 1 + \min(D_{i-1,j}, D_{i,j-1}).$$

When the last characters agree, their two matches in any supersequence can both be moved to the later matched position, so sharing that final position is harmless. For different last characters, the two alternatives exhaust an optimum. Induction on prefix lengths proves the recurrence.

Backtrack from $(|a|, |b|)$. On equal characters record that character and move diagonally; otherwise follow a minimum predecessor and record the removed last character. At a boundary, append the remaining prefix in reverse order, then reverse the entire record. Time and storage are $O((|a| + 1)(|b| + 1))$.

If L is the longest common subsequence length, the answer is also $|a| + |b| - L$. Shared matched positions form a common subsequence, so at most L positions can be shared. Conversely, align a longest common subsequence and insert the intervening characters of both strings to attain this bound.

Example ALG07. Counting and Sampling Interleaving Assignments

Given strings a, b, t with $|t| = |a| + |b|$, assign every position of t to one input while preserving the order within each input. Distinct source assignments count separately even when their characters coincide. Count legal assignments and, when the count is positive, sample one uniformly.

Solution. Let $C_{i,j}$ count completions after using the first i characters of a and j of b. Put $m = |a|$, $n = |b|$, and $C_{m,n} = 1$. With characters indexed from 1, compute backwards:

$$C_{i,j} = \mathbf{1}_{\{i<m,\ a_{i+1}=t_{i+j+1}\}} C_{i+1,j} + \mathbf{1}_{\{j<n,\ b_{j+1}=t_{i+j+1}\}} C_{i,j+1}.$$

Out-of-range branches contribute zero. The two branches have different sources, so their counts add even if the characters agree. The total is $C_{0,0}$; a length mismatch gives zero immediately.

At a state with positive count, select the next source with probability equal to that branch's count divided by $C_{i,j}$, then move to its successor. For any complete assignment, multiplying these probabilities telescopes to $1/C_{0,0}$, proving uniformity. The boundary also covers the unique empty assignment.

There are $O((m + 1)(n + 1))$ states and recovery takes $m + n$ steps. Counts are exact integers; the state-operation bound does not assume constant bit cost for arbitrarily large integer additions. Counts reduced modulo an integer cannot supply exact sampling weights.

Example ALG08. Weighted Interval Selection with a Cardinality Limit

There are n tasks. Task i occupies a half-open interval $[s_i, e_i)$ with $s_i < e_i$ and has possibly negative reward w_i. Given an integer $K \geq 0$, choose at most K pairwise nonoverlapping tasks; one may start exactly when another ends. Find the maximum reward in $O(n \log n + nK)$ time.

Solution. Sort by end time and let $p(i)$ be the largest predecessor index with $e_j \leq s_i$, or 0 if none exists. Since $s_i < e_i$, every such predecessor appears before i. Binary search on sorted end times finds all predecessors in $O(n \log n)$.

Let $F(i, k)$ be the best reward from the first i tasks using at most k. Initialize $F(0, k) = F(i, 0) = 0$ and use

$$F(i, k) = \max\{F(i - 1, k),\ w_i + F(p(i), k - 1)\}.$$

Excluding i gives the first case. If i is included, all other chosen tasks must end before it starts, giving the second. Conversely, an optimal compatible predecessor set can be joined to i, proving attainability as well as the bound.

Replace K by $\min(K, n)$ and return $F(n, K)$. Negative-reward tasks can be skipped, so zero is always attainable. A full $O(n(K + 1))$ table supports backtracking. If only the value is needed, process k by layers with two arrays of length $n + 1$.

Example ALG09. Scheduling Tasks with Individual Cooldowns

Day i offers a task available only that day, with real reward r_i. After performing it, rest for the next $L_i \geq 0$ complete days, where L_i is an integer; no other task may be performed during rest. Tasks may be skipped, for $i = 0, \ldots, T - 1$. Maximize total reward and recover the selected dates in $O(T)$ time.

Solution. Let V_i be the best reward starting on day i, with $V_T = 0$. After choosing today's task, the next eligible day is

$$q_i = \min(T, i + L_i + 1).$$

Compute backwards:

$$V_i = \max\{V_{i+1}, r_i + V_{q_i}\}, \qquad i = T - 1, \ldots, 0.$$

The alternatives skip today or perform the task and resume after its rest period. They cover every legal schedule, and joining either action to an optimal successor schedule is feasible. Backward induction proves optimality.

Store the chosen branch. Starting at 0, advance one day after a skip; after a chosen task, output i and jump to q_i. Indices strictly increase, so recovery visits at most T states. Time and space are $O(T)$. Skipping ensures that nonpositive rewards need not reduce the answer; empty input gives zero. Variable successor gaps prevent a fixed-size recent-state buffer in general.

Example ALG10. Repairing Parentheses with the Fewest Insertions

Given a string containing only opening and closing parentheses, insert characters without deleting or changing any original character. Minimize the insertions needed to make the entire string valid, and output a repair attaining the minimum. Use linear time.

Solution. Scan left to right with the number b of unmatched opening parentheses. Output an opening parenthesis unchanged and increment b. On a closing parenthesis, if $b > 0$, output it and decrement b; otherwise insert an opening parenthesis immediately before it and count one insertion. At the end append b closing parentheses.

Every output prefix has nonnegative balance and the final balance is zero, so the repair is valid and retains all original characters in order.

For optimality, let $m \le 0$ be the minimum original prefix balance (including the empty prefix) and s the final original balance. Every repair needs at least $L = -m$ inserted openings, or the most deficient prefix remains invalid. If ℓ, r openings and closings are inserted, final balance requires $r = s + \ell$. The total is therefore $s + 2\ell \ge s + 2L$. The algorithm inserts exactly L openings, leaving balance $s + L$, and appends that many closings. It attains the lower bound in $O(T)$ time and constant space apart from its output.

Example ALG11. Water Retention on a Two-Dimensional Terrain

A rectangular grid has nonnegative ground height h_v and unit area at each cell. Neighbors share an edge. Water can drain freely outside, and every boundary cell connects to the exterior. After sustained rain reaches equilibrium, find the total retained water. Give an algorithm and explain why row and column maxima alone are insufficient.

Solution. Define the escape level of cell v by

$$\ell_v = \min_{\gamma:\, v \text{ to boundary}} \max_{u \in \gamma} h_u.$$

Paths include their endpoints, so $\ell_v \geq h_v$. The largest height on a path is its drainage threshold; taking the lowest threshold among all escape paths gives the stable water level. Total water is $\sum_v (\ell_v - h_v)$.

Solve this minimax path problem with a min-heap. Initialize every boundary cell at distance h_v and interior cells at $+\infty$. Extract the unsettled cell v with smallest distance and relax each neighbor u by

$$d_u \leftarrow \min\{d_u, \max(d_v, h_u)\}.$$

Ignore obsolete heap entries. Extending a path never reduces its maximum height, so the usual shortest-path greedy argument applies. More explicitly, a lower escape path would contain a first unsettled cell adjacent to a settled one; relaxation from that settled cell would already have supplied a no-larger candidate, contradicting the minimum heap extraction.

There are only a constant number of edges per cell, giving $O(RC \log(RC+1))$ time and $O(RC)$ space. With one row or one column, every cell is on the boundary and retention is zero. Drainage paths may turn, so barriers measured only along straight rows or columns can overestimate retention.

Example ALG12. Largest and Total All-One Squares

For a binary matrix, find both the largest area of an axis-aligned all-one square and the total number of such squares. Different locations or side lengths count separately. Use $O(RC)$ time and $O(C)$ extra space.

Solution. Let $d_{i,j}$ be the largest side length of an all-one square with lower-right corner (i, j). A zero cell gives $d_{i,j} = 0$; a one cell gives

$$d_{i,j} = 1 + \min(d_{i-1,j}, d_{i,j-1}, d_{i-1,j-1}),$$

with out-of-grid values zero. A side-k square requires side-$(k-1)$ squares at all three neighboring corners, proving necessity. Conversely, those three smaller squares together with the current cell cover the entire side-k square, proving sufficiency.

The largest area is $(\max d_{i,j})^2$. At a fixed lower-right corner, feasible side lengths are exactly $1, \ldots, d_{i,j}$, so the total count is

$$\sum_{i,j} d_{i,j}.$$

Every square has one lower-right corner, avoiding double counting. Scan rows with a length-C array and one saved old upper-left value: the old current column is the upper neighbor and the updated previous column is the left neighbor. Both answers are zero for an all-zero or empty matrix.

Example CA01. Weighted Sampling from a Stream

Items arrive sequentially with positive weights w_i. The total number of items is unknown. Only one candidate item and a constant number of numerical values may be stored. Design an algorithm such that after any $t \geq 1$ arrivals, item i is the candidate with probability $w_i / \sum_{j=1}^{t} w_j$. Each update may use an independent continuous uniform random variable. Prove the claim and explain why replacing the candidate whenever the new item is heavier fails.

Solution. Maintain the cumulative weight W. Select the first item and set $W = w_1$. When item t arrives, first update $W \leftarrow W + w_t$, then replace the candidate with probability w_t / W. Otherwise retain it. Each update takes constant arithmetic time and stores only the candidate identifier and cumulative weight.

For the inductive proof, let W_{t-1} denote the previous total. An old item $i < t$ survives precisely when it was selected previously and the new update does not replace it. Independence of the fresh random variable gives

$$\mathbb{P}(\text{candidate} = i) = \frac{w_i}{W_{t-1}} \left(1 - \frac{w_t}{W_t}\right) = \frac{w_i}{W_t}.$$

The new item is selected with probability w_t / W_t directly. The base case $t = 1$ is immediate, proving the invariant at every fixed length. It also applies when the ending length is fixed externally or independent of the sampling randomness.

For weights $1, 2, 3$, the second update selects item 2 with probability $2/3$ and the third replaces with probability $1/2$. Final probabilities are $1/6, 1/3, 1/2$. Deterministic replacement by a heavier item always retains the last item in a strictly increasing stream, so it cannot have the required distribution.

The proof applies to a fixed sequence of weights. Random weights independent of the sampling randomness can be handled by conditioning on the entire sequence. If future weights adapt to publicly revealed candidate choices, that conditioning argument is no longer valid without further analysis. The space bound counts stored values; their exact bit lengths may grow.

Method. The candidate represents all past weight through a probability invariant. A new item competes against the entire accumulated weight, not merely against the weight of the current representative.

Example CA02. The Median of Two Sorted Arrays

Two nondecreasing real arrays A, B have lengths m, n with $m + n > 0$. Random access is allowed, but fully merging the arrays is not. Find their combined median, averaging the central two elements when the total length is even. Give an $O(\log(\min\{m, n\} + 2))$-time, $O(1)$-extra-space algorithm covering empty arrays, ties, and very unequal lengths.

Solution. Swap the arrays if necessary so that $m \le n$. If $m = 0$, read the central one or two elements of B directly. Otherwise put

$$h = \left\lfloor \frac{m+n+1}{2} \right\rfloor.$$

Take i elements from the beginning of A and $j = h - i$ from the beginning of B as the left half. The feasible search range is

$$\max(0, h-n) \le i \le \min(m, h).$$

The four boundary elements are

$$L_A = A_{i-1}\ (i > 0), \quad R_A = A_i\ (i < m),$$
$$L_B = B_{j-1}\ (j > 0), \quad R_B = B_j\ (j < n).$$

Use $-\infty$ for a missing left boundary and $+\infty$ for a missing right boundary; these sentinels are used only in comparisons.

A partition is correct exactly when $L_A \le R_B$ and $L_B \le R_A$. Internal sortedness then guarantees that every left-half element is at most every right-half element. If $L_A > R_B$, too many elements came from A, so decrease i. Larger i cannot work, since L_A is nondecreasing and R_B nonincreasing as i grows. If $L_B > R_A$, increase i by the symmetric argument. Binary search therefore applies.

A correct partition exists: conceptually sort all elements, using any fixed stable order for ties. The first h elements form a prefix of each original array. Every elimination preserves at least one correct partition, so the search terminates. For odd $m + n$, return $\max(L_A, L_B)$. For even $m + n$, return

$$\frac{\max(L_A, L_B) + \min(R_A, R_B)}{2}.$$

In the even, nonempty case both halves contain elements, so the final average uses finite data values. Ties may permit several partitions but all yield the same median. There are at most $m + 1$ candidate integers, proving the time and space bounds.

Method. Search for the number of elements contributed by the shorter array to the left half. Monotonicity of the boundary tests replaces the need to merge the data.

Example CA03. Inversions and Minimum Adjacent Swaps

A real array $a_1, \ldots, a_n$ may contain ties. An inversion is a pair $i < j$ with $a_i > a_j$. Count inversions in $O(n \log n)$ time and prove that their number equals the minimum number of adjacent swaps needed to make the array nondecreasing. Evaluate $(3, 1, 2, 1)$ and explain how ties are handled.

Solution. Augment merge sort with an inversion count. Recursively sort each half and count its internal inversions. During merging, output the left element whenever it is at most the right element. Otherwise output the right element and add the number of unconsumed left elements: all of them are strictly larger. Each cross-half inversion is counted exactly when its right element is output first, so none is missed or duplicated.

Each merge level costs $O(n)$ and there are $O(\log n)$ levels. A merge buffer uses $O(n)$ additional space. Empty and one-element arrays have count zero. Taking the left element first on equality avoids counting ties as inversions. The counter must accommodate values up to $n(n-1)/2$.

For optimality of the swap count, an adjacent swap leaves the total contribution of every other element unchanged: that element lies on the same side of both swapped positions. Only the swapped pair can change its inversion status, so one swap decreases the total count by at most one. Any sorting sequence therefore needs at least the initial inversion count in swaps.

Conversely, any unsorted array contains a strictly decreasing adjacent pair; otherwise all adjacent comparisons would already imply sortedness. Swapping such a pair decreases the inversion count by exactly one. Repeating sorts the array in exactly the initial inversion count, attaining the lower bound. For the example, the inverted index pairs are

$$(1,2),(1,3),(1,4),(3,4),$$

so both the inversion count and the minimum adjacent-swap count are 4.

Method. Divide and conquer assigns each pair a unique counting step. The same count is a potential function that certifies the optimal number of swaps.

5.4 Exercises

Key Ideas and Methods. These exercises test what can be recovered from a summary, when a window is feasible, and why candidates may be discarded. For variance, distinguish the population denominator from the degrees-of-freedom correction. For interval problems, specify indices, duration, and tie rules. A feasible interval establishes only feasibility; optimality also requires ruling out a better one. A linear-time argument must count how often each candidate is processed over the entire scan.

AG4-1. Recovering a Merged Group

Two disjoint groups have a combined count of 5, mean 4, and centered sum of squares 26. Group A is $(1, 4, 7)$. Without reading group B, recover its count, mean, and centered sum of squares, and both combined variance conventions. Can B's two observations be recovered uniquely?

AG4-2. Endpoints of a Longest Stable Interval

For $(4, 2, 3, 5, 8, 7, 6)$, find the longest interval whose range is at most 3. Use zero-based indices and the earliest start among ties. Give the result and a short independent argument excluding longer intervals.

AG4-3. Which Peak Does the Duration Limit Exclude?

Prices are $(120, 80, 100, 70, 90)$. Find maximum drawdown with duration at most 2 and with no duration limit, giving zero-based endpoints. Explain why the historical maximum cannot be used directly for the first problem.

AG4-4. The Shortest Time to a Quarter Drawdown

Prices are $(100, 94, 80, 79, 60)$ and the threshold is $d = 1/4$. Find the shortest qualifying drawdown interval. Exclude shorter possibilities and explain why a successful starting index can be discarded permanently during an end scan.

6

Mixed Exercises

6.1 Worked Examples

Key Ideas and Methods. A mixed problem may become simpler after changing what is counted or optimized. The comparison count in randomized quicksort is a sum of indicators for pairs of keys being compared; these indicators need not be independent for linearity of expectation. Convexity and one-sided derivatives characterize minimizers of asymmetric losses, including distributions with atoms. For optimization involving infinite sums, establish a finite truncation bound before justifying the limiting step.

After finding a candidate, prove its feasibility, show that it attains the bound, and identify the equality conditions. An average-performance bound does not guarantee the same worst-case behavior, and existence of an optimum does not imply uniqueness.

Example CX01. Comparisons in Randomized Quicksort

Randomized quicksort sorts n distinct keys. In each nonempty subproblem, it independently chooses a uniform pivot, compares it once with every other key in that subproblem, then recurses on the two sides. Let C_n count all comparisons. Find $\mathbb{E}C_n$ exactly and explain why expected $O(n \log n)$ work does not mean that every run has this order of cost.

Solution. Number the keys by increasing rank, $1, \ldots, n$. Let I_{ij} indicate whether ranks $i < j$ are ever compared. Every comparison involves a pivot, which is then removed from future subproblems. A pair can therefore be compared at most once, and

$$C_n = \sum_{1 \le i < j \le n} I_{ij}.$$

Consider the first pivot selected from ranks $\{i, i+1, \ldots, j\}$. Pivots outside this interval cannot separate i from j. If the first pivot inside is an interior rank, it separates the two permanently. If it is either endpoint, the endpoints are compared then. Hence

$$\mathbb{P}(I_{ij} = 1) = \frac{2}{j - i + 1}.$$

Uniformity of the first pivot can be seen through an equivalent implementation: give all keys independent continuous random priorities and always choose the

lowest-priority key in a subproblem. This has the same pivot law, and each rank in the interval is equally likely to have its lowest priority.

Linearity of expectation does not require the indicators to be independent. Grouping by $d = j - i$ and writing $H_n = \sum_{k=1}^{n} 1/k$ gives

$$\begin{aligned}\mathbb{E}C_n &= 2\sum_{d=1}^{n-1} \frac{n-d}{d+1} \\ &= 2(n+1)(H_n - 1) - 2(n-1) \\ &= \boxed{2(n+1)H_n - 4n}.\end{aligned}$$

For $n = 1$ the formula yields zero; define $C_0 = 0$ separately. For example, $\mathbb{E}C_4 = 29/6$.

If every pivot happens to be the smallest key in its subproblem, the cost is $(n-1) + \cdots + 1 = n(n-1)/2$. This event has positive probability $1/n!$. The expected order thus averages over random runs and is not a worst-case bound for each run.

Method. Ask when a particular pair of keys meets, rather than tracking the whole random recursion tree. Correlated pair indicators can still be added in expectation.

Example CX02. Balancing Shortage and Excess Inventory

Nonnegative demand D has finite mean and distribution function F. A real order quantity $q \geq 0$ incurs shortage cost $c_u > 0$ per unit and excess cost $c_o > 0$ per unit. Minimize

$$L(q) = c_u\mathbb{E}(D-q)_+ + c_o\mathbb{E}(q-D)_+.$$

Without assuming a continuous demand distribution, characterize every minimizer. For $\mathbb{P}(D = 0, 1, 2) = (1/4, 1/2, 1/4)$, find all minimizers when $(c_u, c_o) = (3, 1)$ and when $(c_u, c_o) = (2, 1)$.

Solution. For each fixed demand, the loss is a convex, piecewise linear function of q, so its expectation is convex. One-sided difference quotients are bounded by $\max(c_u, c_o)$, allowing their limits inside the expectation. Thus

$$\begin{aligned}L'_-(q) &= (c_u + c_o)F(q-) - c_u, \\ L'_+(q) &= (c_u + c_o)F(q) - c_u.\end{aligned}$$

At $q > 0$, convexity makes nonpositive left derivative and nonnegative right derivative necessary and sufficient. Writing $\tau = c_u/(c_u + c_o)$, the full minimizer set is

$$\boxed{\{q \geq 0 : F(q-) \leq \tau \leq F(q)\}}.$$

At the boundary $q = 0$, only the right derivative must be nonnegative. Since nonnegative demand implies $F(0-) = 0 \leq \tau$, the same formula remains valid. Extending the loss formally to $q < 0$ makes it strictly decreasing there, so the constraint loses no additional candidate.

A minimizer exists: $\tau \in (0,1)$, and the finite generalized quantile $\inf\{x : F(x) \geq \tau\}$ satisfies the displayed condition. If F is flat at level τ, an entire interval can minimize the loss; uniqueness must not be assumed.

For the discrete example and costs $(3,1)$, $\tau = 3/4$. The distribution function jumps to $3/4$ at 1 and remains there until 2, yielding all minimizers $[1,2]$, each with expected loss 1. For costs $(2,1)$, $\tau = 2/3$ lies strictly inside the jump at 1. The unique minimizer is then $q = 1$, with minimum loss $3/4$.

Method. Asymmetric absolute loss selects a cost-weighted quantile. The mean is generally not optimal, and atoms or gaps in the distribution determine whether the optimal action is unique.

Example CX03. Maximum Entropy with a Fixed Mean

A nonnegative integer-valued random variable N has mean $\mu > 0$. With natural logarithms, define

$$H(N) = -\sum_{k=0}^{\infty} p_k \log p_k, \qquad 0 \log 0 = 0.$$

Prove a finite entropy bound and identify the unique maximizing distribution and its entropy. You may use the log-sum inequality for finite distributions, but must handle the infinite support without assuming finite entropy in advance.

Solution. Put $r = \mu/(1+\mu) \in (0,1)$ and consider

$$q_k = (1-r)r^k, \qquad k \geq 0.$$

This geometric distribution has mean $r/(1-r) = \mu$, and its entropy is

$$H(q) = -\log(1-r) - \mu \log r = (1+\mu)\log(1+\mu) - \mu \log \mu.$$

For a rigorous general bound, retain states $0, \ldots, M-1$ separately and merge $N \geq M$ into one tail state of probability t_M. The reference distribution assigns probability r^M to this tail. The finite log-sum inequality bounds the grouped entropy by

$$H_M \leq -\sum_{k<M} p_k \log q_k - t_M \log(r^M).$$

The right-hand side is

$$-(1-t_M)\log(1-r) - \log r \left(\sum_{k<M} k p_k + M t_M \right).$$

As $M \to \infty$, $t_M \to 0$ and the parenthesized quantity is $\mathbb{E}\min(N, M) \uparrow \mu$. Also $H_M = -\sum_{k<M} p_k \log p_k - t_M \log t_M$ tends to $H(N)$, even if the latter is initially allowed to be infinite, since $-t \log t \to 0$. Consequently

$$H(N) \leq (1+\mu)\log(1+\mu) - \mu \log \mu < \infty.$$

To prove uniqueness, the difference between the grouped cross-entropy bound and H_M is the grouped relative entropy $D_M \geq 0$. Its limit is $H(q) - H(N)$. If

$p \neq q$, select j with $p_j \neq q_j$. For every $M > j$, merge all groups other than state j. The log-sum inequality gives

$$D_M \geq p_j \log \frac{p_j}{q_j} + (1 - p_j) \log \frac{1 - p_j}{1 - q_j} > 0.$$

This positive lower bound does not depend on M, excluding equality. Thus the geometric distribution uniquely maximizes entropy.

Method. On an infinite state space, first aggregate the tail and then pass to the limit. This avoids presupposing finite entropy when manipulating relative entropy.

6.2 Exercises

Key Ideas and Methods. Translate each constraint into a structure that can be computed. Congruence conditions suggest residue classes or roots-of-unity filters. Rotational symmetry can reduce conditional moments to averages over a sphere, with attention to the sampling measure. Patterns in repeated trials can be tracked by finite states recording the relevant suffix. Variance calculations must retain covariance terms caused by shared variables or dependence over time.

Integer problems often require separating prime exponents from the remaining factors. Division in modular arithmetic is valid only when the divisor is invertible. Asymptotic counts must account for boundary terms and repeated entries. Finite enumeration can check small cases; a general conclusion still needs a recurrence, identity, or invariant that proves it.

AA01. Local Independence with a Global Parity Constraint

Generate $n - 1$ independent fair bits, $n \geq 3$, and set B_n equal to their sum modulo 2. Prove that every set of $n - 1$ bits is independent and fair, whereas all n bits are not independent. Find the probability that they are all zero.

AA02. The Number of Successes Modulo Three

Toss $n \geq 0$ fair coins independently and let S count heads. Find $\mathbb{P}(S \equiv r \pmod 3)$ for $r = 0, 1, 2$. You may use $\omega = e^{2\pi i/3}$, but justify the root-of-unity filter. Evaluate $n = 8, r = 0$.

AA19. Return Probability under Adjacent Swaps

Three distinct cards start in order 123. At each step, independently swap positions 1 and 2 or positions 2 and 3 with equal probability. Find the probability of returning to 123 after $t \geq 0$ steps and explain why it has no single long-time limit.

AA22. Isolated Marks in a Circular Arrangement

Randomly arrange m A's and n B's on $N = m+n \geq 3$ labeled circular positions, uniformly over type arrangements. A B is isolated if both neighbors are A. Find the expected isolated-B count, then the corresponding expectation on a line counting only nonendpoint centers. Evaluate $m = 4, n = 3$.

AA03. A Projection of a Uniform Disk Point

Choose (X, Y) uniformly by area in a disk of radius $R > 0$. Find the density of X, $\mathbb{E}|X|$, and $\operatorname{Var} X$, and explain why X is not uniform on $[-R, R]$.

AA07. Linear Separation of Two Classes on a Circle

Choose $m + n$ independent uniform points on the unit circle. Label the first $m \geq 2$ as A and the remaining $n \geq 2$ as B. Find the probability that a line strictly separates all A points from all B points. Evaluate $m = n = 3$.

AA09. A Gaussian Ratio in the Positive Quadrant

Let X, Y be independent standard normals. Conditional on $X > 0, Y > 0$, define $Z = X/Y$. Find its distribution, density, and median, and decide whether its conditional mean is finite.

AA10. Dependence of Squared Coordinates at Fixed Energy

Let X, Y, Z be independent standard normals and $S = X^2 + Y^2 + Z^2$. Find $\mathbb{E}[X^2Y^2 \mid S]$, $\operatorname{Var}(X^2 \mid S)$, and $\operatorname{Cov}(X^2, Y^2 \mid S)$. Derive the fourth moments from rotation symmetry instead of citing a sphere-moment formula.

AA13. The Closest Gap among Three Arrival Times

Three arrival times are independent and uniform on $[0, L]$, with $L > 0$. Let D be the smallest time difference between any two arrivals. Find its complete distribution function and mean; distances to the interval endpoints are not included.

AA16. The Minimum and Range of an Exponential Sample

Let $X_1, \ldots, X_n$ be independent exponentials of rate $\lambda > 0$, with $n \geq 2$. Put $U = \min_i X_i$, $V = \max_i X_i$, and $D = V - U$. Prove that U, D are independent, find the distribution of D, and compute $\mathbb{E}[V \mid U]$.

AA17. Correlation after Squaring Gaussian Variables

Let (X, Y) be jointly standard normal with correlation $\rho \in [-1, 1]$. Find $\mathbb{E}[Y^2 \mid X]$, $\operatorname{Cov}(X^2, Y^2)$, and $\operatorname{Corr}(X^2, Y^2)$. Can the squared variables' correlation identify the original sign?

AA20. Reciprocal Distance to a Uniform Sphere Point

In three dimensions, let X be uniform by area on the sphere of radius $R > 0$. A fixed point a is distance $r \geq 0$ from the center, and $D = |X - a|$. Find $\mathbb{E}(1/D)$. For $r > 0$, also find $\mathbb{E}(1/D^2)$ and analyze $r = R$.

AA06. The First Repeated Adjacent Outcome

Independent trials succeed with probability $p \in (0, 1)$; put $q = 1 - p$. Starting before any trials, stop when two successive outcomes first agree. Find the mean total trial count and the probability of stopping on two successes.

AA08. Variance of an Adjacent Pattern Count

Toss a coin with head probability $p \in (0,1)$ independently $n \geq 2$ times. Let N count adjacent occurrences of HT at all starting positions. Find its mean and variance, and evaluate $n = 10$, $p = 1/2$.

AA12. Temporal Correlation along a Cycle Walk

Consider a symmetric nearest-neighbor walk on a cycle of $n \geq 3$ vertices, with X_0 uniform over the vertices. Put $Y_t = \cos(2\pi X_t/n)$ and $S_T = \sum_{t=0}^{T-1} Y_t$, $T \geq 1$. Find $\mathrm{Cov}(Y_0, Y_k)$, $\mathrm{Var}\, S_T$, and $\lim_{T\to\infty} \mathrm{Var}\, S_T/T$.

AA04. Inverting the Trailing-Zero Count

For a nonnegative integer K, find every nonnegative integer n for which $n!$ has exactly K trailing decimal zeros, with $0! = 1$. Avoid computing factorials. Prove that a nonempty solution set consists of five consecutive integers, and solve the cases $K = 24$ and $K = 29$.

AA05. The Last Nonzero Digit of a Binomial Coefficient

Design an algorithm for the last nonzero decimal digit of $\binom{n}{k}$, for integers $0 \leq k \leq n$, without first constructing the full coefficient. Separate powers of 2 and 5 from the unit factors. Explain why direct division modulo 10 is invalid, and evaluate $\binom{20}{7}$.

AA11. Visibility between Random Lattice Points

Choose two distinct points uniformly from $\{0,\dots,M\} \times \{0,\dots,N\}$, with $M, N \geq 1$. They are visible to each other if their connecting segment has no other integer lattice point in its interior. Give a finite-sum formula for the probability and evaluate $M = N = 2$.

AA14. The Asymptotic Number of Integer-Sided Triangles

Let $T(N)$ count positive integer triples $x \le y \le z$ with $x + y + z = N$ and $x + y > z$. Prove $T(N) = N^2/48 + O(N)$ and list all solutions for $N = 12$. Explain why dividing the ordered count by 6 is generally not exact.

AA15. Algebraic Powers Close to Integers

Let $\alpha = (3+\sqrt{5})/2$. Construct an integer recurrence giving the nearest integer L_n to α^n for every $n \ge 1$. Prove uniqueness and find the exact error and parity pattern. Find the smallest n for which the error is below 10^{-6}.

AA21. Fixing Both Digit Sum and Digit Product

Find the smallest positive decimal integer with digit product 72 and digit sum 14. Then change the digit sum to 18 and find the new minimum. Explain why a digit 1 cannot always be deleted in this problem.

AA18. Transcendental Logarithms and an Irrational Ratio

Prove that $\log 2$ is transcendental and that $\log 2/\log 3$ is irrational. Explain why the ratio of two transcendental numbers need not be transcendental. Here log is natural logarithm. You may use the Hermite–Lindemann theorem: if z is nonzero and algebraic, then e^z is transcendental.

Solutions to the Exercises

Probability

PG1-1.

Let E denote at least one head and C the coin type. Conditional on type i, the complement of E is n consecutive tails, so its likelihood is $L_i = 1 - (1 - p_i)^n$. Bayes' formula gives

$$\boxed{w := \mathbb{P}(C = 1 \mid E) = \frac{\pi L_1}{\pi L_1 + (1 - \pi) L_2}.}$$

The next toss is independent of E conditional on the type. Thus

$$\boxed{\mathbb{P}(H_{n+1} \mid E) = w p_1 + (1 - w) p_2.}$$

For $n > 1$, the record combines many sequences and has likelihood L_i, generally not the single-head likelihood p_i. As $n \to \infty$, both likelihoods approach one and the posterior returns to the prior: either coin eventually produces a head with probability approaching one.

PG1-2.

Put $s = \alpha + \beta$ and let E be the conditioning event. The joint density that A rings at u while B has not rung is αe^{-su}. Hence

$$\mathbb{P}(E) = \int_0^t \alpha e^{-su} \, \mathrm{d}u = \frac{\alpha}{s}(1 - e^{-st}),$$

and division gives

$$\boxed{f_{T|E}(u) = \frac{s e^{-su}}{1 - e^{-st}}, \quad 0 < u < t.}$$

Integration by parts yields

$$\mathbb{E}(T \mid E) = \frac{-t e^{-st} + (1 - e^{-st})/s}{1 - e^{-st}} = \boxed{\frac{1}{s} - \frac{t}{e^{st} - 1}}.$$

As $t \to \infty$, the density becomes exponential with rate s and the mean tends to $1/s$. Knowing the winning clock alone does not change the first ringing time; the additional deadline truncates it.

PG1-3.

Put $d = 1 - a - b$. After removing the common multinomial coefficient, the likelihoods under $T = 1$ and $T = 0$ are $a^h b^z d^e$ and $b^h a^z d^e$. Therefore

$$\boxed{O := \frac{\mathbb{P}(T = 1 \mid \text{record})}{\mathbb{P}(T = 0 \mid \text{record})} = \frac{\pi}{1 - \pi} \left(\frac{a}{b}\right)^{h - z}.}$$

The posterior probability is $O/(1+O)$. The factor d^e cancels, so for fixed h, z the number of missing reports is irrelevant in this model.

If missing probabilities under the two truths are d_1, d_0, each missing report instead contributes likelihood ratio d_1/d_0. The probabilities of nonmissing outputs must also be recomputed under the revised model. Missingness can therefore be informative and cannot simply be discarded.

PG1-4.

Let $S = (X+Z)/\sqrt{2}$, $D = (X-Z)/\sqrt{2}$, and $b = a/\sqrt{2}$. The variables S, D are independent standard normals, and selection restricts only $|S| < b$, an event of probability $2\Phi(b) - 1$. The conditional mean is zero by symmetry. Integration by parts gives

$$v := \mathrm{Var}(S \mid |S| < b) = \frac{\int_{-b}^{b} s^2 \phi(s)\,\mathrm{d}s}{2\Phi(b)-1} = 1 - \frac{2b\phi(b)}{2\Phi(b)-1}.$$

The variance of D remains one, and it remains independent of the truncated S. Thus

$$\boxed{\mathrm{Corr}(X, Z \mid |X+Z| < a) = \frac{v-1}{v+1}.}$$

As $b \downarrow 0$, the bound $|S| < b$ implies $0 \le v \le b^2$, so the correlation tends to -1. As $b \to \infty$, truncation disappears, $v \to 1$, and the correlation tends to zero.

PG2-1.

The first draw cannot repeat, and the fourth must repeat, so $T \in \{2, 3, 4\}$. The probability that the first two types differ is

$$\mathbb{P}(T > 2) = 1 - \left[(1/2)^2 + 2(1/4)^2\right] = \frac{5}{8}.$$

Three distinct draws require one of each type in one of 3! orders, giving

$$\mathbb{P}(T > 3) = 6 \cdot \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{3}{16}.$$

Since $\mathbb{P}(T > 0) = \mathbb{P}(T > 1) = 1$, the tail sum yields

$$\boxed{\mathbb{E}T = 1 + 1 + \frac{5}{8} + \frac{3}{16} = \frac{45}{16}, \qquad \mathbb{P}(T = 4) = \frac{3}{16}.}$$

The last equality uses $T \le 4$; it is not the probability of any specified type appearing.

PG2-2.

Each red run has exactly one black-to-red boundary. Any directed circular adjacency has a black start and red end with probability $nm/[N(N-1)]$. Summing over all N boundaries gives

$$\boxed{\mathbb{E}(\text{red runs}) = \frac{mn}{N-1}.}$$

If all red cards are consecutive, specifying the start of the unique red run determines the entire color arrangement. There are N choices, producing distinct labeled arrangements because both colors occur. Since there are $\binom{N}{m}$ arrangements in total,

$$\boxed{\mathbb{P}(\text{one red run}) = \frac{N}{\binom{N}{m}}.}$$

For $m = 1$ or $n = 1$, this probability is one, as it should be.

PG2-3.

For each triple A, let I_A indicate that its objects share a bucket. Then $\mathbb{E}I_A = s_3$, so $\mathbb{E}C = \binom{m}{3}s_3$. Two disjoint triples are independent. Triples sharing two objects collide together exactly when all four objects share a bucket, giving covariance $s_4 - s_3^2$. If they share one object, all five must share a bucket, giving covariance $s_5 - s_3^2$.

There are $6\binom{m}{4}$ unordered pairs sharing two objects: choose their common pair among four objects. There are $15\binom{m}{5}$ pairs sharing one object: choose that object in five ways, then partition the remaining four into two unordered pairs in three ways. Each unordered pair contributes twice to the variance expansion. Thus

$$\boxed{\begin{aligned} \mathbb{E}C &= \binom{m}{3}s_3, \\ \operatorname{Var} C &= \binom{m}{3}s_3(1 - s_3) + 12\binom{m}{4}(s_4 - s_3^2) \\ &\quad + 30\binom{m}{5}(s_5 - s_3^2). \end{aligned}}$$

With $\binom{m}{r} = 0$ for $r > m$, the formula includes $m = 3, 4$.

PG2-4.

Let Y_i equal 1 if ordinary card i precedes every target, -1 if it follows every target, and zero otherwise. Reversing the permutation gives $\mathbb{E}Y_i = 0$. Among this card and the K targets, each extreme position has probability $1/(K + 1)$, so $\mathbb{E}Y_i^2 = 2/(K + 1)$.

For $i \neq j$, the probability that both precede all targets is $2/[(K+1)(K+2)]$, as is the probability that both follow them. The probability that specified i is before and j after is $1/[(K + 1)(K + 2)]$, and likewise in reverse. Hence

$$\mathbb{E}(Y_iY_j) = \frac{2}{(K + 1)(K + 2)}.$$

Expanding $D = \sum_i w_iY_i$ and using $2\sum_{i<j} w_iw_j = (\sum_i w_i)^2 - \sum_i w_i^2$ gives

$$\boxed{\mathbb{E}D = 0, \qquad \operatorname{Var} D = \frac{2[(K + 1)\sum_i w_i^2 + (\sum_i w_i)^2]}{(K + 1)(K + 2)}.}$$

The algebra does not require positive weights. For $m = 0$, all sums and the variance are zero.

PG3-1.

The simplex volume gives the waiting-time tail:

$$\mathbb{P}(N > k) = \mathbb{P}(U_1 + \cdots + U_k \le 1) = \frac{1}{k!}, \quad k \ge 0.$$

Thus $\mathbb{E}N = \sum_{k\ge 0} 1/k! = e$. Writing the cost as a nonnegative series permits interchange of expectation and summation:

$$\mathbb{E}C = \sum_{j\ge 1} \mathbb{E}[U_j^2 \mathbf{1}\{N \ge j\}].$$

The event $\{N \ge j\}$ says that the first $j-1$ increments sum to at most one. It depends only on the past and is independent of the fresh U_j. Each term is therefore $\mathbb{E}U_j^2 \mathbb{P}(N \ge j) = \mathbb{P}(N \ge j)/3$, so

$$\boxed{\mathbb{E}C = \frac{1}{3}\mathbb{E}N = \frac{e}{3}.}$$

The stopping time does depend on the increment sequence. Only the decision whether draw j occurs is independent of that draw's as-yet unseen increment.

PG3-2.

Put $t = -\log x$ and $b = \log 2$. The negative log-length is $nb + G_n$, where $G_n = \sum_{i=1}^{n}(-\log U_i)$ has density $s^{n-1}e^{-s}/(n-1)!$. Since $G_n > 0$ almost surely, crossing has occurred whenever $nb \ge t$. Thus $N \le L := \lceil t/b \rceil$ almost surely.

For $n \ge 1$, set $a_n = t - nb$. If $a_n \le 0$, the tail is zero; otherwise,

$$\boxed{\mathbb{P}(N > n) = \mathbb{P}(G_n \le a_n) = 1 - e^{-a_n} \sum_{j=0}^{n-1} \frac{a_n^j}{j!}.}$$

The final expression follows by repeated integration by parts of the gamma density. Including $\mathbb{P}(N > 0) = 1$ gives

$$\boxed{\mathbb{E}N = 1 + \sum_{n=1}^{L-1} \left[1 - e^{-(t-nb)} \sum_{j=0}^{n-1} \frac{(t-nb)^j}{j!} \right].}$$

For $x = 1/4$, $L = 2$. Crossing on the first draw means $U_1/2 < 1/4$, which has probability $1/2$; otherwise crossing occurs on the second. Hence $\mathbb{P}(N = 1) = \mathbb{P}(N = 2) = 1/2$ and $\mathbb{E}N = 3/2$.

PG3-3.

For $0 < s < 1$, summing the densities of all positive partial sums gives $u(s) = \sum_{k\ge 1} s^{k-1}/(k-1)! = e^s$. A first entry into $(a, b]$ at s may follow $S_0 = 0$ or a

positive partial sum $t \le a$. Since $s - t \in (0, 1)$, the next-increment density is one. The first-entry density is therefore

$$1 + \int_0^a e^t \, \mathrm{d}t = e^a, \quad a < s \le b.$$

Put $d = b - a$. Exactly $k \ge 1$ entries have ordered positions $a < s_1 < \cdots < s_k \le b$. Internal transition densities are one, and the probability of jumping past b after s_k is $1 - b + s_k$. For $s_k = a + v$, the earlier $k - 1$ positions have simplex volume $v^{k-1}/(k-1)!$. Consequently,

$$\begin{aligned} \mathbb{P}(K = k) &= \frac{e^a}{(k-1)!} \int_0^d v^{k-1}(1 - d + v) \, \mathrm{d}v \\ &= \boxed{e^a \left(\frac{d^k}{k!} - \frac{d^{k+1}}{(k+1)!} \right)}. \end{aligned}$$

Summing over $k \ge 1$ telescopes to $e^a d$, so $\boxed{\mathbb{P}(K = 0) = 1 - e^a d}$. This is nonnegative because $d \le 1 - a$ and $e^a(1 - a) \le 1$. Finally, integrate the renewal density to obtain

$$\boxed{\mathbb{E}K = \int_a^b u(s) \, \mathrm{d}s = e^b - e^a.}$$

PG3-4.

Future shrinkage factors are independent of the observed history $\mathcal{F}_n$. The normalized representation in Problem P24 gives

$$Q = Q_n + (1 - R_n)D',$$

where D' is independent of $\mathcal{F}_n$, with mean one and variance $1/2$. Hence

$$\boxed{\widehat{Q} = \mathbb{E}(Q \mid \mathcal{F}_n) = Q_n + 1 - R_n,} \qquad \boxed{\operatorname{Var}(Q \mid \mathcal{F}_n) = \frac{(1 - R_n)^2}{2}.}$$

For any square-integrable predictor Z based on the history,

$$\mathbb{E}(Q - Z)^2 = \mathbb{E} \operatorname{Var}(Q \mid \mathcal{F}_n) + \mathbb{E}(\widehat{Q} - Z)^2.$$

The cross term vanishes by conditional expectation, proving optimality of $\widehat{Q}$. Since $1 - R_n = (1 - x) \prod_{i=1}^n U_i$,

$$\boxed{\mathbb{E}(Q - \widehat{Q})^2 = \frac{(1 - x)^2}{2 \cdot 3^n}.}$$

The partial sum omits a positive tail with conditional mean $1 - R_n$. Adding that mean removes the predictable bias.

PG4-1.

Introduce $S = U + V$. Then $U = WS$, $V = (1 - W)S$, with Jacobian s. The unit-square constraints become

$$0 < w < 1, \qquad 0 < s < \frac{1}{\max(w, 1 - w)}.$$

Integrating out s gives

$$\boxed{f_W(w) = \int_0^{1/\max(w,1-w)} s \, \mathrm{d}s = \frac{1}{2\max(w, 1 - w)^2}, \quad 0 < w < 1.}$$

The density is not constant, so the distribution is not uniform. Exchanging U, V replaces W by $1 - W$, implying $\mathbb{E}W = 1/2$. For the second moment,

$$\begin{aligned}\mathbb{E}W^2 &= \frac{1}{2}\int_0^{1/2} \frac{w^2}{(1 - w)^2} \, \mathrm{d}w + \frac{1}{2}\int_{1/2}^1 1 \, \mathrm{d}w \\ &= \left(\frac{3}{4} - \log 2\right) + \frac{1}{4} = 1 - \log 2.\end{aligned}$$

Thus $\boxed{\operatorname{Var} W = 3/4 - \log 2}$. Symmetry of the two shares determines the mean, not a constant density.

PG4-2.

For $0 < l < h < 1$, the joint density of the extremes is $n(n - 1)(h - l)^{n-2}$: select the minimum and maximum observations, then place the other $n-2$ between them. Set $r = h - l$ and $c = (l + h)/2$. The inverse Jacobian is one and $0 < r < 2\min(c, 1 - c)$. Hence

$$\boxed{f_C(c) = \int_0^{2\min(c,1-c)} n(n - 1)r^{n-2} \, \mathrm{d}r = n[2\min(c, 1 - c)]^{n-1}, \quad 0 < c < 1.}$$

Symmetry about $1/2$ gives $\mathbb{E}C = 1/2$. Using that symmetry and substituting $u = 2c$,

$$\begin{aligned}\operatorname{Var} C &= 2\int_0^{1/2} (1/2 - c)^2 n(2c)^{n-1} \, \mathrm{d}c \\ &= \frac{n}{4}\int_0^1 (1 - u)^2 u^{n-1} \, \mathrm{d}u = \boxed{\frac{1}{2(n + 1)(n + 2)}}.\end{aligned}$$

The last integral follows by expanding $(1-u)^2$, giving $1/n-2/(n+1)+1/(n+2)$.

PG4-3.

Let the spacings be $D_1, \ldots, D_n$. Given them, the random point falls in spacing i with probability D_i, not $1/n$. For any nonnegative function g,

$$\mathbb{E}g(L) = \mathbb{E}\sum_i D_i g(D_i) = n\mathbb{E}[D_1 g(D_1)].$$

One spacing has tail $\mathbb{P}(D_1 > l) = (1-l)^{n-1}$ and density $(n-1)(1-l)^{n-2}$. Therefore

$$\boxed{f_L(l) = n(n-1)l(1-l)^{n-2}, \quad 0 < l < 1.}$$

This is the Beta$(2, n-1)$ density. Beta integration gives

$$\boxed{\mathbb{E}L = \frac{2}{n+1}, \qquad \mathbb{E}(1/L) = n(n-1)\int_0^1 (1-l)^{n-2}\,\mathrm{d}l = n.}$$

Choosing a spacing uniformly by index gives mean $\mathbb{E}(\sum_i D_i/n) = 1/n$, strictly smaller than $2/(n+1)$. A random point is more likely to select a longer spacing.

PG4-4.

The vector's direction is uniform on the circle. Projections along two directions separated by θ have opposite signs in two sectors of total angle 2θ. Thus $\mathbb{P}(D_1 = 1) = \mathbb{P}(D_2 = 1) = \theta/\pi$.

Let V_0, V_1, V_2 be the actual projection values. A trigonometric identity gives

$$V_1 = \frac{V_0 + V_2}{2\cos\theta}.$$

Since $\cos\theta > 0$, equal signs of the two outer projections force the middle sign to agree. Two successive sign changes are therefore impossible: $\mathbb{P}(D_1 = D_2 = 1) = 0$. Zero projections have probability zero. Combining with the marginals,

	$D_2 = 0$	$D_2 = 1$
$D_1 = 0$	$1 - 2\theta/\pi$	θ/π
$D_1 = 1$	θ/π	0

Consequently,

$$\boxed{\mathrm{Cov}(D_1, D_2) = -\frac{\theta^2}{\pi^2}, \qquad \mathbb{P}(A = B = C) = 1 - \frac{2\theta}{\pi}.}$$

The adjacent comparisons share a middle projection, so their sign changes are not independent.

Section Exercises

OP02.

Write Z for B's head count in the first $2n$ tosses and B for the indicator of a head on the final toss. Then $Y = Z + B$, with B independent and equally likely to be 0 or 1. Put $D = Z - 2X$. Complement every outcome in A's tosses and B's first $2n$ tosses. This preserves probability and maps

$$D \longmapsto (2n - Z) - 2(n - X) = -D.$$

Thus $\mathbb{P}(D > 0) = \mathbb{P}(D < 0)$. A positive D guarantees success; a negative D cannot be made strictly positive by adding $B \le 1$; and $D = 0$ gives success exactly when $B = 1$. Consequently,

$$\mathbb{P}(Y > 2X) = \mathbb{P}(D > 0) + \frac{1}{2}\mathbb{P}(D = 0) = \boxed{\frac{1}{2}}.$$

The symmetry is that of the difference $Z - 2X$; it does not assert that Z and $2X$ have the same distribution.

OP30.

Let F mean that the double-headed coin was selected, and let H denote two heads. The prior probabilities and likelihoods are

$$\mathbb{P}(F) = \mathbb{P}(F^c) = \frac{1}{2}, \qquad \mathbb{P}(H \mid F) = 1, \quad \mathbb{P}(H \mid F^c) = \frac{1}{4}.$$

The law of total probability gives $\mathbb{P}(H) = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} = \frac{5}{8}$. Bayes' formula therefore yields

$$\mathbb{P}(F \mid H) = \frac{\mathbb{P}(F)\mathbb{P}(H \mid F)}{\mathbb{P}(H)} = \frac{1/2}{5/8} = \boxed{\frac{4}{5}}.$$

Both observations come from the same selected coin, so the coin type must not be resampled between tosses.

OP18.

Regard each connected component that has not yet closed as an open chain. Initially there are n such chains, each with two free ends. With k open chains there are $2k$ free ends. Joining the two ends of one chain closes a loop; joining ends of different chains merges them. Either operation reduces the number of open chains by one. The process therefore visits $k = n, n-1, \dots, 1$ exactly once each.

Of the $\binom{2k}{2}$ possible endpoint pairs, exactly k join the two ends of the same chain. Thus the conditional probability of creating a loop at that step is always

$$\frac{k}{\binom{2k}{2}} = \frac{1}{2k-1},$$

regardless of the previous chain configuration. If I_k indicates a loop at that step, the final loop count is $C = \sum_{k=1}^{n} I_k$. Linearity of expectation gives

$$\boxed{\mathbb{E}C = \sum_{k=1}^{n} \frac{1}{2k-1}.}$$

Independence of the loop indicators is not needed for this expectation.

OP25.

The four endpoints of any two chords are distinct almost surely. Fix their four unlabeled positions. Independent identical sampling makes every assignment of endpoint labels to these positions equally likely. The predetermined pairing therefore induces each of the three unordered pairings of these positions with equal probability. Only the pairing that joins alternating endpoints produces an interior crossing. Hence each pair of chords intersects with probability $1/3$.

Let I_{ij} indicate an interior crossing of chords i and j. Coincident endpoints, coincident chords, and triple intersections have probability zero under continuous sampling. For example, after fixing an intersection of two chords and one endpoint of a third chord, at most one position for its other endpoint makes it pass through that intersection. Thus the number of distinct crossings is almost surely

$$K = \sum_{1 \le i < j \le n} I_{ij}.$$

Linearity of expectation, without any independence assumption on the indicators, gives

$$\boxed{\mathbb{E}K = \binom{n}{2}\frac{1}{3} = \frac{n(n-1)}{6}.}$$

The sampling rule is essential: a random chord here is defined by two independent uniform endpoints.

OP26.

Count the non-aces left after the last ace. For each non-ace i, let I_i indicate that it appears after all four aces. The relative order of that card and the four aces is uniform, so the specified non-ace is last among those five cards with probability $1/5$. Hence $\mathbb{E}I_i = 1/5$.

There are 48 non-aces. If R is the number of cards after the last ace, then

$$R = \sum_{i=1}^{48} I_i, \qquad \mathbb{E}R = \frac{48}{5}.$$

Since $T + R = 52$, the answer is

$$\boxed{\mathbb{E}T = 52 - \frac{48}{5} = \frac{212}{5} = 42.4.}$$

This is the stopping position including the fourth ace. The number of cards strictly before that ace would be smaller by one.

OP22.

While the pattern is still valid and the current value is x, let $f(x)$ be the expected number of additional draws when the next comparison must be an increase;

let $g(x)$ be the corresponding expectation when a decrease is required. One draw is always made, and continuation occurs only if that comparison succeeds. Thus

$$f(x) = 1 + \int_x^1 g(y)\,\mathrm{d}y, \qquad g(x) = 1 + \int_0^x f(y)\,\mathrm{d}y.$$

The expectations are finite. If the first n values obey the pattern, every disjoint pair $(U_1, U_2), (U_3, U_4), \ldots$ must increase, so the survival probability is at most $2^{-\lfloor n/2 \rfloor}$. The same geometric bound controls the remaining draws from any state and justifies the integral equations.

Differentiating gives $f' = -g$ and $g' = f$, hence $f'' + f = 0$. The boundary conditions $f(1) = 1$ and $g(0) = 1$, or $f'(0) = -1$, yield

$$f(x) = A\cos x - \sin x, \qquad A = \frac{1 + \sin 1}{\cos 1} = \sec 1 + \tan 1.$$

After the first draw, the next comparison must be an increase. Therefore,

$$\begin{aligned} \mathbb{E}T &= 1 + \int_0^1 f(x)\,\mathrm{d}x \\ &= A\sin 1 + \cos 1 = \boxed{\sec 1 + \tan 1}. \end{aligned}$$

The trigonometric arguments are in radians.

OP24.

Start more generally from $R_0 = x \in [0, 1)$. The process can be represented as

$$1 - R_k = (1 - x)\prod_{j=1}^{k} U_j, \qquad U_j \overset{\text{iid}}{\sim} \mathrm{Unif}(0, 1).$$

The finite products $Q_n = \prod_{k=1}^{n} R_k$ decrease and lie in $[0, 1]$, so they converge to a limit Q. Set $g(x) = \mathbb{E}[Q \mid R_0 = x]$ and define $g(1) = 1$. Bounded convergence makes the requested limit equal to $g(0)$.

The right boundary condition needs justification. The inequality $0 \le 1 - \prod a_k \le \sum(1 - a_k)$ gives

$$0 \le 1 - g(x) \le \sum_{k=1}^{\infty} \mathbb{E}(1 - R_k) = (1 - x)\sum_{k=1}^{\infty} 2^{-k} = 1 - x.$$

Hence $g(x) \to 1$ as $x \uparrow 1$; the solution identically zero on $[0, 1)$ is excluded.

Conditioning on the first draw, with bounded convergence for the subsequent infinite product, gives

$$(1 - x)g(x) = \int_x^1 t g(t)\,\mathrm{d}t.$$

The integral of a bounded function first shows that g is continuous on $[0, 1)$, after which differentiation is valid. We obtain

$$-g(x) + (1 - x)g'(x) = -xg(x), \qquad g'(x) = g(x).$$

Thus $g(x) = Ce^x$. The established condition $g(1-) = 1$ implies $C = e^{-1}$, and therefore

$$\boxed{\lim_{n\to\infty} \mathbb{E}\prod_{k=1}^{n} R_k = e^{-1}.}$$

OP35.

Conditional on $H = h$, the remaining $n-1$ observations, before sorting, behave as independent uniform draws on $(0, h)$. Thus, for $0 \le t \le h$,

$$\mathbb{P}(L > t \mid H = h) = \left(1 - \frac{t}{h}\right)^{n-1}.$$

Integrating the conditional survival function yields

$$\mathbb{E}[L \mid H = h] = \int_0^h \left(1 - \frac{t}{h}\right)^{n-1} \mathrm{d}t = \frac{h}{n}.$$

It follows that $\mathbb{E}L = \mathbb{E}H/n$ and

$$\mathbb{E}[LH] = \mathbb{E}[H\mathbb{E}(L \mid H)] = \frac{1}{n}\mathbb{E}H^2, \qquad \mathrm{Cov}(L, H) = \frac{1}{n}\,\mathrm{Var}(H).$$

Replacing every observation by $1 - X_i$ preserves the joint distribution and changes the minimum to $1 - H$. Hence $L \overset{d}{=} 1 - H$, so $\mathrm{Var}(L) = \mathrm{Var}(H) > 0$. Consequently,

$$\boxed{\mathrm{Corr}(L, H) = \frac{1}{n}.}$$

Conditioning on $H = h$ uses the version determined by the continuous densities, not a ratio of probabilities for a null event.

OP36.

Let the piece lengths be $D_1, \ldots, D_n$, with total length one. Positive lengths form a nondegenerate polygon if and only if the largest is strictly smaller than the sum of the others. Thus the event of interest is

$$\max_i D_i < \frac{1}{2}.$$

Necessity follows from the strict triangle inequality along the other sides. For sufficiency, a convex construction also rules out self-intersections. Write L for the largest length and S for the total, with $2L < S$. On a circle of radius $r \ge L/2$, the smaller central angle for side D_i is

$$\theta_i(r) = 2 \arcsin \frac{D_i}{2r}.$$

If the angle sum at $r = L/2$ is at least 2π, continuity and its decrease to zero give a radius where the sum is exactly 2π. Otherwise use the major arc $2\pi - \theta_L(r)$ for the longest side. We then need the other angles to sum to $\theta_L(r)$. Their sum minus $\theta_L(r)$ is negative at $r = L/2$, but for large r it is asymptotic to $(S - 2L)/r > 0$. Continuity again gives the required radius. In either case, take these positive arcs

successively around the circle and join neighboring vertices. This gives a convex polygon with the specified side lengths and no self-intersections.

The ordered uniform cut points have constant joint density $(n-1)!$. Transforming to spacings gives a constant density on the simplex $D_i > 0$, $\sum D_i = 1$, so the piece lengths are exchangeable. The first piece exceeds $1/2$ exactly when all cut points lie to the right of $1/2$. Hence

$$\mathbb{P}(D_i > 1/2) = \mathbb{P}(D_1 > 1/2) = 2^{-(n-1)}.$$

Two pieces cannot both exceed $1/2$, and equality has probability zero. Therefore,

$$\boxed{\mathbb{P}(\text{a polygon can be formed}) = 1 - \frac{n}{2^{n-1}}.}$$

This gives $1/4$ for $n = 3$ and $1/2$ for $n = 4$.

OP37.

Sort the points as $X_{(1)} < \cdots < X_{(N)}$ and define the interior gaps

$$D_i = X_{(i+1)} - X_{(i)}, \qquad 1 \le i \le N-1.$$

Mutual nearest neighbors must be adjacent. Let I_i indicate that the endpoints of adjacent edge i choose one another. The first edge qualifies precisely when $D_1 < D_2$, and the last edge has the analogous condition. An interior edge, $2 \le i \le N-2$, qualifies precisely when

$$D_i < D_{i-1}, \qquad D_i < D_{i+1}.$$

All uniform spacings, including the two boundary spacings, are exchangeable and have zero probability of ties. Each end edge therefore qualifies with probability $1/2$, while an interior gap is smallest among the relevant three with probability $1/3$.

Each point has a unique nearest neighbor, so distinct mutual pairs cannot share an endpoint. Every pair contributes exactly two good points. Thus, if G is the number of good points,

$$G = 2\sum_{i=1}^{N-1} I_i, \quad \mathbb{E}G = 2\left(\frac{1}{2} + \frac{1}{2} + \frac{N-3}{3}\right) = \boxed{\frac{2N}{3}}.$$

For $N = 3$ the interior sum is empty and the same formula applies. The expected number of pairs, rather than points, is $N/3$.

Stochastic Processes

RS1.

Use four nonterminal prefixes: the empty prefix, H, HT and HTH. Write their remaining means as e_0, e_1, e_2, e_3. When H follows HTH, the longest useful suffix is H, not the empty string. First-step equations are

$$e_0 = 2 + e_1, \quad e_1 = 2 + e_2, \quad e_2 = 1 + \tfrac{1}{2}e_3 + \tfrac{1}{2}e_0, \quad e_3 = 1 + \tfrac{1}{2}e_1.$$

The first two equations give $e_1 = e_0 - 2$, $e_2 = e_0 - 4$, and the last gives $e_3 = e_0/2$. Substitution yields $e_0 - 4 = 1 + 3e_0/4$, hence $\boxed{e_0 = 20}$. Each block of four tosses equals HTHT with probability $1/16$, ensuring finite mean.

RS2.

The stationary geometric ratio solves $\frac{1}{2}r^2 - r + \frac{1}{4} = 0$. Its root in $(0,1)$ is $r = 1 - 1/\sqrt{2}$, so

$$\boxed{\mathbb{P}_\pi(X \geq 2) = \left(1 - \frac{1}{\sqrt{2}}\right)^2}, \qquad \boxed{\mathbb{E}_0\tau_0^+ = \frac{1}{1-r} = \sqrt{2}}.$$

Clearing or attempting service on an empty queue counts as a positive-time return, so the mean can be less than 2.

RS3.

A two-step switch occurs exactly when one of the two steps switches. Thus $2\alpha(1-\alpha) = 3/8$, giving

$$\boxed{\alpha = 1/4 \quad \text{or} \quad \alpha = 3/4}.$$

Both parameters give the same two-step transition matrix. With the same initial law, all joint distributions at even times are consequently identical, so such data cannot distinguish them. One-step transition information can: one model favors persistence and the other favors alternation.

RS4.

Weights 3 and 2 for pending A and B tasks decrease in conditional expectation by 1 per processed task, proving a finite expected total. The means satisfy $a = 1 + b$, $b = 1 + a/3$, giving $a = 3$, $b = 2$. If second moments are u, v, independence of the two B subtrees gives

$$u = 1 + 2b + v + b^2 = 9 + v, \qquad v = 1 + \tfrac{2}{3}a + \tfrac{1}{3}u = 3 + \tfrac{1}{3}u.$$

These recursions may first be applied to trees truncated by generation. Their means are bounded by 3 and 2, and the second-moment recursion is bounded by 18 and 9. Monotone convergence proves finite second moments. Solving then yields

$$\boxed{\mathbb{E}T = 3, \quad \mathbb{E}T^2 = 18, \quad \operatorname{Var} T = 9}.$$

RH1.

In R11, $\rho = 1/2$ gives $r = 2 - \sqrt{3}$, hence

$$\boxed{\mathbb{E}T = 2 + 2\sqrt{3}}.$$

Whether step t must be performed is determined by the past and is independent of its reset decision. Therefore the total reset count K satisfies $\mathbb{E}K = (1/2)\mathbb{E}T = \boxed{1+\sqrt{3}}$. The pathwise cost is $T + 2K$, giving

$$\boxed{\mathbb{E}C = 4 + 4\sqrt{3}}.$$

RH2.

The drawdown gap has a self-loop at zero, so a two-absorbing-boundary time formula does not apply. In R16, $d = 3, D = 12$, giving

$$\boxed{\mathbb{E}\tau = 12, \qquad \operatorname{Var}\tau = 92}, \qquad \boxed{\mathbb{P}(M_\tau \geq 4) = \left(\frac{3}{4}\right)^4 = \frac{81}{256}}.$$

Each increase in the maximum requires winning a fresh boundary race with distances 1 and 3.

RH3.

The denominator in R13 is $D = 6(1/4) + 1/2 = 2$, so

$$\boxed{h_3^+ = \frac{3/4 + 1/2}{2} = \frac{5}{8}}, \qquad \boxed{h_3^- = \frac{3/4}{2} = \frac{3}{8}}.$$

With an unobserved equally likely initial direction, average the two conditional laws to obtain success probability $\boxed{1/2}$. This averaging does not automatically make the position process memoryless.

RH4.

The probabilities that the first 0, 1, 2 or 3 draws are distinct are $1, 1, 2/3, 2/9$. Thus

$$\mathbb{E}N = 1 + 1 + \frac{2}{3} + \frac{2}{9} = \frac{26}{9}.$$

The mean cost per draw is $7/3$. The past-measurable continuation-event argument of R18 gives

$$\boxed{\mathbb{E}C = \frac{7}{3}\frac{26}{9} = \frac{182}{27}}.$$

There are exactly $N-1$ distinct types before the first repeat, and the last draw adds none. Their mean count is therefore $\boxed{17/9}$.

RC1.

The up and down payoffs are 0 and $1/2$. Exact replication solves

$$2\Delta + B = 0, \qquad \tfrac{1}{2}\Delta + B = \tfrac{1}{2},$$

giving $\boxed{\Delta = -1/3, B = 2/3, V = 1/3}$. If $\Delta \geq 0$, covering the down state requires $\frac{1}{2}\Delta + B \geq 1/2$, hence initial cost $\Delta + B \geq 1/2 + \Delta/2 \geq 1/2$. Cash $1/2$ covers both outcomes, so $\boxed{V^{\text{super}} = 1/2}$.

RC2.

After an initial H, the next H probability is $2/3$, so the expected gain from continuing is $1/3$. After an initial T, it is $-1/3$, so stop. Toss initially, continue after H and stop after T. The initial value is

$$\boxed{V = \tfrac{1}{2}(1 + \tfrac{1}{3}) + \tfrac{1}{2}(-1) = \frac{1}{6}}.$$

Stopping immediately gives zero and is inferior. The second-stage comparison includes all feasible actions.

RC3.

Replacement pays net $7/8 - X$, so replace when $X < 7/16$, with either action at equality. Put $a = 7/16$. Integration yields

$$\boxed{V = \int_0^a (7/8 - x)\,\mathrm{d}x + \int_a^1 x\,\mathrm{d}x = \frac{1}{2} + \frac{49}{256} = \frac{177}{256}}.$$

The constant continuation value of an independent redraw does not apply; here the replacement value depends on the observed X.

RC4.

For all-pass testing, the indices are $c/(1-p) = (2, 8, 4/3)$. The order is $(3, 1, 2)$, with cost

$$\boxed{1 + \tfrac{1}{4} \cdot 1 + \tfrac{1}{4} \cdot \tfrac{1}{2} \cdot 2 = \frac{3}{2}}.$$

For at-least-one-pass testing, the indices are $c/p = (2, 8/3, 4)$. The order is $(1, 2, 3)$, with cost

$$\boxed{1 + \tfrac{1}{2} \cdot 2 + \tfrac{1}{2} \cdot \tfrac{1}{4} \cdot 1 = \frac{17}{8}}.$$

Each later cost is weighted by the probability that earlier tests have not yet settled the decision.

RG1.

The variables $1, W_1, W_1^2 - 1$ are orthogonal in square-integrable space; the last two have variances 1 and 2. R25 gives $\mathbb{E}I = 1$, $\operatorname{Cov}(I, W_1) = 1/2$ and $\operatorname{Var} I = 2e - 5$. Differentiating normal exponential moments also gives

$$\mathbb{E}[e^{W_t - t/2}(W_1^2 - 1)] = t^2, \qquad \operatorname{Cov}(I, W_1^2 - 1) = \int_0^1 t^2\,\mathrm{d}t = \frac{1}{3}.$$

Thus the best predictor and residual error are

$$\boxed{\widehat{I} = 1 + \frac{1}{2}W_1 + \frac{1}{6}(W_1^2 - 1)}.$$

$$\boxed{\mathbb{E}(I - \widehat{I})^2 = 2e - 5 - \frac{1}{4} - \frac{1}{18} = 2e - \frac{191}{36}}.$$

The quadratic term reduces error by another $1/18$, although a finite-degree polynomial still differs from the full conditional mean.

RG2.

Substitute $T = 2, b = 0, a = 1$ into R27. At time 1, the mean is $6/8 = 3/4$ and the variance is $1 - 1/2 - 3/8 = 1/8$, so

$$\boxed{W_1 \mid (W_2 = 0, A = 1) \sim N(3/4, 1/8)}.$$

The two other times have conditional covariance

$$\boxed{\frac{1}{2} - \frac{(1/2)(3/2)}{2} - \frac{3(1/2)(3/2)(3/2)(1/2)}{8} = -\frac{11}{128}}.$$

The area constraint makes their deviations partly compensate each other.

RG3.

The conditional reflection formula with $T = 2, b = 0$ gives maximum tail $\mathbb{P}(M \geq a) = e^{-a^2}$ for $a \geq 0$. Therefore

$$\boxed{\mathbb{P}(M \geq 1 \mid W_2 = 0) = e^{-1}}, \qquad \boxed{\mathbb{E}(M \mid W_2 = 0) = \int_0^\infty e^{-a^2}\,\mathrm{d}a = \frac{\sqrt{\pi}}{2}}.$$

RG4.

In R28, $k = \sqrt{2\lambda} = 1$, giving

$$\boxed{u(1) = \frac{\sinh 1}{\sinh 2} = \frac{1}{2\cosh 1}}, \qquad \boxed{\mathbb{E}(\tau \mid \text{success}) = 2\coth 2 - \coth 1 = \tanh 1}.$$

The simplification uses $\sinh 2 = 2\sinh 1 \cosh 1$ and $\cosh 2 = \cosh^2 1 + \sinh^2 1$. Without failure, success has probability $1/2$ and conditional mean duration $(4 - 1)/3 = 1$. Failure lowers the success probability and selects faster paths among those that succeed.

Section Exercises

OR01.

Partition the tosses into disjoint blocks of length n. Each block consists entirely of heads with probability 2^{-n}, independently of the other blocks. Thus T_n is bounded by n times a geometric random variable and has finite mean.

Let e_k be the expected number of additional tosses when the current run of heads has length k, for $0 \leq k \leq n$. The terminal condition is $e_n = 0$. A tail resets the run to zero, so

$$e_k = 1 + \tfrac{1}{2}e_{k+1} + \tfrac{1}{2}e_0, \qquad 0 \leq k < n.$$

Substitution backwards from $e_n = 0$ gives

$$e_0 = \left(1 + \frac{e_0}{2}\right)\sum_{j=0}^{n-1} 2^{-j} = \left(1 + \frac{e_0}{2}\right)(2 - 2^{1-n}).$$

Rearranging yields $2^{-n}e_0 = 2 - 2^{1-n}$ and therefore

$$\boxed{\mathbb{E}T_n = 2^{n+1} - 2}.$$

A run requires retaining the current progress: a tail destroys all consecutive heads accumulated so far.

OR07.

First suppose $0 < p < 1$ and write $q = 1 - p$. Let X_t count the umbrellas at the departure location for trip t. The state space is $\{0, 1, \ldots, n\}$. The next trip departs from the other location, so

$$P(0, n) = 1, \qquad P(i, n - i) = q, \quad P(i, n - i + 1) = p \quad (i \geq 1).$$

This finite chain is irreducible. For $i < n$, a trip without an umbrella followed by a trip carrying one reaches $i + 1$. For $i \geq 2$, the reverse sequence reaches $i - 1$; from 1, the chain can also reach 0 through n. All these paths have positive probability.

Consider

$$\pi_0 = \frac{q}{n + q}, \qquad \pi_i = \frac{1}{n + q} \quad (1 \leq i \leq n).$$

The inflow to 0 is $q\pi_n = \pi_0$. For $1 \leq j < n$, the inflow is $q\pi_{n-j} + p\pi_{n-j+1} = \pi_j$; the inflow to n is $\pi_0 + p\pi_1 = \pi_n$. The probabilities sum to 1, so this is the unique stationary distribution.

Getting wet requires state 0 and rain on the current trip. The long-run reward theorem for a finite irreducible chain therefore gives the almost-sure limiting frequency

$$\boxed{p\pi_0 = \frac{p(1 - p)}{n + 1 - p}}.$$

The limit does not depend on k. If $p = 0$, no trip is wet. If $p = 1$, only the first trip can be wet: every subsequent departure location has an umbrella. The limiting frequency is again 0, as the same formula predicts.

OR11.

Write $q = 1 - p$ and $u = u_1$. Since a downward jump has size 1, a walk from n to 0 must first visit $n - 1, \ldots, 1$ in order. Applying the strong Markov property and translation invariance at these descending hitting times gives

$$u_n = u^n.$$

There may be arbitrarily many rightward jumps between successive descending records; the leftward jumps need not be consecutive. Conditioning on the first step from 1 gives

$$u = p + qu_3 = p + qu^3,$$

and hence

$$(u - 1)(qu^2 + qu - p) = 0.$$

The positive root of the quadratic factor is

$$r = \frac{\sqrt{1 + 4p/q} - 1}{2}.$$

If $p \geq 2/3$, then $r \geq 1$ and the other quadratic root is negative. Thus the only possible value in $[0, 1]$ is 1.

If $p < 2/3$, the mean increment is $2 - 3p > 0$, so the strong law implies that the walk tends to positive infinity almost surely. Consequently $u \neq 1$. Indeed, if every integer's left neighbor were reached almost surely, repeated use of the strong Markov property would imply that every negative integer is visited almost surely, contradicting convergence to positive infinity. We must therefore choose $u = r < 1$. Thus

$$\boxed{u_n = \begin{cases} \left(\dfrac{\sqrt{1 + 4p/(1-p)} - 1}{2} \right)^n, & 0 < p < 2/3, \\ 1, & 2/3 \leq p < 1. \end{cases}}$$

A first-step equation may have multiple roots in the probability range. Drift, or the minimal nonnegative solution property, identifies the correct root.

OR16.

From every unabsorbed state, $a + b$ consecutive rightward steps force absorption. Such a block has probability $2^{-(a+b)}$. Hence τ has a geometric tail and all its moments are finite.

The symmetric independent increments make the following processes martingales:

$$S_n, \qquad S_n^2 - n, \qquad S_n^3 - 3nS_n,$$
$$S_n^4 - 6nS_n^2 + 3n^2 + 2n.$$

These identities follow by expanding $(S_n \pm 1)^j$ and averaging. Apply optional stopping first at the bounded time $\tau \wedge N$, then let $N \to \infty$. The stopped position is bounded in absolute value by $\max(a, b)$, and τ has finite moments, so dominated convergence justifies the limit.

Set $r = \mathbb{P}(S_\tau = b)$. The first two martingales give

$$br - a(1 - r) = 0, \qquad \mathbb{E}\tau = \mathbb{E}S_\tau^2 = b^2 r + a^2(1 - r).$$

Therefore

$$\boxed{r = \frac{a}{a + b}}, \qquad \boxed{\mathbb{E}\tau = ab}.$$

To obtain the second moment, write $A = \mathbb{E}[\tau \mathbf{1}_{\{S_\tau = b\}}]$ and $B = \mathbb{E}[\tau \mathbf{1}_{\{S_\tau = -a\}}]$. The cubic martingale yields

$$A + B = ab, \qquad bA - aB = \frac{\mathbb{E}S_\tau^3}{3} = \frac{ab(b - a)}{3}.$$

Solving these two linear equations gives

$$A = \frac{ab(2a+b)}{3(a+b)}, \qquad B = \frac{ab(a+2b)}{3(a+b)}.$$

Consequently,

$$\mathbb{E}(\tau S_\tau^2) = b^2 A + a^2 B = \frac{ab(a^2+ab+b^2)}{3},$$

$$\mathbb{E}S_\tau^4 = ab(a^2 - ab + b^2).$$

Substitute these values into the stopped quartic identity

$$0 = \mathbb{E}S_\tau^4 - 6\mathbb{E}(\tau S_\tau^2) + 3\mathbb{E}\tau^2 + 2\mathbb{E}\tau$$

to obtain

$$\boxed{\mathbb{E}\tau^2 = \frac{ab(a^2+3ab+b^2-2)}{3}}.$$

OR20.

Let $V(a,b)$ be the maximum expected additional score when a red and b black cards remain. Stopping immediately adds 0, and the boundary values are

$$V(a,0) = a, \qquad V(0,b) = 0.$$

For $a, b > 0$, the value of drawing one more card is

$$C(a,b) = \frac{a[1+V(a-1,b)] + b[-1+V(a,b-1)]}{a+b}.$$

The Bellman equation is therefore

$$\boxed{V(a,b) = \max\{0, C(a,b)\}}.$$

Evaluate states in increasing order of $a+b$. Backward induction proves optimality: once the successors use optimal policies, continuing is worth exactly $C(a,b)$, and comparison with the stopping value 0 gives the best current decision. The deck shrinks after every draw, so there is no infinite-horizon or integrability issue.

Starting with 26 cards of each color, the current cumulative score is $s = b - a$ and the number remaining is $N = a + b$. One optimal rule is as follows. Stop if $N = 0$. Continue if $N > 0$ and $s \le 0$. For positive scores, continue precisely under the conditions in the table; otherwise stop:

s	1	2	3	4	5
Continue if	$N \ge 5$	$N \ge 10$	$N \ge 19$	$N \ge 30$	$N \ge 43$

Stop whenever $s \ge 6$. Only reachable states are included, so N and s have the same parity. At $s = 1, N = 3$, continuing and stopping both have value 0; the table chooses to stop.

Exact rational evaluation of the recurrence gives

$$\boxed{V(26,26) = \frac{41984711742427}{15997372030584} \approx 2.624475549}.$$

The recurrence also determines the decision comparison at every state. The following code computes the exact initial value without rounding:

```
from fractions import Fraction as F
V = {}
for a in range(27):
    for b in range(27):
        if a == 0: V[a,b] = F(0)
        elif b == 0: V[a,b] = F(a)
        else:
            c = (a*(1+V[a-1,b])
                 + b*(-1+V[a,b-1]))/(a+b)
            V[a,b] = max(F(0), c)
print(V[26,26])
```

The advantage comes from information about the remaining deck. The successive draws are not independent gambles with a fixed success probability of $1/2$.

OR22.

Let V be the optimal value before the initial free roll. At an observed face j, stopping pays j, while restarting has value $V-1$. The candidate Bellman equation is

$$V = \frac{1}{6}\sum_{j=1}^{6} \max\{j, V-1\}.$$

The value $V = 4$ satisfies it, since

$$\frac{3+3+3+4+5+6}{6} = 4.$$

To verify an upper bound for all strategies, let V_m be the value with at most m rolls. We have $V_1 = 7/2 \le 4$ and

$$V_{m+1} = \frac{1}{6}\sum_{j=1}^{6} \max\{j, V_m - 1\}.$$

The right side is increasing in V_m, so induction gives $V_m \le 4$. Consider a strategy stopping at roll N with $\mathbb{E}N < \infty$, and force it to stop by roll m. The resulting payoff converges to the original payoff. Faces are bounded and fees are controlled by N, so dominated convergence implies that its expected payoff is at most 4. If $\mathbb{E}N = \infty$, the final face is at most 6 and the expected net payoff is negative infinity, which cannot improve the value.

Thus reroll on 1 or 2, accept 4, 5, or 6, and take either action on 3. For example, stop at the first face at least 3. The accepted face is uniform on 3, 4, 5, 6, with mean $9/2$. The number of rolls is geometric with success probability $2/3$, so expected reroll fees are $3/2 - 1 = 1/2$. Hence

$$\boxed{\text{Maximum expected payoff} = 9/2 - 1/2 = 4}.$$

This policy terminates almost surely and attains the upper bound, proving optimality.

OR25.

Integration by parts for tW_t gives

$$TW_T = \int_0^T W_t\,dt + \int_0^T t\,dW_t.$$

Consequently,

$$X = \int_0^T (T-t)\,dW_t, \qquad Y = \int_0^T t\,dW_t.$$

Both integrands are deterministic and square-integrable, so both integrals have mean 0. The Itô isometry and its cross form give

$$\operatorname{Var} X = \int_0^T (T-t)^2 dt = \frac{T^3}{3}, \qquad \operatorname{Var} Y = \int_0^T t^2 dt = \frac{T^3}{3},$$

$$\operatorname{Cov}(X,Y) = \int_0^T t(T-t)dt = \frac{T^3}{6}.$$

The cross form also follows by applying the isometry to the sum of the two integrands and expanding the square. Since $T > 0$, both variances are nonzero. Therefore

$$\boxed{\operatorname{Corr}(X,Y) = \frac{T^3/6}{T^3/3} = \frac{1}{2}}.$$

OR26.

Define

$$\tau = \inf\{t \geq 0 : B_t^{(1)} = -1\}.$$

The reflection principle gives $\mathbb{P}(\tau \leq t) = 2\Phi(-1/\sqrt{t})$, where Φ is the standard normal distribution function. Hence $\tau < \infty$ almost surely, with density

$$f_\tau(t) = \frac{1}{\sqrt{2\pi t^3}} e^{-1/(2t)}, \qquad t > 0.$$

The time τ depends only on the first coordinate and is independent of the entire second-coordinate process. Conditional on $\tau = t$, therefore, $B_\tau^{(2)} \sim N(0,t)$. Set $Z = B_\tau^{(2)}$. Mixing the normal densities yields

$$f_Z(z) = \int_0^\infty \frac{1}{2\pi t^2} e^{-(1+z^2)/(2t)} dt = \frac{1}{\pi(1+z^2)}.$$

The integrand is nonnegative, so Tonelli's theorem justifies the mixture calculation; substitution $u = 1/t$ evaluates the last integral. Thus Z has the standard Cauchy distribution, and

$$\boxed{\mathbb{P}(1+Z>0) = \int_{-1}^\infty \frac{dz}{\pi(1+z^2)} = \frac{3}{4}}.$$

The independence is between the stopping time and the entire second-coordinate process. Evaluation at this random time produces the mixture above, not a normal distribution with fixed variance.

Statistics and Linear Algebra

AG1-1.

Since $\mathrm{Cov}(W, Z) = \mathrm{Var}(T)$, the latent variance is 3. Then $\mathrm{Var}(U) = 5 - 3 = 2$ and $\mathrm{Var}(V) = 4 - 3 = 1$, both nonnegative. The instrumental-variable ratio gives $\beta = 6/3 = 2$, whereas the ordinary slope is $6/5$. Independent centered normal variables with variances $3, 2, 1$, together with independent centered response noise, realize the specified moments. The model necessarily imposes $\mathrm{Cov}(W, Y) = \mathrm{Cov}(Z, Y) = \beta\,\mathrm{Var}(T)$. Values 6 and 5 therefore contradict it; changing measurement-noise variances cannot repair the contradiction. This is a testable model restriction, not a reason to choose one ratio arbitrarily.

AG1-2.

A25 bounds both probabilities by $1/(1 + 2^2) = 1/5$. The non-strict event attains it with probability $1/5$ at 2 and $4/5$ at $-1/2$. The mean is $2/5 - 2/5 = 0$, and the second moment is $4/5 + 1/5 = 1$. The strict event cannot attain the bound: equality would also force equality for the non-strict event, hence the distribution just found, whose strict-tail probability is zero. For $h > 0$, let $b = 2 + h$ and put probability $1/(1 + b^2)$ at b and $b^2/(1 + b^2)$ at $-1/b$. Its mean is zero and variance one. Its strict-tail probability tends to $1/5$ as $h \downarrow 0$.

AG1-3.

Ignoring a constant, the objective is $\sum_j (b_j - z_j)^2/2 + \sum_j b_j/2$. With budget multiplier $\nu \geq 0$, coordinatewise minimization under nonnegativity gives

$$b_j = (z_j - 1/2 - \nu)_+.$$

At $\nu = 0$ this is $(5/2, 1/2, 0)$, whose sum is 3, violating the budget. The budget must bind. Taking $\nu = 1/2$ gives $(2, 0, 0)$ with sum 2. The first coordinate has zero adjusted gradient; the second has zero right derivative at zero; the third has positive right derivative. All optimality conditions hold. The quadratic part is strictly convex, so the solution is unique even though the second coordinate is exactly at the threshold. The common multiplier couples the coordinates.

AG1-4.

Take $c = (1, 1)^\top$, $d = 0$. In A06, $e = 1$ and $q = 1 + 1/4 = 5/4$, so

$$\widehat{\beta}_c = (2, -1)^\top - \frac{4}{5}(1, 1/4)^\top = (6/5, -6/5)^\top, \qquad \Delta\mathrm{RSS} = 4/5.$$

For a direct check, put $\beta = (t, -t)$. The extra sum of squares is $(t - 2)^2 + 4(1 - t)^2$, whose derivative $10t - 12$ vanishes uniquely at $t = 6/5$. Substitution gives $16/25 +$

$4/25 = 4/5$. Equal-sized corrections to the two coefficients would therefore not be optimal.

Section Exercises

OA02.

Centering does not change slopes, so take all means to be zero. Write the joint regression as

$$Y = bX_1 + \beta_2 X_2 + u, \qquad \mathrm{Cov}(u, X_1) = \mathrm{Cov}(u, X_2) = 0.$$

Set $\gamma = \mathrm{Cov}(X_1, X_2)/\mathrm{Var}(X_1)$ and let $r = X_2 - \gamma X_1$ be the residual from regressing X_2 on X_1. The first normal equation implies that the simple-regression slope of Y on X_1 is $b + \beta_2\gamma$. Thus

$$e = \beta_2 r + u.$$

Since r is uncorrelated with X_1 and $X_2 = r + \gamma X_1$,

$$\mathrm{Cov}(r, X_2) = \mathrm{Var}(r) = \mathrm{Var}(X_2)(1 - \rho^2).$$

Consequently,

$$\boxed{\beta_1 = \frac{\mathrm{Cov}(e, X_2)}{\mathrm{Var}(X_2)} = (1 - \rho^2)\beta_2.}$$

Residualizing only the response leaves $\mathrm{Var}(X_2)$ in the denominator of the second slope. Regressing e on r instead uses $\mathrm{Var}(r)$ and recovers β_2. The two slopes agree when $\rho = 0$. At $|\rho| = 1$, the two joint-regression coefficients are generally not separately identifiable, which explains the restriction in the question.

OA03.

Write $A = \{X + Y > 1\}$. The original joint density is 1 on the unit square, and $\mathbb{P}(A) = 1/2$, so the conditional density is 2 on the upper triangle. Given $X = x$ and A, the variable Y is uniform on $(1 - x, 1)$. Hence

$$\mathbb{E}[Y \mid X = x, A] = 1 - \frac{x}{2}, \qquad 0 < x < 1.$$

The conditional mean is already affine. The residual $u = Y - (1 - X/2)$ satisfies $\mathbb{E}[u \mid X, A] = 0$, so it is orthogonal to both the constant and X. The regression therefore has intercept 1 and slope

$$\boxed{-\frac{1}{2}}.$$

For a moment check, the conditional density $f_{X|A}(x) = 2x$ gives

$$\mathbb{E}[X \mid A] = \frac{2}{3}, \quad \mathbb{E}[X^2 \mid A] = \frac{1}{2}, \quad \mathrm{Var}(X \mid A) = \frac{1}{18}.$$

The conditional-mean formula implies $\mathrm{Cov}(X, Y \mid A) = -\mathrm{Var}(X \mid A)/2 = -1/36$. Dividing by the conditional variance again gives $-1/2$. Independence holds before selection and cannot be carried over to the conditional population.

OA05.

Let $c = X^\top y/n$. Expanding the objective difference gives

$$L(\beta) - L(0) = \frac{\|X\beta\|^2}{2n} - c^\top \beta + \lambda\|\beta\|_1.$$

If $\|c\|_\infty \le \lambda$, the inequality $c^\top\beta \le \|c\|_\infty\|\beta\|_1$ makes this difference nonnegative, so zero is optimal. Conversely, suppose $|c_j| > \lambda$ for some coordinate. Take $\beta = t\,\mathrm{sign}(c_j)e_j$ with $t > 0$. Then

$$L(\beta) - L(0) = \frac{t^2\|X_j\|^2}{2n} - t(|c_j| - \lambda).$$

For sufficiently small t, the negative linear term dominates the quadratic term. Thus the objective decreases, proving that the required condition is

$$\boxed{\lambda \ge \frac{\|X^\top y\|_\infty}{n}}.$$

Under this condition, zero is also unique when $\lambda > 0$. A second minimizer with $X\beta \ne 0$ would, by strict convexity in the fitted value, have a midpoint with zero that improves the objective. A nonzero vector with $X\beta = 0$ strictly increases the penalty. When $\lambda = 0$, the condition becomes $X^\top y = 0$, and the full minimizer set is $\ker X$; zero is then unique exactly when X has full column rank.

OA30.

The relation is

$$\boxed{\lambda_i \le \mu_i \le \lambda_{i+1}}, \qquad i = 1, \dots, n-1.$$

Let $H = \{x \in \mathbb{R}^n : x_k = 0\}$. Inserting a zero in coordinate k identifies the Rayleigh quotient of B with that of A restricted to H. Write $R_A(x) = x^\top Ax/\|x\|^2$. The min–max principle gives

$$\lambda_i = \min_{\dim S = i} \max_{0 \ne x \in S} R_A(x).$$

The formula for μ_i minimizes the same quantity only over subspaces $S \subset H$. Restricting the choices cannot decrease the minimum, so $\mu_i \ge \lambda_i$.

For the other bound, let E be spanned by eigenvectors of A corresponding to its first $i+1$ eigenvalues. Since H has codimension one, $\dim(E \cap H) \ge i$. Choose an i-dimensional subspace S of this intersection. Every Rayleigh quotient on S is at most λ_{i+1}. Using this candidate in the min–max formula for μ_i proves $\mu_i \le \lambda_{i+1}$.

Repeated eigenvalues are allowed, so strict inequalities cannot hold in general. For the identity matrix, equality holds on both sides.

Mathematical Reasoning

AG2-1.

Removing the known diagonal's contribution at vertices $1, 3$ leaves zero residual parity at every vertex. The two incident cycle bits must agree at each vertex, so propagation around the cycle makes all four bits zero or all four one. There are exactly two solutions. Four independent fair bits have 16 equally likely assignments, giving probability $2/16 = 1/8$. Observing $12 = 0$ leaves only the all-zero solution; the remaining unknown graph is a path, consistent with A19's forest criterion.

AG2-2.

The column and two diagonal slopes are $a, a + 1, 1 - a$, all of which must be nonzero modulo 5. Excluding $a = 0, 1, 4$ leaves $a = 2, 3$. Each permits all five shifts b, for 10 arrangements. Since 9 is divisible by 3, the square-sum necessary condition in A14 fails. No arrangement exists, and using a non-affine rule cannot avoid that obstruction.

AG2-3.

The upper endpoint of the jump is $3/8 + 1/4 = 5/8$, so the conditional law is $\mathrm{Unif}(3/8, 5/8)$. Half of that interval lies above $1/2$, giving joint probability $(1/4)(1/2) = 1/8$. The original variable takes values in the countable set $\{r_j\}$, so $h(X)$ almost surely belongs to the countable image $\{h(r_j)\}$. A continuous uniform law assigns probability zero to every countable set, contradicting probability one here. Density of the atom locations on the real line does not change this argument.

AG2-4.

The deleted member held three types, each of which changes from three holders to two. The other four types still have three holders. Thus there are 3 two-holder and 4 three-holder types, and none disappears. No item held by a remaining member is deleted, so each still has three. The unique item shared by each remaining pair is also retained. Independently, let the two counts be x, y. Person–item incidences and shared pairs give $2x + 3y = 6 \cdot 3 = 18$ and $x + 3y = \binom{6}{2} = 15$. Subtracting yields $x = 3$, then $y = 4$, agreeing with the construction.

AG3-1.

Uniform hiding makes every search succeed with probability $1/2$, while uniform choice of search pair covers every location with probability $1/2$. Hence the value is $1/2$. For hiding probabilities p_i, the average success probability of the four searches is $1/2$. Their maximum can be at most $1/2$ only if all are equal. Thus all optimal hiding strategies are $p_1 = p_3 = a$, $p_2 = p_4 = 1/2 - a$, where $0 \le a \le 1/2$. For the searcher, the average coverage of the four locations is

also $1/2$. Minimum coverage is at least $1/2$ exactly when every location has that coverage. Writing q_{ij} for the probability of searching edge ij, all solutions are

$$q_{12} = q_{34} = b, \qquad q_{23} = q_{41} = 1/2 - b, \qquad 0 \le b \le 1/2.$$

The conditions are necessary and sufficient, including the interval endpoints.

AG3-2.

Pile sizes $1, 2, 3$ are respectively losing, winning, losing: one stone must be taken at a loss; two can leave one; from three, taking one leaves a winning state and taking all loses immediately. From $4, 5, 6, 7$, removing $3, 4, 3, 4$ respectively leaves 1 or 3. Thus only $1, 3$ lose among the first seven positive positions. The losing residues are $1, 3 \pmod 7$. Removing $1, 3, 4$ from either cannot reach another losing residue; a move that empties the pile is itself losing. From the other residues $0, 2, 4, 5, 6$, remove respectively $4, 1, 3, 4, 3$ to leave a positive losing position. These moves are legal even at the smallest positive representatives. Strong induction proves the classification. Since $20 \equiv 6 \pmod 7$, remove 3 to leave $17 \equiv 3$. The other moves leave winning states $19, 16$, so this is the unique winning first move. Under A09's normal-play rule, remove 4 to leave $16 \equiv 2$; that first move is also uniquely winning.

AG3-3.

With both servers occupied, no-deviation conditions become

$$k \le \frac{9-k}{2} + \delta, \qquad \frac{8-k}{2} + \delta \le k + 1,$$

equivalently $6 + 2\delta \le 3k \le 9 + 2\delta$. For $k = 3$ to be an equilibrium, one needs $0 \le \delta \le 3/2$. In this range neither empty-server configuration is an equilibrium. For $0 < \delta < 3/2$, the only integer solution is $k = 3$. At $\delta = 0$, occupancy $k = 2$ is also an equilibrium; at $\delta = 3/2$, so is $k = 4$. Outside the range, $k = 3$ itself violates a no-deviation condition. Pure equilibria always exist: for $0 \le \delta < 15/2$, take $k = \lceil 2 + 2\delta/3 \rceil$; for $\delta \ge 15/2$, all eight users choosing the slow server is an equilibrium. Thus the desired fees are exactly $0 < \delta < 3/2$. Total service cost at that occupancy remains $43/2$: fees change incentives, not the actual service cost of a fixed occupancy.

AG3-4.

Customers split at $(a + b)/2$. Integrating over the resulting intervals gives

$$C(a, b) = \frac{a^2}{2} + \frac{(b-a)^2}{4} + \frac{(1-b)^2}{2}.$$

The gradient is $((3a - b)/2, (3b - a - 2)/2)$, with unique stationary point $a = 1/4, b = 3/4$. The Hessian has eigenvalues $1, 2$, so the objective is strictly convex. This feasible stationary point is the unique ordered global optimum, and substitution gives $C = 1/8$. Under competition, separated providers cannot be in equilibrium: the left one can move toward the right one and increase its share. Coincidence must be at the uniform median $1/2$; elsewhere, a small deviation to-

ward the side containing more demand earns more than half. At the midpoint, no deviation improves on half. Thus the unique pure equilibrium is $(1/2, 1/2)$, with mean distance $\int_0^1 |t - 1/2|\, \mathrm{d}t = 1/4$, twice the cooperative minimum.

Section Exercises

OA17.

Suppose there are N types in total, and type j occurs in r_j collections, where $1 \le r_j \le 4$. Counting all memberships gives

$$\sum_{j=1}^{N} r_j = 40.$$

Now add the sizes of the six pairwise intersections. Type j is counted $\binom{r_j}{2}$ times, so

$$\sum_{j=1}^{N} \binom{r_j}{2} \le 36, \qquad \sum_{j=1}^{N} r_j^2 \le 112.$$

By Cauchy–Schwarz,

$$40^2 \le N \sum_{j=1}^{N} r_j^2 \le 112N, \qquad N \ge \frac{100}{7}.$$

Since N is an integer, $N \ge 15$.

Label fifteen types by 1 through 15 and take

$$\begin{aligned} A &= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}, \\ B &= \{1, 2, 3, 4, 5, 6, 11, 12, 13, 14\}, \\ C &= \{1, 2, 7, 8, 9, 10, 11, 12, 13, 14\}, \\ D &= \{3, 4, 5, 7, 8, 9, 11, 12, 13, 15\}. \end{aligned}$$

Each collection has ten elements, every pairwise intersection has exactly six elements, and their union contains all fifteen types. The minimum is therefore $\boxed{15}$.

OA22.

If $x > y$, then $3x - 2y = x + 2(x - y) \ge 4$. If $x = y$, integrality excludes $x = 1$, so $3x - 2y = x \ge 2$. It remains to consider $y = x + d$ with $d \ge 1$.

Set

$$P = (x + 1)(x + 2) \cdots (x + d).$$

The given fraction is $x!P/(P + 1)$. Since P and $P + 1$ are coprime, integrality is equivalent to $P + 1 \mid x!$.

Suppose, for a contradiction, that $x - 2d \le 1$, or $x \le 2d + 1$. Let p be any prime divisor of $P+1$. Divisibility gives $p \le x$. If $p \le d$, one of the d consecutive

integers $x+1,\dots,x+d$ is divisible by p, contradicting $p \nmid P$. Hence $p > d$. If also $p \le y/2$, then

$$x < 2d+2 \le 2p \le y.$$

Thus $2p$ is a factor in the product P, again a contradiction. Every prime divisor must therefore lie in

$$\frac{y}{2} < p \le x.$$

If $d \ge x$, this interval is empty, contradicting $P+1 > 1$. Otherwise the number of integers in the interval is

$$x - \left\lfloor \frac{x+d}{2} \right\rfloor = \left\lceil \frac{x-d}{2} \right\rceil \le d,$$

where the last bound follows from $x \le 2d+1$ and $d \ge 1$. Moreover, $p > y/2 > x/2$, so each such prime occurs only once in $x!$. Since $P+1 \mid x!$, the integer $P+1$ is a product of at most d distinct primes, each at most x. Consequently,

$$P+1 \le x^d < (x+1)\cdots(x+d) = P,$$

a contradiction. Hence $3x-2y \ge 2$. For $(x,y) = (4,5)$, the given fraction equals 20 and $3x-2y = 2$, so the constant is sharp.

OA28.

The total weight is $\sum_{n=0}^{\infty} 2^{-n} = 2$. Thus the series converges at every x, with a tail bound that is uniform in x.

First let x be irrational. Given $\varepsilon > 0$, choose N such that $\sum_{n>N} 2^{-n} < \varepsilon$. None of the finitely many points $r_0,\dots,r_N$ equals x. Hence some $\delta > 0$ ensures that, whenever $|y-x| < \delta$, each of these rational points is on the same side of y as it is of x. The first $N+1$ terms remain unchanged, so

$$|f(y) - f(x)| \le \sum_{n>N} 2^{-n} < \varepsilon.$$

The function is continuous at every irrational point.

Now take $x = r_k$ and separate its term:

$$f(y) = 2^{-k}\mathbf{1}_{\{y>r_k\}} + g(y).$$

The same finite-prefix and tail argument shows that g is continuous at r_k, because its prefix contains no point equal to r_k. The strict inequality in the definition gives $f(r_k) = g(r_k)$. Therefore

$$\boxed{\lim_{y\uparrow r_k} f(y) = f(r_k), \qquad \lim_{y\downarrow r_k} f(y) = f(r_k) + 2^{-k}.}$$

Each rational point has an upward jump of size 2^{-k} and is left-continuous but not continuous. The continuity points are exactly the irrationals. Dense jump locations cause no contradiction: their tail weights can be made uniformly small.

OA09.

For a positive integer r, let $\ell(r)$ be the largest power of two dividing r, equivalently the value of its least significant nonzero binary digit. Consider a general position (r, m) with r stones remaining and a current limit of m. We prove

$$(r, m) \text{ is losing} \quad \Longleftrightarrow \quad m < \ell(r).$$

Use induction on the number of stones. If $m \geq \ell(r)$, take $\ell(r)$ stones. This either wins immediately or leaves $r - \ell(r)$, whose least significant nonzero bit is strictly larger than $\ell(r)$. The next limit is $\ell(r)$, so the opponent receives a losing position.

If $m < \ell(r)$, every legal move a satisfies $1 \leq a < \ell(r)$. Write $a = 2^s q$ with q odd. Because r is divisible by a higher power of two, $r - a$ has exactly s factors of two. Thus

$$\ell(r - a) = \ell(a) \leq a.$$

The opponent's limit is a, so the resulting position is winning. Every legal move therefore gives a winning position to the opponent, completing the induction.

Initially the position is $(n, n - 1)$. It is losing exactly when $\ell(n) = n$, which means that n is a power of two. Hence

$$\boxed{\text{First player wins} \iff n \notin \{2^k : k \geq 1\}.}$$

In a winning initial position, take $\ell(n)$ stones. On each subsequent turn, take the least significant nonzero bit of the remaining count. The position criterion ensures that this move is legal and eventually takes the last stone.

OA13.

The first player chooses 6. Every later legal number must avoid the prime factors 2 and 3, leaving only

$$5, 7, 11, 13, 17, 19, 23, 29$$

as possible prime factors. The product of any two distinct primes in this list is at least $5 \cdot 7 = 35 > 30$. Thus every later legal number uses exactly one prime from the list; for example, 25 uses only the prime 5. Once a number has been selected, its prime factor cannot be used again.

Each move therefore consumes exactly one of the eight available primes. Conversely, if a prime p has not yet been used, the number p itself remains a legal choice. The game cannot end while such a prime remains. There are consequently exactly eight further moves after the initial 6, regardless of the players' later choices.

There are nine moves in total, so the first player makes the last move and wins. Both parts of the argument matter: every move consumes one prime, and every unused prime guarantees another legal move.

Algorithms and Programming

AG4-1.

Group A has $n_A = 3$, $\mu_A = 4$, and $M_A = 9 + 0 + 9 = 18$. Thus B has two observations, sum $5 \cdot 4 - 3 \cdot 4 = 8$, and mean 4. Equal group means give no between-group contribution, so $M_B = 26 - 18 = 8$. The combined population and sample variances are $26/5$ and $26/4 = 13/2$.

Write B's observations as $4-u, 4+u$. Their centered sum of squares is $2u^2 = 8$, so $|u| = 2$. The unordered pair is uniquely $\{2, 6\}$, but its original order is unrecoverable. Moment summaries do not generally identify the original observations; this uniqueness relies on B having exactly two values.

AG4-2.

Interval $[0, 3]$ contains $(4, 2, 3, 5)$, with range 3 and length 4. Interval $[3, 6]$ also has range 3 and length 4, so the earliest-start rule chooses $[0, 3]$.

The three length-5 windows have ranges 6, 6, and 5. All fail. Every longer interval contains a length-5 interval and cannot have a smaller range, so no longer feasible answer exists. The two-deque scan gives the same result, while this exclusion argument independently certifies optimality.

AG4-3.

With limit 2, the best candidates by end are: $1 - 80/120 = 1/3$ at end 1; $1 - 100/120 = 1/6$ at end 2; $1 - 70/100 = 3/10$ at end 3, where only prices 80 and 100 are eligible; and $1 - 90/100 = 1/10$ at end 4. The optimum is $1/3$ on $[0, 1]$.

Without the limit, the drop from 120 to 70 gives $5/12$ on $[0, 3]$, the global maximum. Its duration is 3 and it is infeasible for the first problem. Keeping an expired peak would incorrectly substitute the unconstrained answer.

AG4-4.

The fall from 80 to 60 is exactly $1/4$, on $[2, 4]$ with duration 2. Consecutive drops are $6/100$, $14/94$, $1/80$, and $19/79$, each strictly less than $1/4$. No duration-1 interval works, so duration 2 is optimal.

For a fixed start i, once its first qualifying end j has been processed, every later end gives a longer interval. A start already dominated by a later, no-lower price can also be removed safely. These deletion rules depend on the shortest-duration objective; they cannot be transferred without proof to maximizing the drawdown ratio.

Section Exercises

OA31.

Store the window in a circular buffer of length N, so the oldest value can be retrieved when the window is full. Let $k = \min(t, N)$ and maintain

$$S = \sum_{\text{window}} x_i, \qquad Q = \sum_{\text{window}} x_i^2.$$

Before the window fills, a new value b gives $S' = S + b$, $Q' = Q + b^2$, and $k' = k + 1$. Once it is full, let a be the outgoing value and update

$$S' = S - a + b, \qquad Q' = Q - a^2 + b^2, \qquad k' = N.$$

Return

$$\mu' = \frac{S'}{k'}, \qquad v' = \frac{Q'}{k'} - (\mu')^2.$$

Correctness follows from $\sum(x_i - \mu)^2 = Q - k\mu^2$. Each update uses a constant number of arithmetic operations, so it takes $O(1)$ time in the unit-cost arithmetic model. The buffer uses $O(N)$ space. For $N = 1$, the variance is always zero.

When values are large relative to their spread, $Q/k - \mu^2$ can suffer from floating-point cancellation. An alternative maintains the centered sum of squares $M = \sum(x_i - \mu)^2$. For a full window,

$$\mu' = \mu + \frac{b - a}{N},$$
$$M' = M + (b - a)(b + a - \mu - \mu'), \qquad v' = M'/N.$$

The second formula follows by expanding $M = Q - N\mu^2$ and using $N(\mu' - \mu) = b - a$. During initialization, put $\delta = b - \mu$ and use

$$\mu' = \mu + \frac{\delta}{k + 1}, \qquad M' = M + \delta(b - \mu').$$

Initialize the first value directly with $\mu = b, M = 0$. Centered updates reduce repeated subtraction of two large second moments, but rounding errors can still accumulate; recomputing from the buffer may occasionally be appropriate. If a sample variance with denominator $k - 1$ is wanted, return $M/(k - 1)$ for $k > 1$. Whether it is unbiased depends on the sampling model.

OA32.

Maintain a deque of candidate indices. Indices increase strictly from front to back, while their values decrease strictly. On arrival of x_t:

1. Remove all front indices smaller than $t - N + 1$; these have expired.
2. While the deque is nonempty and its back value is at most x_t, remove the back index.
3. Append t and return the value at the front.

Removing a back index is safe. If $i < t$ and $x_i \le x_t$, then in every later window containing both values, i is no better than t, while t expires later. Thus i is no

longer needed as a maximum candidate. Expired indices are irrelevant to the current window. The remaining values are decreasing, so the front is the current maximum. For equal values, keeping the newer index extends its useful lifetime. If the earliest maximizing index is also required, replace "at most" in step 2 by "strictly less than."

Each index is inserted once and removed at most once. Processing T values therefore costs $O(T)$ operations, giving $O(1)$ amortized time per arrival and $O(N)$ space. A single very large value can nevertheless remove almost the entire deque, so the worst-case time for one update is $O(N)$, not $O(1)$. For $N = 1$, the output is simply the newly arrived value.

OA33.

Initialize the running peak $H = p_1$, its index $i_H = 1$, and the answer $D = 0$. For $j = 2, \ldots, n$, first compute $d = 1 - p_j/H$. If $d > D$, set $D = d$ and save the interval (i_H, j). Then, if $p_j > H$, set $H = p_j$ and $i_H = j$. If no positive decline occurs, return $D = 0$ without a positive-drawdown interval.

Before processing j, the invariant is $H = \max_{i<j} p_i$. For fixed positive p_j, the quantity $1 - p_j/p_i$ increases with positive p_i. The greatest decline ending at j is therefore $1 - p_j/H$. The algorithm checks the best starting point for every endpoint and takes the maximum with zero, proving global optimality. Computing the decline before updating the peak keeps the start strictly earlier than the end.

Each price is processed once, giving $O(n)$ time and $O(1)$ auxiliary space. Updating only when $d > D$ retains the earliest endpoint attaining the maximum. Not replacing a tied peak retains its earliest index.

The denominator of a drawdown is the preceding peak. A fall from 100 to 60 has a drawdown of 40%. The expression $H/p_j - 1$ instead measures the gain needed to recover from the trough to the peak; here it equals $2/3$. With cumulative peaks $H_j = \max_{i \le j} p_i$, an equivalent formula is

$$\boxed{D = \max_{1 \le j \le n} \left(1 - \frac{p_j}{H_j}\right).}$$

Mixed Exercises

AA01.

The completed vector always has even total parity. Every vector satisfying this constraint has one possible initial $(n-1)$-bit sequence and hence probability $2^{-(n-1)}$; other vectors have probability zero.

Specify any $n - 1$ coordinates. The remaining coordinate is determined uniquely by even parity. The specified event therefore has probability $2^{-(n-1)}$, the product of $n - 1$ fair-bit probabilities, proving independence of that subset.

The all-zero probability is also $2^{-(n-1)}$, whereas independence of all n fair bits would give 2^{-n}. Thus global independence fails. The parity constraint is detectable only when all coordinates are considered together.

AA02.

For every integer k,

$$\mathbf{1}_{\{k\equiv r\ (3)\}} = \frac{1}{3}\sum_{j=0}^{2} \omega^{j(k-r)}.$$

If the congruence holds, all terms are 1; otherwise they are the three cube roots of unity and sum to zero. Independence gives

$$\mathbb{E}\omega^{S} = \left(\frac{1+\omega}{2}\right)^{n} = 2^{-n}e^{in\pi/3}.$$

The two nonconstant filter terms are conjugates after the appropriate phase adjustment. Therefore

$$\boxed{\mathbb{P}(S \equiv r\ (3)) = \frac{1}{3} + \frac{2^{1-n}}{3}\cos\left(\frac{n\pi}{3} - \frac{2\pi r}{3}\right).}$$

The formula also covers $n = 0$. For $n = 8, r = 0$, the cosine is $-1/2$, yielding $1/3 - 1/768 = 85/256$. Directly summing the binomial probabilities at $0, 3, 6$ gives $(1 + 56 + 28)/256$, confirming the value.

AA19.

Every swap changes permutation parity, so return is impossible after an odd number of steps. Group steps in pairs. Repeating the same swap has probability $1/2$ and leaves the order unchanged; using the two different swaps produces the two three-cycles, each with probability $1/4$.

The two-step chain on the three even permutations has diagonal probability $1/2$ and off-diagonal probabilities $1/4$. If u_m is the return probability after m pairs,

$$u_{m+1} = \frac{1}{2}u_m + \frac{1}{4}(1-u_m) = \frac{1}{4} + \frac{1}{4}u_m, \qquad u_0 = 1.$$

Subtracting the fixed point $1/3$ yields

$$\boxed{\mathbb{P}(\text{return at step } t) = \begin{cases} 0, & t \text{ odd}, \\ \dfrac{1}{3} + \dfrac{2}{3}\,4^{-m}, & t = 2m. \end{cases}}$$

Even times approach $1/3$, whereas odd times remain zero. Finiteness and symmetry alone do not remove periodicity.

AA22.

At a fixed circular center, the three relevant positions are distinct. The probability of B at the center and A at both neighbors is

$$\frac{n}{N}\frac{m}{N-1}\frac{m-1}{N-2}.$$

There are N centers, so the circular expectation is

$$\boxed{\frac{nm(m-1)}{(N-1)(N-2)}}.$$

On a line only $N-2$ centers are eligible, each with the same pattern probability. The expectation becomes

$$\boxed{\frac{nm(m-1)}{N(N-1)}}.$$

Both vanish when $m < 2$ or $n = 0$. At $(m, n) = (4, 3)$ the answers are $6/5$ and $6/7$. The center indicators may be dependent; linearity of expectation does not require independence.

AA03.

The vertical slice at coordinate x has length $2\sqrt{R^2 - x^2}$. Dividing by total area gives

$$f_X(x) = \frac{2}{\pi R^2}\sqrt{R^2 - x^2}\,\mathbf{1}_{\{|x|<R\}}.$$

Central slices are longer, making the projection more concentrated near zero. Symmetry yields $\mathbb{E}X = 0$, and

$$\mathbb{E}|X| = \frac{4}{\pi R^2}\int_0^R x\sqrt{R^2 - x^2}\,\mathrm{d}x = \frac{4R}{3\pi}.$$

For the second moment, rotation symmetry gives $\mathbb{E}X^2 = \mathbb{E}Y^2 = \mathbb{E}(X^2 + Y^2)/2$. Squared radius is uniform on $[0, R^2]$, hence $\mathbb{E}(X^2 + Y^2) = R^2/2$ and

$$\boxed{\operatorname{Var} X = R^2/4}.$$

The slice-length weighting explains why uniform area does not give a uniform coordinate.

AA07.

Points are distinct almost surely. Separation is possible exactly when the A points form one contiguous circular block and B points form its complement. For sufficiency, choose a chord with endpoints in the two gaps between types, avoiding sample points; its line separates the blocks. For necessity, a line intersects the circle at most twice, and each side corresponds to one arc. More than two type changes around the circle are therefore impossible.

After fixing a numbered circular starting position, the A positions are uniform among $\binom{m+n}{m}$ subsets. A contiguous A block can start in $m+n$ places, all distinct because both types occur. Thus

$$\boxed{\mathbb{P}(\text{separable}) = \frac{m+n}{\binom{m+n}{m}}}.$$

For $m = n = 3$ this is $6/20 = 3/10$. Only the circular type order matters; the actual angular gaps do not.

AA09.

The standard planar Gaussian has a uniform direction. Conditional on the positive quadrant, its angle Θ is uniform on $(0, \pi/2)$ and $Z = \cot\Theta$. For $z > 0$,

$$F_Z(z) = \mathbb{P}(\Theta \geq \arctan(1/z)) = \frac{2}{\pi}\arctan z,$$

and $F_Z(z) = 0$ for $z \leq 0$. Differentiating,

$$f_Z(z) = \frac{2}{\pi(1+z^2)} \quad (z > 0).$$

Since $F_Z(1) = 1/2$, the median is 1. However,

$$\int_0^\infty z f_Z(z)\, \mathrm{d}z = \frac{2}{\pi}\int_0^\infty \frac{z}{1+z^2}\, \mathrm{d}z = \infty.$$

Finite moments of both numerator and denominator do not ensure a finite ratio mean: a denominator close to zero creates a heavy tail.

AA10.

Conditional on $S = s > 0$, direction remains uniform. Exchange symmetry gives $\mathbb{E}[X^2 \mid S = s] = s/3$. Set

$$a = \mathbb{E}[X^4 \mid S = s], \qquad b = \mathbb{E}[X^2Y^2 \mid S = s].$$

Sign symmetry annihilates mixed odd powers. The rotated coordinate $(X+Y)/\sqrt{2}$ has the same conditional distribution as X, so

$$a = \frac{2a + 6b}{4}, \qquad a = 3b.$$

Expanding $s^2 = (X^2 + Y^2 + Z^2)^2$ gives $s^2 = 3a + 6b = 15b$. Consequently,

$$\mathbb{E}[X^2Y^2 \mid S] = \frac{S^2}{15}, \quad \mathrm{Var}(X^2 \mid S) = \frac{4S^2}{45}, \quad \mathrm{Cov}(X^2, Y^2 \mid S) = -\frac{2S^2}{45}.$$

These identities hold almost surely. With total energy fixed, an increase in one squared coordinate must be offset by the others, explaining negative conditional dependence without contradicting unconditional independence.

AA13.

Order the times as $U_1 < U_2 < U_3$, strict almost surely. Then $D = \min(U_2 - U_1, U_3 - U_2)$. For $0 \le d \le L/2$, the event $D > d$ requires both internal gaps to exceed d. Translate

$$v_1 = U_1, \qquad v_2 = U_2 - d, \qquad v_3 = U_3 - 2d.$$

The region becomes $0 < v_1 < v_2 < v_3 < L - 2d$, of volume $(L - 2d)^3/6$. Multiplying by the ordered density $6/L^3$ yields

$$F_D(d) = \begin{cases} 0, & d < 0, \\ 1 - (1 - 2d/L)^3, & 0 \le d \le L/2, \\ 1, & d > L/2. \end{cases}$$

Tail integration gives

$$\mathbb{E}D = \int_0^{L/2} (1 - 2d/L)^3 \, \mathrm{d}d = \frac{L}{8}.$$

The translation subtracts two constrained internal gaps, not four gaps including endpoints; this explains the factor 2.

AA16.

The minimum has a unique index J almost surely. Fix $J = j$ and write the other residuals as $r_i = X_i - U > 0$. Their joint density with the minimum and its index is

$$\lambda^n \exp\left[-\lambda\left(nu + \sum_{i \ne j} r_i\right)\right] = \frac{1}{n}\,(n\lambda e^{-n\lambda u}) \prod_{i \ne j} \lambda e^{-\lambda r_i}.$$

It factors into a uniform index probability, the density of $U \sim \mathrm{Exp}(n\lambda)$, and $n - 1$ independent rate-λ residual densities. The residuals are independent of U. Their maximum is D, so

$$\mathbb{P}(D \le d) = \begin{cases} 0, & d < 0, \\ (1 - e^{-\lambda d})^{n-1}, & d \ge 0. \end{cases}$$

Tail integration with $y = 1 - e^{-\lambda d}$ gives

$$\mathbb{E}D = \frac{1}{\lambda} \int_0^1 \frac{1 - y^{n-1}}{1 - y} \, \mathrm{d}y = \frac{1}{\lambda} \sum_{k=1}^{n-1} \frac{1}{k}.$$

Writing $H_{n-1} = \sum_{k=1}^{n-1} 1/k$, we obtain $\mathbb{E}[V \mid U] = U + H_{n-1}/\lambda$. Independence follows from exact factorization of the exponential density, not merely from a zero covariance.

AA17.

For $|\rho| < 1$, write $Y = \rho X + \sqrt{1 - \rho^2}\, Z$ with an independent standard normal Z. Conditional on X, the cross term has mean zero, giving

$$\mathbb{E}[Y^2 \mid X] = \rho^2 X^2 + 1 - \rho^2.$$

Using $\mathbb{E}X^2 = 1$ and $\mathbb{E}X^4 = 3$,

$$\mathbb{E}[X^2Y^2] = \rho^2\mathbb{E}X^4 + (1-\rho^2)\mathbb{E}X^2 = 1 + 2\rho^2.$$

Thus the covariance is $2\rho^2$. Each squared variable has variance $3 - 1 = 2$, so

$$\boxed{\operatorname{Corr}(X^2, Y^2) = \rho^2}.$$

At $|\rho| = 1$, $Y = \rho X$ almost surely and the identities follow directly. Squaring loses the sign, since ρ and $-\rho$ give the same result. The nonlinear conditional mean is explicit, but includes a constant term that cannot be discarded.

AA20.

Assume first that $r > 0$. With the polar axis pointing toward a, $U = \cos\Theta$ is uniform on $[-1, 1]$, and

$$D^2 = R^2 + r^2 - 2RrU.$$

Thus D has density $d/(2Rr)$ on $|R - r| < d < R + r$. Therefore

$$\mathbb{E}\frac{1}{D} = \frac{R + r - |R - r|}{2Rr} = \boxed{\frac{1}{\max(R, r)}}.$$

If $r = 0$, then $D = R$ and the result remains valid. On the sphere, zero distance occurs only on a probability-zero event, and the first reciprocal moment is still integrable.

For $r > 0$ and $r \neq R$,

$$\mathbb{E}\frac{1}{D^2} = \frac{1}{2Rr}\log\frac{R + r}{|R - r|}.$$

At $r = R$, the lower endpoint is zero and the integral is proportional to $\int_0^{2R} \mathrm{d}d/d$, which diverges. A geometric singularity may preserve a first moment while destroying the second.

AA06.

A fresh pair agrees with probability $p^2 + q^2 > 0$, giving a geometric tail. Let e_S, e_F be the remaining means when the last outcome was success or failure. First-step equations are

$$e_S = 1 + qe_F, \qquad e_F = 1 + pe_S.$$

They give $e_S = (1 + q)/(1 - pq)$ and $e_F = (1 + p)/(1 - pq)$. Including the first trial,

$$\boxed{\mathbb{E}T = 1 + pe_S + qe_F = \frac{2 + pq}{1 - pq}}.$$

Let a_S, a_F be the corresponding probabilities of eventual SS absorption. From $a_S = p + qa_F$ and $a_F = pa_S$,

$$\boxed{\mathbb{P}(\text{stop on SS}) = pa_S + qa_F = \frac{p^2(1 + q)}{1 - pq}}.$$

Interchanging p, q gives the FF probability, and the two sum to 1. For fair trials the mean is 3 and either absorption type has probability $1/2$. The last outcome determines which next outcome stops the process, so one undifferentiated transient state is insufficient.

AA08.

Let I_i indicate H at trial i and T at trial $i+1$, for $1 \le i \le n-1$, and put $a = p(1-p)$. Then $N = \sum I_i$, with each indicator having mean a and variance $a(1-a)$.

Starts separated by at least two use disjoint trials and are independent. Adjacent starts cannot both be HT, since their shared trial cannot be both H and T. Hence

$$\operatorname{Cov}(I_i, I_{i+1}) = -a^2.$$

There are exactly $n-2$ neighboring indicator pairs, yielding

$$\boxed{\mathbb{E}N = (n-1)a, \qquad \operatorname{Var} N = (n-1)a - (3n-5)a^2}.$$

At $n = 10, a = 1/4$, the mean is $9/4$ and the variance is $11/16$. A binomial model ignores overlap and overstates the variance.

AA12.

The uniform law is stationary, so $\mathbb{E}Y_t = 0$ and $\mathbb{E}Y_t^2 = 1/2$ at every time. Set $\lambda = \cos(2\pi/n) \in (-1, 1)$. The cosine addition formula gives

$$\mathbb{E}[Y_{t+1} \mid X_t] = \lambda Y_t.$$

Iterating conditional expectations yields $\mathbb{E}[Y_k \mid X_0] = \lambda^k Y_0$, hence

$$\boxed{\operatorname{Cov}(Y_0, Y_k) = \tfrac{1}{2}\lambda^k}.$$

Expanding the variance using stationarity,

$$\boxed{\operatorname{Var} S_T = \frac{1}{2}\left[T + 2\sum_{k=1}^{T-1}(T-k)\lambda^k\right]}.$$

After division by T, absolute summability of $|\lambda|^k$ dominates $(1 - k/T)\mathbf{1}_{\{k<T\}}\lambda^k$, giving

$$\lim_{T\to\infty} \frac{\operatorname{Var} S_T}{T} = \frac{1}{2} + \frac{\lambda}{1-\lambda} = \frac{1+\lambda}{2(1-\lambda)}.$$

For $n = 6, T = 4$, $\lambda = 1/2$ and the finite variance is $33/8$, rather than the value 2 obtained under an incorrect independence assumption.

AA04.

Factorials contain at least as many factors of 2 as of 5, so the trailing-zero count is

$$z(n) = \sum_{j\ge 1} \left\lfloor \frac{n}{5^j} \right\rfloor.$$

It is nondecreasing and $z(5K) \geq K$. Binary-search the integer range $[0, 5K]$ for the first n_* with $z(n_*) \geq K$. If the value exceeds K, a jump has skipped the target and no solution exists. If equality holds, return the corresponding constant block.

For each $m \geq 0$, all floors are constant on $5m, \ldots, 5m+4$, whereas advancing to $5m+5$ adds at least one factor of 5. Thus every attained value belongs to exactly one block of five consecutive integers. A successful search returns $n_*, \ldots, n_*+4$. This also covers $K = 0$, giving 0 through 4.

Repeated division by 5 evaluates $z(n)$ in $O(\log(n+2))$ integer operations. Binary search uses $O(\log^2(K+2))$ such operations overall. Since $z(100) = 20 + 4 = 24$, the first solution set is 100 through 104. But $z(120) = 28$ and $z(125) = 25 + 5 + 1 = 31$, so 29 is unattainable.

AA05.

Use

$$\binom{n}{k} = \frac{(n-k+1)\cdots n}{1\cdots k},$$

replacing k by $\min(k, n-k)$ if useful. Strip every power of 2 and 5 from each numerator and denominator factor. Accumulate their net exponents a, b as numerator minus denominator totals. All remaining factors are units modulo 10: multiply numerator units and inverses of denominator units to obtain $u \in \{1, 3, 7, 9\}$. Their inverses are respectively $1, 7, 3, 9$, verified by multiplication.

Integrality of the binomial coefficient implies $a, b \geq 0$. Put $z = \min(a, b)$. The desired digit is

$$\boxed{u\, 2^{a-z} 5^{b-z} \pmod{10}}.$$

Only the stripped unit factors are inverted. An original denominator containing 2 or 5 has no inverse modulo 10, so ordinary modular division would be invalid. Repeated stripping computes the exponents, and periodicity or binary exponentiation computes the final powers. An empty product at $k = 0$ gives 1.

For example, $\binom{20}{7} = 77520$ has net exponents $a = 4, b = 1$ and unit part $u = 9$. Removing one factor of 10 leaves $9 \cdot 2^3 \equiv 2 \pmod{10}$. The full number is shown only as a check, not as an algorithmic requirement.

AA11.

A displacement (a, b) gives $\gcd(|a|, |b|) - 1$ interior lattice points. Dividing the displacement by its gcd gives the primitive integer direction, and all lattice points on the segment occur at integer multiples of this basic step. Visibility is therefore equivalent to gcd 1.

Horizontally only adjacent points qualify, contributing $M(N+1)$ pairs; vertical pairs contribute $N(M+1)$. For positive absolute displacement a, b, each of the

two slope directions has $(M+1-a)(N+1-b)$ pairs. Thus the number of visible unordered pairs is

$$V = M(N+1) + N(M+1) + 2 \sum_{\substack{1\le a\le M,\ 1\le b\le N \\ \gcd(a,b)=1}} (M+1-a)(N+1-b).$$

Divide by $\binom{(M+1)(N+1)}{2}$ for the probability. With $M = N = 2$, axis directions contribute 12 pairs, and positive primitive displacements $(1,1)$, $(1,2)$, $(2,1)$ contribute 16 more. The probability is $28/36 = 7/9$.

AA14.

Without ordering restrictions there are $\binom{N-1}{2}$ positive triples. If the third coordinate violates the strict triangle condition, then $z \ge x + y$. Put $s = x + y$; for $2 \le s \le \lfloor N/2 \rfloor$, there are $s - 1$ positive ordered pairs. The bad count for that coordinate is therefore $\binom{\lfloor N/2 \rfloor}{2}$.

With positive coordinates, two coordinates cannot each be at least the sum of the others, so the three bad events are disjoint. The ordered valid count is

$$A(N) = \binom{N-1}{2} - 3\binom{\lfloor N/2 \rfloor}{2} = \frac{N^2}{8} + O(N).$$

Distinct-sided triangles have six permutations. Equal-coordinate solutions number only $O(N)$, and their orbit sizes are 3 or 1. Consequently

$$T(N) = \frac{A(N)}{6} + O(N) = \frac{N^2}{48} + O(N).$$

For $N = 12$, the largest side satisfies $4 \le z < 6$. The solutions are $(4,4,4)$, $(2,5,5)$, and $(3,4,5)$. Their permutation counts are 1, 3, and 6, totaling 10 ordered triples. This demonstrates why division by 6 is not an exact count.

AA15.

The conjugate $\beta = (3 - \sqrt{5})/2$ lies in $(0, 1/2)$, and both numbers solve $z^2 - 3z + 1 = 0$. Define

$$L_0 = 2, \quad L_1 = 3, \quad L_n = 3L_{n-1} - L_{n-2}.$$

Induction or the characteristic roots give $L_n = \alpha^n + \beta^n \in \mathbb{Z}$. Thus $L_n - \alpha^n = \beta^n \in (0, 1/2)$: it is the unique nearest integer with exact error β^n. Also $\lfloor \alpha^n \rfloor = L_n - 1$.

Modulo 2, the recurrence is $L_n \equiv L_{n-1} + L_{n-2}$. Starting from $(0, 1)$ gives the repeating pattern $(0, 1, 1)$, so L_n is even exactly when 3 divides n.

The error decreases strictly. Solving $\beta^n < 10^{-6}$ gives $n > 6 \log 10/(-\log \beta) \approx 14.35$, hence the minimum is 15. Directly, $\beta^{14} \approx 1.4072 \times 10^{-6}$ and $\beta^{15} \approx 5.3749 \times 10^{-7}$.

AA21.

A nonzero product excludes digit 0. Digits 5 and 7 are also impossible because neither divides 72. One digit cannot have product 72. A two-digit product of 72 requires digits 8 and 9, whose sum 17 satisfies neither target.

The complete sorted three-digit multisets with product 72 are

$$(1, 8, 9),\ (2, 4, 9),\ (2, 6, 6),\ (3, 3, 8),\ (3, 4, 6).$$

Enumerate the minimum digit as 1, 2, or 3 to obtain this list. Minimum digit 4 would require two digits at least 4 with product 18, impossible; minimum at least 5 already makes the product at least 125.

Sum 14 leaves $(2, 6, 6)$ and $(3, 3, 8)$, whose smallest arrangements are 266 and 338. The minimum is 266. Sum 18 leaves only $(1, 8, 9)$, giving 189. Every integer with at least four digits exceeds these three-digit solutions. Deleting the 1 in 189 preserves the product but changes the sum from 18 to 17, violating the constraint.

AA18.

If $\log 2$ were algebraic, it would be a nonzero algebraic number. The given theorem would make $e^{\log 2} = 2$ transcendental, contradicting that 2 is an integer. Thus $\log 2$ is transcendental; the same argument applies to $\log 3$ and $\log 4$.

If $\log 2/\log 3 = a/b$ were rational, choose positive integers a, b. Cross-multiplication and exponentiation give $2^b = 3^a$. Unique prime factorization rules this out: the left side has only prime factor 2, and the right only prime factor 3. The ratio is therefore irrational.

A ratio of transcendental numbers can nevertheless be rational: $\log 4/\log 2 = 2$. The proof above establishes irrationality of $\log 2/\log 3$, not its transcendence directly from the given theorem. These conclusions require different arguments and must not be interchanged.